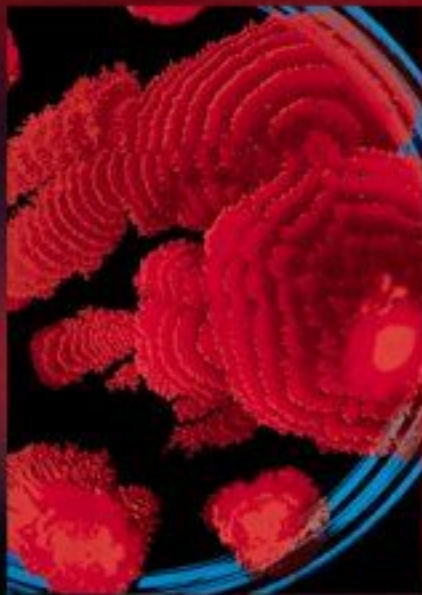


Eighth Edition

Microbiological Applications

Laboratory Manual in General Microbiology



Benson

Preface

This eighth edition of *Microbiological Applications* differs from the previous edition in that it has acquired four new exercises and dropped three experiments. It retains essentially the same format throughout, however. In response to requests for more emphasis on laboratory safety, three new features have been incorporated into the text. In addition, several experiments have been altered to improve simplicity and reliability.

The three exercises that were dropped pertain to flagellar staining, bacterial conjugation, and nitrification in soil. All of these exercises were either difficult to perform, unreliable, or of minimal pedagogical value.

To provide greater safety awareness in the laboratory, the following three features were added: (1) an introductory laboratory protocol, (2) many cautionary boxes dispersed throughout the text, and (3) a new exercise pertaining to aseptic technique.

The three-page laboratory protocol, which follows this preface, replaces the former introduction. It provides terminology, safety measures, an introduction to aseptic technique, and other rules that apply to laboratory safety.

To alert students to potential hazards in performing certain experiments, caution boxes have been incorporated wherever they are needed. Although most of these cautionary statements existed in previous editions, they were not emphasized as much as they are in this edition.

Exercise 8 (Aseptic Technique) has been structured to provide further emphasis on culture tube handling. In previous editions it was assumed that students would learn these important skills as experiments were performed. With the risk of being redundant, six pages have been devoted to the proper handling of culture tubes when making inoculation transfers.

Although most experiments remain unchanged, there are a few that have been considerably altered. Exercise 27 (Isolation of Anaerobic Phototrophic Bacteria), in particular, is completely new. By using the Winogradsky column for isolating and identifying the phototrophic sulfur bacteria, it has been possible to greatly enrich the scope of this experiment. Another exercise that has been altered somewhat is Exercise 48, which pertains to oxidation and fermentation tests that are used for identifying unknown bacteria.

The section that has undergone the greatest reorganization is Part 10 (Microbiology of Soil). In the previous edition it consisted of five exercises. In this edition it has been expanded to seven exercises. A more complete presentation of the nitrogen cycle is offered in Exercise 58, and two new exercises (Exercises 61 and 62) are included that pertain to the isolation of denitrifiers.

In addition to the above changes there has been considerable upgrading of graphics throughout the book. Approximately thirty-five illustrations have been replaced. Several critical color photographs pertaining to molds and physiological tests were also replaced to bring about more faithful color representation.

I am greatly indebted to my editors, Jean Fornango and Jim Smith, who made the necessary contacts for critical reviews. As a result of their efforts the following individuals have provided me with excellent suggestions for improvement of this manual: Barbara Collins at California Lutheran University, Thousand Oaks, CA; Alfred Brown of Auburn University, Auburn, AL; Lester A. Scharlin at El Camino College, Torrance, CA; and Hershell Hanks at Collin County Community College, Plano, TX.

Laboratory Protocol

Welcome to the exciting field of microbiology! The intent of this laboratory manual is to provide you with basic skills and tools that will enable you to explore a vast microbial world. Its scope is incredibly broad in that it includes a multitude of viruses, bacteria, protozoans, yeasts, and molds. Both beneficial and harmful ones will be studied. Although an in-depth study of any single one of these groups could constitute a full course by itself, we will be able to barely get acquainted with them.

To embark on this study it will be necessary for you to learn how to handle cultures in such a way that they are not contaminated or inadvertently dispersed throughout the classroom. This involves learning aseptic techniques and practicing preventive safety measures. The procedures outlined here address these two aspects. **It is of paramount importance that you know all the regulations that are laid down here as Laboratory Protocol.**

Scheduling During the first week of this course your instructor will provide you with a schedule of laboratory exercises arranged in the order of their performance. Before attending laboratory each day, check the schedule to see what experiment or experiments will be performed and prepare yourself so that you understand what will be done.

Each laboratory session will begin with a short discussion to brief you on the availability of materials and procedures. *Since the preliminary instructions start promptly at the beginning of the period, it is extremely important that you are not late to class.*

Personal Items When you first enter the lab, place all personal items such as jackets, bags, and books in some out of the way place for storage. Don't stack them on your desktop. Desk space is minimal and must be reserved for essential equipment and your laboratory manual. The storage place may be a drawer, locker, coatrack, or perimeter counter. Your instructor will indicate where they should be placed.

Attire A lab coat or apron must be worn at all times in the laboratory. It will protect your clothing from accidental contamination and stains in the lab. When leaving the laboratory, remove the coat or apron. In

addition, long hair must be secured in a ponytail to prevent injury from Bunsen burners and contamination of culture material.

TERMINOLOGY

Various terms such as sterilization, disinfection, germicides, sepsis, and aseptic techniques will be used here. To be sure that you understand exactly what they mean, the following definitions are provided.

Sterilization is a process in which all living microorganisms, including viruses, are destroyed. The organisms may be killed with steam, dry heat, or incineration. If we say an article is sterile, we understand that it is completely free of all living microorganisms. Generally speaking, when we refer to sterilization as it pertains here to laboratory safety, we think, primarily, in terms of steam sterilization with the autoclave. The ultimate method of sterilization is to burn up the infectious agents or *incinerate* them. All biological wastes must ultimately be incinerated for disposal.

Disinfection is a process in which vegetative, nonsporing microorganisms are destroyed. Agents that cause disinfection are called **disinfectants** or **germicides**. Such agents are used only on inanimate objects because they are toxic to human and animal tissues.

Sepsis is defined as the growth (multiplication) of microorganisms in tissues of the body. The term **asepsis** refers to any procedure that prevents the entrance of infectious agents into sterile tissues, thus preventing infection. **Aseptic techniques** refer to those practices that are used by microbiologists to exclude all organisms from contaminating media or contacting living tissues. **Antiseptics** are chemical agents (often dilute disinfectants) that can be safely applied externally to human tissues to destroy or inhibit vegetative bacteria.

ASEPTIC TECHNIQUES

When you start handling bacterial cultures as in Exercises 9 and 10, you will learn the specifics of aseptic techniques. Some of the basic things you will do are as follows:

Laboratory Protocol

Hand Washing Before you start working in the lab, wash your hands with a liquid detergent and dry them with paper toweling. At the end of the period, before leaving the laboratory, wash them again.

Tabletop Disinfection. The first chore of the day will be to sponge down your desktop with a disinfectant. This process removes any dust that may be present and minimizes the chances of bacterial contamination of cultures that you are about to handle.

Your instructor will indicate where the bottles of disinfectant and sponges are located. At the end of the period before leaving the laboratory, perform the same procedure to protect students that may occupy your desk in the next class.

Bunsen Burner Usage When using a Bunsen burner to flame loops, needles, and test tubes, follow the procedures outlined in Exercise 8. Inoculating loops and needles should be heated until they are red-hot. Before they are introduced into cultures, they must be allowed to cool down sufficiently to prevent killing organisms that are to be transferred.

If your burner has a pilot on it and you plan to use the burner only intermittently, use it. If your burner lacks a pilot, turn off the burner when it is not being used. Excessive unnecessary use of Bunsen burners in a small laboratory can actually raise the temperature of the room. More important is the fact that unattended burner flames are a constant hazard to hair, clothing, and skin.

The proper handling of test tubes, while transferring bacteria from one tube to another, requires a certain amount of skill. Test-tube caps must never be placed down on the desktop while you are making inoculations. Techniques that enable you to make transfers properly must be mastered. Exercise 8 pertains to these skills.

Pipetting Transferring solutions or cultures by pipette must always be performed with a mechanical suction device. Under no circumstances is pipetting by mouth allowed in this laboratory.

Disposal of Cultures and Broken Glass The following rules apply to culture and broken glass disposal:

1. Petri dishes must be placed in a plastic bag to be autoclaved.
2. Unneeded test-tube cultures must be placed in a wire basket to be autoclaved.
3. Used pipettes must be placed in a plastic bag for autoclaving.
4. Broken glass should be swept up into a dustpan and placed in a container reserved for broken

glass. Don't try to pick up the glass fragments with your fingers.

5. Contaminated material must never be placed in a wastebasket.

ACCIDENTAL SPILLS

All accidental spills, whether chemical or biological, must be reported immediately to your instructor. Although the majority of microorganisms used in this laboratory are nonpathogens, some pathogens will be encountered. It is for this reason that we must treat all accidental biological spills as if pathogens were involved.

Chemical spills are just as important to report because some agents used in this laboratory may be carcinogenic; others are poisonous; and some can cause dermal damage such as blistering and depigmentation.

Decontamination Procedure Once your instructor is notified of an accidental spill, the following steps will take place:

1. Any clothing that is contaminated should be placed in an autoclavable plastic bag and autoclaved.
2. Paper towels, soaked in a suitable germicide, such as 5% bleach, are placed over the spill.
3. Additional germicide should be poured around the edges of the spill to prevent further aerosolization.
4. After approximately 20 minutes, the paper towels should be scraped up off the floor with an autoclavable squeegee into an autoclavable dust pan.
5. The contents of the dust pan are transferred to an autoclavable plastic bag, which may itself be placed in a stainless steel bucket or pan for transport to an autoclave.
6. All materials, including the squeegee and dustpan, are autoclaved.

ADDITIONAL IMPORTANT REGULATIONS

Here are a few additional laboratory rules:

1. Don't remove cultures, reagents, or other materials from the laboratory unless you have been granted specific permission.
2. Don't smoke or eat food in the laboratory.
3. Make it a habit to keep your hands away from your mouth. Obviously, labels are never moistened with the tongue; use tap water or self-adhesive labels instead.

Laboratory Protocol

4. Always clean up after yourself. Gram-stained slides that have no further use to you should be washed and dried and returned to a slide box. Coverslips should be cleaned, dried, and returned. Staining trays should be rinsed out and returned to their storage place.
5. Return all bulk reagent bottles to places of storage.
6. Return inoculating loops and needles to your storage container. Be sure that they are not upside down.
7. If you have borrowed something from someone, return it.
8. Do not leave any items on your desk at the end of the period.
9. Do not disturb another class at any time. Wait until the class is dismissed.
10. Treat all instruments, especially microscopes, with extreme care. If you don't understand how a piece of equipment functions, ask your instructor.
11. Work cooperatively with other students in group-assigned experiments, but do your own analyses of experimental results.

PART 1 Microscopy

Although there are many kinds of microscopes available to the microbiologist today, only four types will be described here for our use: the brightfield, darkfield, phase-contrast, and fluorescence microscopes. If you have had extensive exposure to microscopy in previous courses, this unit may not be of great value to you; however, if the study of microorganisms is a new field of study for you, there is a great deal of information that you need to acquire about the proper use of these instruments.

Microscopes in a college laboratory represent a considerable investment and require special care to prevent damage to the lenses and mechanicals. The fact that a laboratory microscope may be used by several different individuals during the day and moved around from one place to another results in a much greater chance for damage and wear to occur than if the instrument were used by only one individual.

The complexity of some of the more expensive microscopes also requires that certain adjustments be made periodically. Knowing how to make these adjustments to get the equipment to perform properly is very important. An attempt is made in the five exercises of this unit to provide the necessary assistance in getting the most out of the equipment.

Microscopy should be as fascinating to the beginner as it is to the professional of long standing; however, only with intelligent understanding can the beginner approach the achievement that occurs with years of experience.

1

Brightfield Microscopy

A microscope that allows light rays to pass directly through to the eye without being deflected by an intervening opaque plate in the condenser is called a **brightfield microscope**. This is the conventional type of instrument encountered by students in beginning courses in biology; it is also the first type to be used in this laboratory.

All brightfield microscopes have certain things in common, yet they differ somewhat in mechanical operation. An attempt will be made in this exercise to point out the similarities and differences of various makes so that you will know how to use the instrument that is available to you. Before attending the first laboratory session in which the microscope will be used, read over this exercise and answer all the questions on the Laboratory Report. Your instructor may require that the Laboratory Report be handed in prior to doing any laboratory work.

CARE OF THE INSTRUMENT

Microscopes represent considerable investment and can be damaged rather easily if certain precautions are not observed. The following suggestions cover most hazards.

Transport When carrying your microscope from one part of the room to another, use both hands when holding the instrument, as illustrated in figure 1.1. If it is carried with only one hand and allowed to dangle at your side, there is always the danger of collision with furniture or some other object. And, incidentally, *under no circumstances should one attempt to carry two microscopes at one time.*

Clutter Keep your workstation uncluttered while doing microscopy. Keep unnecessary books, lunches, and other unneeded objects away from your work area. A clear work area promotes efficiency and results in fewer accidents.

Electric Cord Microscopes have been known to tumble off of tabletops when students have entangled a foot in a dangling electric cord. Don't let the light cord on your microscope dangle in such a way as to hazard foot entanglement.

Lens Care At the beginning of each laboratory period check the lenses to make sure they are clean. At the end of each lab session be sure to wipe any immersion oil off the immersion lens if it has been used. More specifics about lens care are provided on page 5.

Dust Protection In most laboratories dustcovers are used to protect the instruments during storage. If one is available, place it over the microscope at the end of the period.

COMPONENTS

Before we discuss the procedures for using a microscope, let's identify the principal parts of the instrument as illustrated in figure 1.2.

Framework All microscopes have a basic frame structure, which includes the **arm** and **base**. To this framework all other parts are attached. On many of the older microscopes the base is not rigidly attached to the arm as is the case in figure 1.2; instead, a pivot point is present that enables one to tilt the arm backward to adjust the eyepoint height.

Stage The horizontal platform that supports the microscope slide is called the **stage**. Note that it has a clamping device, the **mechanical stage**, which is used for holding and moving the slide around on the

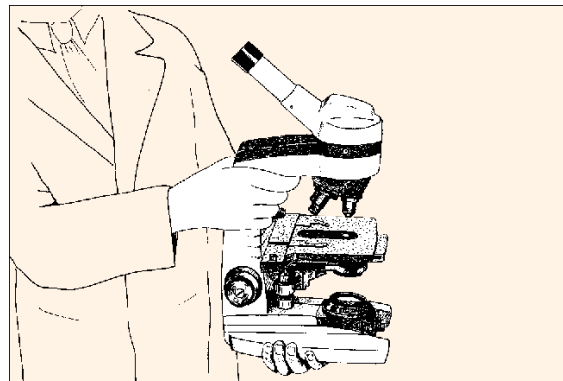


Figure 1.1 The microscope should be held firmly with both hands while carrying it.

Brightfield Microscopy • Exercise 1

stage. Note, also, the location of the **mechanical stage control** in figure 1.2.

Light Source In the base of most microscopes is positioned some kind of light source. Ideally, the lamp should have a **voltage control** to vary the intensity of light. The microscope in figure 1.2 has a knurled wheel on the right side of its base to regulate the voltage supplied to the light bulb. The microscope base in figure 1.4 has a knob (the left one) that controls voltage.

Most microscopes have some provision for reducing light intensity with a **neutral density filter**. Such a filter is often needed to reduce the intensity of light below the lower limit allowed by the voltage control. On microscopes such as the Olympus CH-2, one can simply place a neutral density filter over the light source in the base. On some microscopes a filter is built into the base.

Lens Systems All microscopes have three lens systems: the oculars, the objectives, and the condenser.

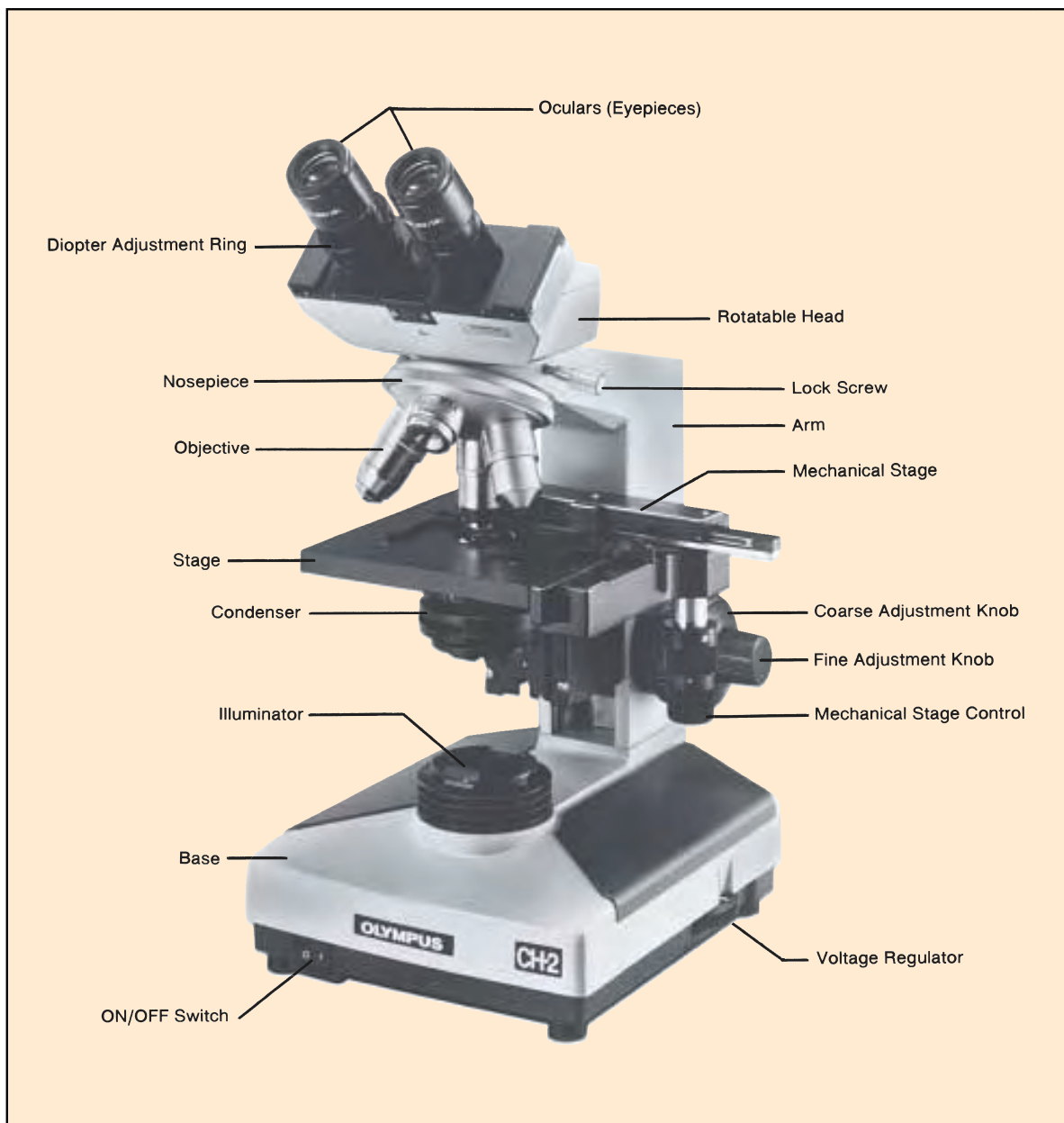


Figure 1.2 The compound microscope

Courtesy of the Olympus Corporation, Lake Success, N.Y.

Exercise 1 • Brightfield Microscopy

Figure 1.3 illustrates the light path through these three systems.

The **ocular**, or eyepiece, is a complex piece, located at the top of the instrument, that consists of two or more internal lenses and usually has a magnification of $10\times$. Although the microscope in figure 1.2 has two oculars (binocular), a microscope often has only one.

Three or more **objectives** are usually present. Note that they are attached to a rotatable **nosepiece**, which makes it possible to move them into position over a slide. Objectives on most laboratory microscopes have magnifications of $10\times$, $45\times$, and $100\times$, designated as **low power**, **high-dry**, and **oil immersion**, respectively. Some microscopes will have a fourth objective for rapid scanning of microscopic fields that is only $4\times$.

The third lens system is the **condenser**, which is located under the stage. It collects and directs the light from the lamp to the slide being studied. The condenser can be moved up and down by a knob under the stage. A **diaphragm** within the condenser regulates the amount of light that reaches the slide. Microscopes that lack a voltage control on the light source rely entirely on the diaphragm for controlling light intensity. On the Olympus microscope in figure 1.2 the diaphragm is controlled by turning a knurled ring. On some microscopes a diaphragm lever is present. Figure 1.3 illustrates the location of the condenser and diaphragm.

Focusing Knobs The concentrically arranged **coarse adjustment** and **fine adjustment knobs** on the side of the microscope are used for bringing objects into focus when studying an object on a slide. On some microscopes these knobs are not positioned concentrically as shown here.

Ocular Adjustments On binocular microscopes one must be able to change the distance between the oculars and to make diopter changes for eye differences. On most microscopes the interocular distance is changed by simply pulling apart or pushing together the oculars.

To make diopter adjustments, one focuses first with the right eye only. Without touching the focusing knobs, diopter adjustments are then made on the left eye by turning the knurled **diopter adjustment ring** (figure 1.2) on the left ocular until a sharp image is seen. One should now be able to see sharp images with both eyes.

RESOLUTION

The resolution limit, or **resolving power**, of a microscope lens system is a function of its numerical aperture, the wavelength of light, and the design of the

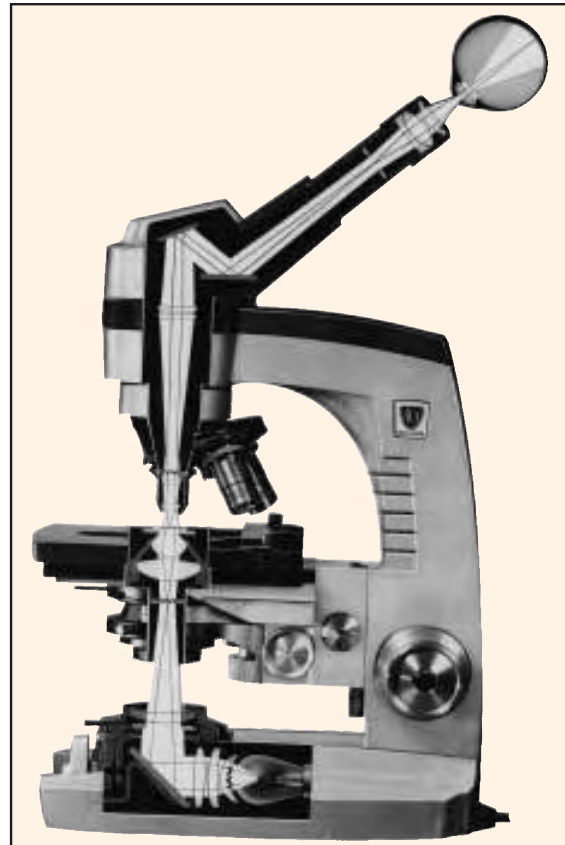


Figure 1.3 The light pathway of a microscope.

condenser. The optimum resolution of the best microscopes with oil immersion lenses is around $0.2\ \mu\text{m}$. This means that two small objects that are $0.2\ \mu\text{m}$ apart will be seen as separate entities; objects closer than that will be seen as a single object.

To get the maximum amount of resolution from a lens system, the following factors must be taken into consideration:

- A **blue filter** should be in place over the light source because the short wavelength of blue light provides maximum resolution.
- The **condenser** should be kept at its highest position where it allows a maximum amount of light to enter the objective.
- The **diaphragm** should not be stopped down too much. Although stopping down improves contrast, it reduces the numerical aperture.
- **Immersion oil** should be used between the slide and the $100\times$ objective.

Of significance is the fact that, as magnification is increased, the resolution must also increase. Simply increasing magnification by using a $20\times$ ocular won't increase the resolution.

Brightfield Microscopy • Exercise 1

LENS CARE

Keeping the lenses of your microscope clean is a constant concern. Unless all lenses are kept free of dust, oil, and other contaminants, they are unable to achieve the degree of resolution that is intended. Consider the following suggestions for cleaning the various lens components:

Cleaning Tissues Only lint-free, optically safe tissues should be used to clean lenses. Tissues free of abrasive grit fall in this category. Booklets of lens tissue are most widely used for this purpose. Although several types of boxed tissues are also safe, *use only the type of tissue that is recommended by your instructor.*

Solvents Various liquids can be used for cleaning microscope lenses. Green soap with warm water works very well. Xylene is universally acceptable. Alcohol and acetone are also recommended, but often with some reservations. Acetone is a powerful solvent that could possibly dissolve the lens mounting cement in some objective lenses if it were used too liberally. When it is used it should be used sparingly. Your instructor will inform you as to what solvents can be used on the lenses of your microscope.

Oculars The best way to determine if your eyepiece is clean is to rotate it between the thumb and forefinger as you look through the microscope. A rotating pattern will be evidence of dirt.

If cleaning the top lens of the ocular with lens tissue fails to remove the debris, one should try cleaning the lower lens with lens tissue and blowing off any excess lint with an air syringe or gas cannis-

ter. *Whenever the ocular is removed from the microscope, it is imperative that a piece of lens tissue be placed over the open end of the microscope as illustrated in figure 1.5.*

Objectives Objective lenses often become soiled by materials from slides or fingers. A piece of lens tissue moistened with green soap and water, or one of the acceptable solvents mentioned above, will usually remove whatever is on the lens. Sometimes a cotton swab with a solvent will work better than lens tissue. At any time that the image on the slide is unclear or cloudy, assume at once that the objective you are using is soiled.

Condenser Dust often accumulates on the top surface of the condenser; thus, wiping it off occasionally with lens tissue is desirable.

PROCEDURES

If your microscope has three objectives you have three magnification options: (1) low-power, or $100\times$ total magnification, (2) high-dry magnification, which is $450\times$ total with a $45\times$ objective, and (3) $1000\times$ total magnification with a $100\times$ oil immersion objective. Note that the total magnification seen through an objective is calculated by simply multiplying the power of the ocular by the power of the objective.

Whether you use the low-power objective or the oil immersion objective will depend on how much magnification is necessary. Generally speaking, however, it is best to start with the low-power objective and progress to the higher magnifications as your study progresses. Consider the following suggestions for setting up your microscope and making microscopic observations.



Figure 1.4 On this microscope, the left knob controls voltage. The other knob is used for moving a neutral density filter into position.

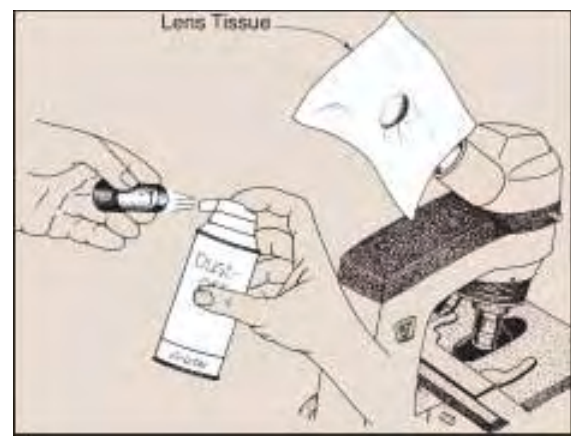


Figure 1.5 When oculars are removed for cleaning, cover the ocular opening with lens tissue. A blast from an air syringe or gas cannister removes dust and lint.

Exercise 1 • Brightfield Microscopy

Viewing Setup If your microscope has a rotatable head, such as the ones being used by the two students in figure 1.6, there are two ways that you can use the instrument. Note that the student on the left has the arm of the microscope *near* him, and the other student has the arm *away from* her. With this type of microscope, the student on the right has the advantage in that the stage is easier to observe. Note, also that when focusing the instrument she is able to rest her arm on the table. The manufacturer of this type of microscope intended that the instrument be used in the way demonstrated by the young lady. If the microscope head is not rotatable, it will be necessary to use the other position.

Low-Power Examination The main reason for starting with the low-power objective is to enable you to explore the slide to look for the object you are planning to study. Once you have found what you are looking for, you can proceed to higher magnifications. Use the following steps when exploring a slide with the low-power objective:

1. Position the slide on the stage with the material to be studied on the *upper* surface of the slide. Figure 1.7 illustrates how the slide must be held in place by the mechanical stage retainer lever.
2. Turn on the light source, using a *minimum* amount of voltage. If necessary, reposition the slide so that the stained material on the slide is in the *exact center* of the light source.
3. Check the condenser to see that it has been raised to its highest point.
4. If the low-power objective is not directly over the center of the stage, rotate it into position. Be sure that as you rotate the objective into position it clicks into its locked position.
5. Turn the coarse adjustment knob to lower the objective until it stops. A built-in stop will prevent the objective from touching the slide.



Figure 1.6 The microscope position on the right has the advantage of stage accessibility.

6. While looking down through the ocular (or oculars), bring the object into focus by turning the fine adjustment focusing knob. Don't readjust the coarse adjustment knob. If you are using a binocular microscope it will also be necessary to adjust the interocular distance and diopter adjustment to match your eyes.
7. Manipulate the diaphragm lever to reduce or increase the light intensity to produce the clearest, sharpest image. Note that as you close down the diaphragm to reduce the light intensity, the contrast improves and the depth of field increases. Stopping down the diaphragm when using the low-power objective does not decrease resolution.
8. Once an image is visible, move the slide about to search out what you are looking for. The slide is moved by turning the knobs that move the mechanical stage.
9. Check the cleanliness of the ocular, using the procedure outlined earlier.
10. Once you have identified the structures to be studied and wish to increase the magnification, you may proceed to either high-dry or oil immersion magnification. However, before changing objectives, *be sure to center the object you wish to observe*.

High-Dry Examination To proceed from low-power to high-dry magnification, all that is necessary is to rotate the high-dry objective into position and open up the diaphragm somewhat. It may be necessary to make a minor adjustment with the fine adjustment knob to sharpen up the image, but *the coarse adjustment knob should not be touched*.

If a microscope is of good quality, only minor focusing adjustments are needed when changing from low power to high-dry because all the objectives will be **parfocalized**. Nonparfocalized micro-

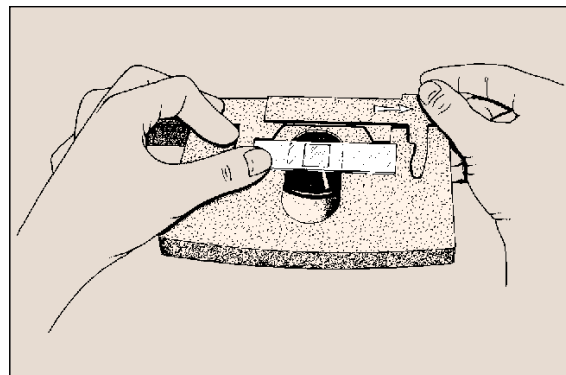


Figure 1.7 The slide must be properly positioned as the retainer lever is moved to the right.

Brightfield Microscopy • Exercise 1

scopes do require considerable refocusing when changing objectives.

High-dry objectives should be used only on slides that have cover glasses; without them, images are usually unclear. When increasing the lighting, be sure to open up the diaphragm first instead of increasing the voltage on your lamp; reason: *lamp life is greatly extended when used at low voltage*. If the field is not bright enough after opening the diaphragm, feel free to increase the voltage. A final point: Keep the condenser at its highest point.

Oil Immersion Techniques The oil immersion lens derives its name from the fact that a special mineral oil is interposed between the lens and the microscope slide. The oil is used because it has the same refractive index as glass, which prevents the loss of light due to the bending of light rays as they pass through air. The use of oil in this way enhances the resolving power of the microscope. Figure 1.8 reveals this phenomenon.

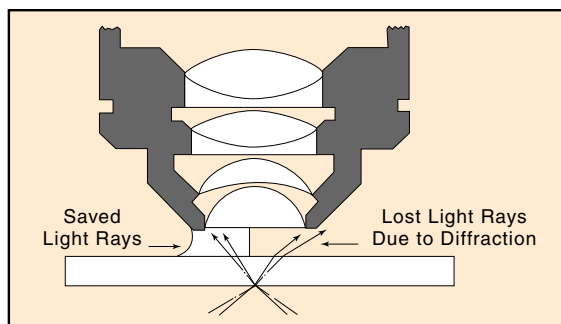


Figure 1.8 Immersion oil, having the same refractive index as glass, prevents light loss due to diffraction.

With parfocalized objectives one can go to oil immersion from either low power or high-dry. On some microscopes, however, going from low power to high power and then to oil immersion is better. Once the microscope has been brought into focus at one magnification, the oil immersion lens can be rotated into position without fear of striking the slide.

Before rotating the oil immersion lens into position, however, a drop of immersion oil must be placed on the slide. An oil immersion lens should never be used without oil. Incidentally, if the oil appears cloudy it should be discarded.

When using the oil immersion lens it is best to open the diaphragm as much as possible. Stopping

down the diaphragm tends to limit the resolving power of the optics. In addition, the condenser must be kept at its highest point. If different colored filters are available for the lamp housing, it is best to use blue or greenish filters to enhance the resolving power.

Since the oil immersion lens will be used extensively in all bacteriological studies, it is of paramount importance that you learn how to use this lens properly. Using this lens takes a little practice due to the difficulties usually encountered in manipulating the lighting. A final comment of importance: At the end of the laboratory period remove all immersion oil from the lens tip with lens tissue.

PUTTING IT AWAY

When you take a microscope from the cabinet at the beginning of the period, you expect it to be clean and in proper working condition. The next person to use the instrument after you have used it will expect the same consideration. A few moments of care at the end of the period will ensure these conditions. Check over this list of items at the end of each period before you return the microscope to the cabinet.

1. Remove the slide from the stage.
2. If immersion oil has been used, wipe it off the lens and stage with lens tissue. (Do not wipe oil off slides you wish to keep. Simply put them into a slide box and let the oil drain off.)
3. Rotate the low-power objective into position.
4. If the microscope has been inclined, return it to an erect position.
5. If the microscope has a built-in movable lamp, raise the lamp to its highest position.
6. If the microscope has a long attached electric cord, wrap it around the base.
7. Adjust the mechanical stage so that it does not project too far on either side.
8. Replace the dustcover.
9. If the microscope has a separate transformer, return it to its designated place.
10. Return the microscope to its correct place in the cabinet.

LABORATORY REPORT

Before the microscope is to be used in the laboratory, answer all the questions on Laboratory Report 1,2 that pertain to brightfield microscopy. Preparation on your part prior to going to the laboratory will greatly facilitate your understanding. Your instructor may wish to collect this report at the *beginning of the period* on the first day that the microscope is to be used in class.

Darkfield Microscopy

2

Delicate transparent living organisms can be more easily observed with darkfield microscopy than with conventional brightfield microscopy. This method is particularly useful when one is attempting to identify spirochaetes in the exudate from a syphilitic lesion. Figure 2.1 illustrates the appearance of these organisms under such illumination. This effect may be produced by placing a darkfield stop below the regular condenser or by replacing the condenser with a specially constructed one.

Another application of darkfield microscopy is in the fluorescence microscope (Exercise 4). Although fluorescence may be seen without a dark field, it is greatly enhanced with this application.

To achieve the darkfield effect it is necessary to alter the light rays that approach the objective in such a way that only oblique rays strike the objects being viewed. The obliquity of the rays must be so extreme that if no objects are in the field, the background is completely light-free. Objects in the field become brightly illuminated, however, by the rays that are reflected up through the lens system of the microscope.

Although there are several different methods for producing a dark field, only two devices will be described here: the star diaphragm and the cardioid condenser. The availability of equipment will determine the method to be used in this laboratory.



Figure 2.1 Transparent living microorganisms, such as the syphilis spirochaete, can be seen much more easily when observed in a dark field.

THE STAR DIAPHRAGM

One of the simplest ways to produce the darkfield effect is to insert a star diaphragm into the filter slot of the condenser housing as shown in figure 2.2. This device has an opaque disk in the center that blocks the central rays of light. Figure 2.3 reveals the effect of this stop on the light rays passing through the condenser. If such a device is not available, one can be made by cutting round disks of opaque paper of different sizes that are cemented to transparent celluloid disks that will fit into the slot. If the microscope normally has a diffusion disk in this slot, it is best to replace it with rigid clear celluloid or glass.

An interesting modification of this technique is to use colored celluloid stops instead of opaque paper. Backgrounds of blue, red, or any color can be produced in this way.

In setting up this type of darkfield illumination it is necessary to keep these points in mind:

1. Limit this technique to the study of large organisms that can be seen easily with low-power magnification. *Good resolution with higher powered objectives is difficult with this method.*
2. Keep the diaphragm wide open and use as much light as possible. If the microscope has a voltage



Figure 2.2 The insertion of a star diaphragm into the filter slot of the condenser will produce a dark field suitable for low magnifications.

Exercise 2 • Darkfield Microscopy

regulator, you will find that the higher voltages will produce better results.

3. Be sure to center the stop as precisely as possible.
4. Move the condenser up and down to produce the best effects.

THE CARDIOID CONDENSER

The difficulty that results from using the star diaphragm or opaque paper disks with high-dry and oil immersion objectives is that the oblique rays are not as carefully metered as is necessary for the higher magnifications. Special condensers such as the cardioid or paraboloid types must be used. Since the cardioid type is the most frequently used type, its use will be described here.

Figure 2.4 illustrates the light path through such a condenser. Note that the light rays entering the lower element of the condenser are reflected first off a convex mirrored surface and then off a second concave surface to produce the desired oblique rays of light. Once the condenser has been installed in the microscope, the following steps should be followed to produce ideal illumination.

Materials:

slides and cover glasses of excellent quality
(slides of 1.15–1.25 mm thickness and
No. 1 cover glasses)

1. Adjust the upper surface of the condenser to a height just below stage level.
2. Place a clear glass slide in position over the condenser.
3. Focus the 10× objective on the top of the condenser until a bright ring comes into focus.
4. Center the bright ring so that it is concentric with the field edge by adjusting the centering screws on the darkfield condenser. If the condenser has a

light source built into it, it will also be necessary to center it as well to achieve even illumination.

5. Remove the clear glass slide.
6. If a funnel stop is available for the oil immersion objective, remove this object and insert this unit. (This stop serves to reduce the numerical aperture of the oil immersion objective to a value that is less than the condenser.)
7. Place a drop of immersion oil on the upper surface of the condenser and place the slide on top of the oil. The following preconditions in slide usage must be adhered to:
 - Slides and cover glasses should be optically perfect. Scratches and imperfections will cause annoying diffractions of light rays.
 - Slides and cover glasses must be free of dirt or grease of any kind.
 - A cover glass should always be used.
8. If the oil immersion lens is to be used, place a drop of oil on the cover glass.
9. If the field does not appear dark and lacks contrast, return to the 10× objective and check the ring concentricity and light source centration. If contrast is still lacking after these adjustments, the specimen is probably too thick.
10. If sharp focus is difficult to achieve under oil immersion, try using a thinner cover glass and adding more oil to the top of the cover glass and bottom of the slide.

LABORATORY REPORT

This exercise may be used in conjunction with Part 2 when studying the various types of organisms. After reading over this exercise and doing any special assignments made by your instructor, answer the questions on the last portion of Laboratory Report 1,2 that pertain to darkfield microscopy.

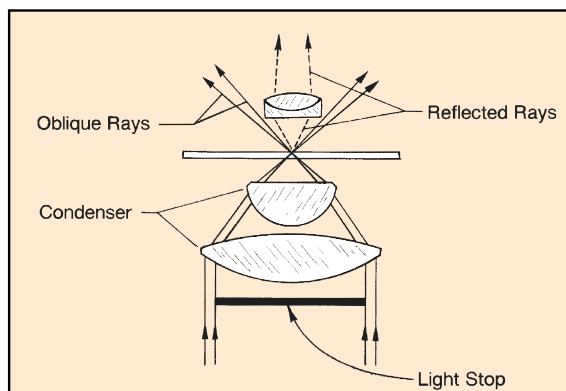


Figure 2.3 The star diaphragm allows only peripheral light rays to pass up through the condenser. This method requires maximum illumination.

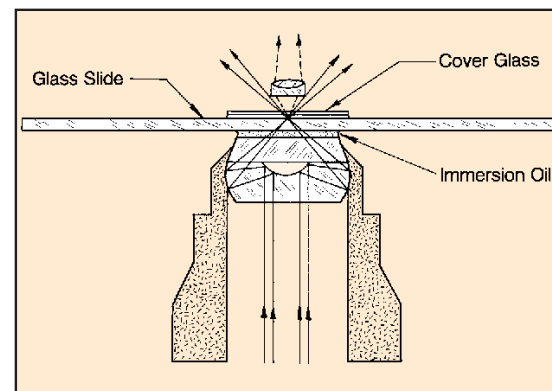


Figure 2.4 A cardioid condenser provides greater light concentration for oblique illumination than the star diaphragm.

Phase-Contrast Microscopy

3

The difficulty that one encounters in trying to examine cellular organelles is that most protoplasmic material is completely transparent and defies differentiation. It is for this reason that stained slides are usually used in brightfield cytological studies. Since the staining of slides results in cellular death, it is obvious that when we study stained microorganisms on a slide, we are observing artifacts rather than living cells.

A microscope that is able to differentiate transparent protoplasmic structures without staining and killing them is the **phase-contrast microscope**. The first phase-contrast microscope was developed in 1933 by Frederick Zernike and was originally referred to as the *Zernike microscope*. It is the instrument of choice for studying living protozoans and other types of transparent cells. Figure 3.1 illustrates the differences between brightfield and phase-contrast images. Note the greater degree of differentiation that can be seen inside cells when they are observed with phase-contrast optics. In this exercise we will study the principles that govern this type of microscope; we will also see how different manufacturers have met the design challenges of these principles.

IMAGE CONTRAST

Objects in a microscopic field may be categorized as being either amplitude or phase objects. **Amplitude objects** (illustration 1, figure 3.2) show up as dark objects under the microscope because the amplitude (intensity) of light rays is reduced as the rays pass through the objects. **Phase objects** (illustration 2, figure 3.2), on the other hand, are completely transparent since light rays pass through them unchanged with respect to amplitude. As some of the light rays pass through phase objects, however, they are retarded by $\frac{1}{4}$ wavelength.

This retardation, known as *phase shift*, occurs with no amplitude diminution; thus, the objects appear transparent rather than opaque. Since most biological specimens are phase objects, lacking in contrast, it becomes necessary to apply dyes of various kinds to cells that are to be studied with a brightfield microscope. To understand how Zernike took advantage of the $\frac{1}{4}$ wavelength phase shift in developing his microscope we must understand the difference between direct and diffracted light rays.



Figure 3.1 Comparison of brightfield and phase-contrast images

Exercise 3 • Phase-Contrast Microscopy

TWO TYPES OF LIGHT RAYS

Light rays passing through a transparent object emerge as either direct or diffracted rays. Those rays that pass straight through unaffected by the medium are called **direct rays**. They are unaltered in amplitude and phase. The balance of the rays that are bent by their slowing through the medium (due to density differences) emerge from the object as **diffracted rays**. It is these rays that are retarded $\frac{1}{4}$ wavelength. Illustration 3, figure 3.2, illustrates these two types of light rays.

An important characteristic of these light rays is that if the direct and diffracted rays of an object can be brought into exact phase, or *coincidence*, with each other, the resultant amplitude of the converged rays is the sum of the two waves. This increase in amplitude will produce increased brightness of the object in the field. On the other hand, if two rays of equal amplitude are in reverse phase ($\frac{1}{2}$ wavelength off), their amplitudes cancel each other to produce a dark object. This phenomenon is called *interference*. Illustration 4, figure 3.2, shows these two conditions.

THE ZERNIKE MICROSCOPE

In constructing his first phase-contrast microscope, Zernike experimented with various configurations of

diaphragms and various materials that could be used to retard or advance the direct light rays. Figure 3.3 illustrates the optical system of a typical modern phase-contrast microscope. It differs from a conventional brightfield microscope by having (1) a different type of diaphragm and (2) a phase plate.

The diaphragm consists of an **annular stop** that allows only a hollow cone of light rays to pass up through the condenser to the object on the slide. The **phase plate** is a special optical disk located at the rear focal plane of the objective. It has a **phase ring** on it that advances or retards the direct light rays $\frac{1}{4}$ wavelength.

Note in figure 3.3 that the direct rays converge on the phase ring to be advanced or retarded $\frac{1}{4}$ wavelength. These rays emerge as solid lines from the object on the slide. This ring on the phase plate is coated with a material that will produce the desired phase shift. The diffracted rays, on the other hand, which have already been retarded $\frac{1}{4}$ wavelength by the phase object on the slide, completely miss the phase ring and are not affected by the phase plate. It should be clear, then, that depending on the type of phase-contrast microscope, the convergence of diffracted and direct rays on the image plane will result in either a brighter image (*amplitude summation*) or a darker

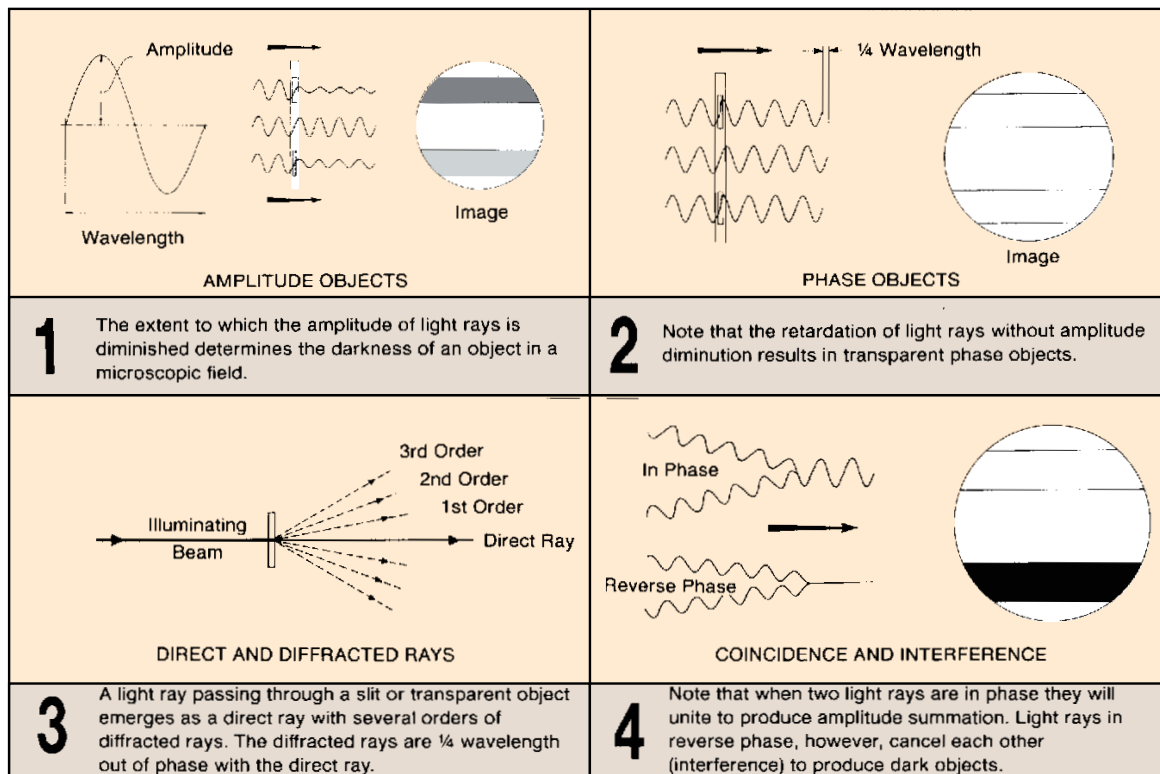


Figure 3.2 The utilization of light rays in phase-contrast microscopy

Phase-Contrast Microscopy • Exercise 3

image (*amplitude interference* or *reverse phase*). The former is referred to as ***bright phase*** microscopy; the latter as ***dark phase*** microscopy. The apparent brightness or darkness, incidentally, is proportional to the square of the amplitude; thus, the image will be four

times as bright or dark as seen through a brightfield microscope.

It should be added here, parenthetically, that the phase plates of some microscopes have coatings to change the phase of the diffracted rays. In any event

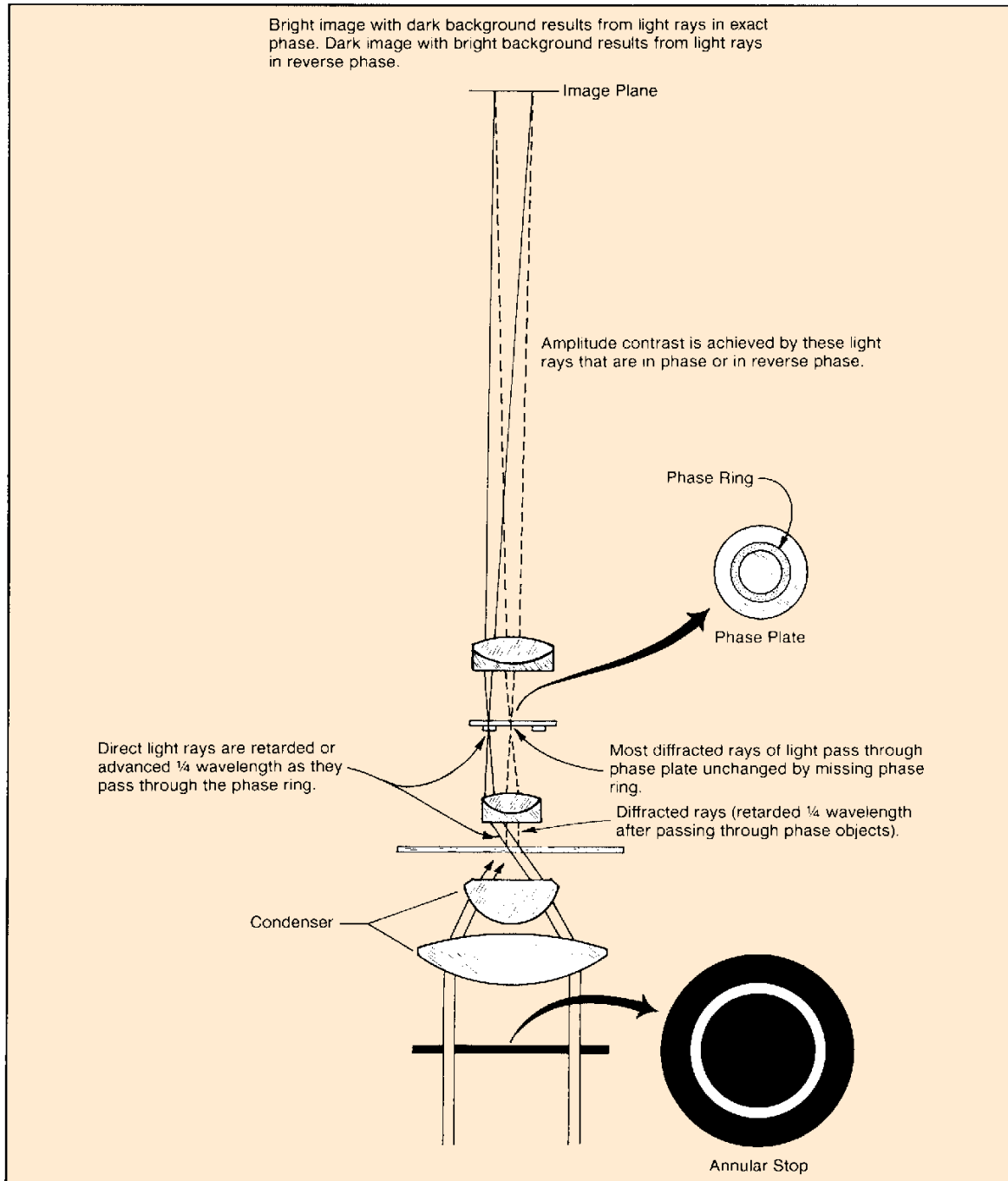


Figure 3.3 The optical system of a phase-contrast microscope

Exercise 3 • Phase-Contrast Microscopy

the end result will be the same: to achieve coincidence or interference of direct and diffracted rays.

MICROSCOPE ADJUSTMENTS

If the annular stop under the condenser of a phase-contrast microscope can be moved out of position, this instrument can also be used for brightfield studies. Although a phase-contrast objective has a phase ring attached to the top surface of one of its lenses, the presence of that ring does not seem to impair the resolution of the objective when it is used in the brightfield mode. It is for this reason that manufacturers have designed phase-contrast microscopes in such a way that they can be quickly converted to brightfield operation.

To make a microscope function efficiently in both phase-contrast and brightfield situations one must master the following procedures:

- lining up the annular ring and phase rings so that they are perfectly concentric,
- adjusting the light source so that maximum illumination is achieved for both phase-contrast and brightfield usage, and
- being able to shift back and forth easily from phase-contrast to brightfield modes. The following suggestions should be helpful in coping with these problems.

Alignment of Annulus and Phase Ring

Unless the annular ring below the condenser is aligned perfectly with the phase ring in the objective, good phase-contrast imagery cannot be achieved. Figure 3.4 illustrates the difference between non-alignment and alignment. If a microscope has only one phase-contrast objective, there will be only one annular stop that has to be aligned. If a microscope has two or more phase objectives, there must be a substage unit with separate annular stops for each phase objective, and alignment procedure must be performed separately for each objective and its annular stop.

Since the objective cannot be moved once it is locked in position, all adjustments are made to the annular stop. On some microscopes the adjustment may be made with tools, as illustrated in figure 3.5. On other microscopes, such as the Zeiss in figure 3.6 which has five phase-contrast objectives, the annular rings are moved into position with special knobs on the substage unit. Since the method of adjustment varies from one brand of microscope to another, one has to follow the instructions provided by the manufacturer. Once the adjustments have been made, they

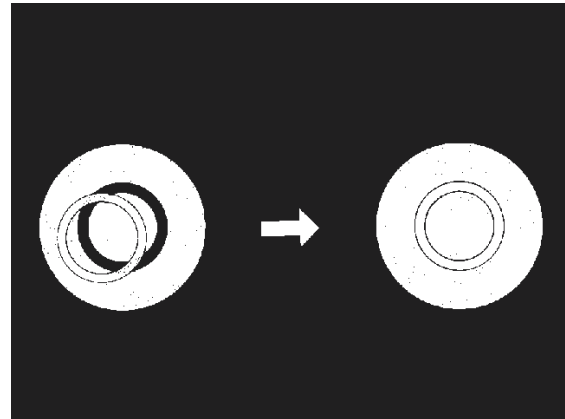


Figure 3.4 The image on the right illustrates the appearance of the rings when perfect alignment of phase ring and annulus diaphragm has been achieved.



Figure 3.5 Alignment of the annulus diaphragm and phase ring is accomplished with a pair of Allen-type screwdrivers on this American Optical microscope.



Figure 3.6 Alignment of the annulus and phase ring on this Zeiss microscope is achieved by adjusting the two knobs as shown.

Phase-Contrast Microscopy • Exercise 3



Figure 3.7 If the ocular of a phase-contrast microscope is replaced with a centering telescope, the orientation of the phase ring and annular ring can be viewed.



Figure 3.8 Some microscopes have an aperture viewing unit that can be used instead of a centering telescope for observing the orientation of the phase ring and annular ring.



Figure 3.9 The annular stop on this American Optical microscope has the annular stop located on a slideway. When pushed in, the annular stop is in position.

are rigidly set and needn't be changed unless someone inadvertently disturbs them.

To observe ring alignment, one can replace the eyepiece with a **centering telescope** as shown in figure 3.7. With this unit in place, the two rings can be brought into sharp focus by rotating the focusing ring on the telescope. Refocusing is necessary for each objective and its matching annular stop. Some manufacturers, such as American Optical, provide an aperture viewing unit (figure 3.8), which enables one to observe the rings without using a centering telescope. Zeiss microscopes have a unit called the *Optovar*, which is located in a position similar to the American Optical unit that serves the same purpose.

Light Source Adjustment

For both brightfield and phase-contrast modes it is essential that optimum lighting be achieved. This is no great problem for a simple setup such as the American Optical instrument shown in figure 3.9. For multiple phase objective microscopes, however, (such as the Zeiss in figure 3.6) there are many more adjustments that need to be made. A few suggestions that highlight some of the problems and solutions follow:

1. Since blue light provides better images for both phase-contrast and brightfield modes, make certain that a blue filter is placed in the filter holder that is positioned in the light path. If the microscope has no filter holder, placing the filter over the light source on the base will help.
2. Brightness of field under phase-contrast is controlled by adjusting the voltage or the iris diaphragm on the base. Considerably more light is required for phase-contrast than for brightfield since so much light is blocked out by the annular stop.
3. The evenness of illumination on some microscopes, such as the Zeiss seen on these pages, can be adjusted by removing the lamp housing from the microscope and focusing the light spot on a piece of translucent white paper. For the detailed steps in this procedure, one should consult the instruction manual that comes with the microscope. Light source adjustments of this nature are not necessary for the simpler types of microscopes.
4. Since each phase-contrast objective must be used with a matching annular stop, make certain that the proper annular stop is being used with the objective that is over the microscope slide. If image quality is lacking, check first to see if the matching annular stop is in position.

Exercise 3 • Phase-Contrast Microscopy

WORKING PROCEDURES

Once the light source is correct and the phase elements are centered you are finally ready to examine slide preparations. Keep in mind that from now on most of the adjustments described earlier should not be altered; however, if misalignment has occurred due to mishandling, it will be necessary to refer back to alignment procedures. The following guidelines should be adhered to in all phase-contrast studies:

- Use only optically perfect slides and cover glasses (no bubbles or striae in the glass).
- Be sure that slides and cover glasses are completely free of grease or chemicals.
- Use wet mount slides instead of hanging drop preparations. The latter leave much to be desired. Culture broths containing bacteria or protozoan suspensions are ideal for wet mounts.
- In general, limit observations to living cells. In most instances stained slides are not satisfactory.

The first time you use phase-contrast optics to examine a wet mount, follow these suggestions:

1. Place the wet mount slide on the stage and bring the material into focus, *using brightfield optics* at low-power magnification.
2. Once the image is in focus, switch to phase optics at the same magnification. Remember, it is necessary to place in position the matching annular stop.
3. Adjust the light intensity, first with the base diaphragm and then with the voltage regulator. In most instances you will need to increase the amount of light for phase-contrast.
4. Switch to higher magnifications, much in the same way you do for brightfield optics, except that you have to rotate a matching annular stop into position.
5. If an oil immersion phase objective is used, add immersion oil to the top of the condenser as well as to the top of the cover glass.
6. Don't be disturbed by the "halo effect" that you observe with phase optics. Halos are normal.

LABORATORY REPORT

This exercise may be used in conjunction with Part 2 in studying various types of organisms. Organelles in protozoans and algae will show up more distinctly than with brightfield optics. After reading this exercise and doing any special assignments made by your instructor, answer the questions on combined Laboratory Report 3–5 that pertain to this exercise.

Fluorescence Microscopy

4

The fluorescence microscope is a unique instrument that is indispensable in certain diagnostic and research endeavors. Differential dyes and immunofluorescence techniques have made laboratory diagnosis of many diseases much simpler with this type of microscope than with the other types described in Exercises 1, 2, and 3. If you are going to prepare and study any differential fluorescence slides that are described in certain exercises in this manual, you should have a basic understanding of the microscope's structure, its capabilities, and its limitations. In addition, it is important that one be aware of the potential of experiencing eye injury if one of these instruments is not used in a safe manner.

A fluorescence microscope differs from an ordinary brightfield microscope in several respects. First of all, it utilizes a powerful mercury vapor arc lamp for its light source. Secondly, a darkfield condenser is usually used in place of the conventional Abbé brightfield condenser. The third difference is that it employs three sets of filters to alter the light that passes up through the instrument to the eye. Some general principles related to its operation will follow an explanation of the principle of fluorescence.

THE PRINCIPLE OF FLUORESCENCE

It was pointed out in the last exercise that light exists as a form of energy propagated in wave form. An interesting characteristic of such an electromagnetic wave is that it can influence the electrons of molecules that it encounters, causing significant interaction. Those electrons within a molecule that are not held too securely may be set in motion by the oscillations of the light beam. Not only are these electrons interrupted from their normal pathways, but they are also forced to oscillate in resonance with the passing light wave. This excitation, caused by such oscillation, requires energy that is supplied by the light beam. When we say that a molecule absorbs light, this is essentially what is taking place.

Whenever a physical body absorbs energy, as in the case of the activated molecule, the energy doesn't

just disappear; it must reappear again in some other form. This new manifestation of the energy may be in the form of a chemical reaction, heat, or light. If light is emitted by the energized molecules, the phenomenon is referred to as **photoluminescence**. In photoluminescence there is always a certain time lapse between the absorption and emission of light. If the time lag is greater than 1/10,000 of a second it is generally called **phosphorescence**. On the other hand, if the time lapse is less than 1/10,000 of a second, it is known as **fluorescence**.

Thus, we see that fluorescence is initiated when a molecule absorbs energy from a passing wave of light. The excited molecule, after a brief period of time, will return to its fundamental energy state after emitting fluorescent light. It is significant that *the wavelength of fluorescence is always longer than the exciting light*. This follows Stokes' law, which applies to liquids but not to gases. This phenomenon is due to the fact that energy loss occurs in the process so that the emitting light has to be of a longer wavelength. This energy loss, incidentally, occurs as a result of the mobilization of the comparatively heavy atomic nuclei of the molecules rather than the displacement of the lighter electrons.

Microbiological material that is to be studied with a fluorescence microscope must be coated with special compounds that possess this quality of fluorescence. Such compounds are called **fluorochromes**. Auramine O, acridine orange, and fluorescein are well-known fluorochromes. Whether a compound will fluoresce will depend on its molecular structure, the temperature, and the pH of the medium. The proper preparation and use of fluorescent materials for microbiological work must take all these factors into consideration.

MICROSCOPE COMPONENTS

Figure 4.2 illustrates, diagrammatically, the light pathway of a fluorescence microscope. The essential components are the light source, heat filter, exciter filter, condenser, and barrier filter. The characteristics and functions of each item follow.

Exercise 4 • Fluorescence Microscopy

Light Source The first essential component of a fluorescence microscope is its bright mercury vapor arc lamp. Such a bulb is preferred over an incandescent one because it produces an ample supply of shorter wavelengths of light (ultraviolet, violet, and blue) that are needed for good fluorescence. To produce the arc in one of these lamps, voltages as high as 18,000 volts are required; thus, a power supply transformer is always used.

The wavelengths produced by these lamps include the ultraviolet range of 200–400 nm, the visible range of 400–780 nm, and the long infrared rays that are above 780 nm.

Mercury vapor arc lamps are expensive and potentially dangerous. Certain precautions must be taken, not only to promote long bulb life, but to protect the user as well. One of the hazards of these bulbs is that they are pressurized and can explode. Another hazard exists in direct exposure of the eyes to harmful rays. Knowledge of these hazards is essential to safe operation. If one follows certain precautionary measures, there is little need for anxiety. However, one should not attempt to use one of these instruments without a complete understanding of its operation.

Heat Filter The infrared rays generated by the mercury vapor arc lamp produce a considerable amount of heat. These rays serve no useful purpose in fluorescence and place considerable stress on the filters within the system. To remove these rays, a heat-absorbing filter is the first element in front of the condensers. Ultraviolet rays, as well as most of the visible spectrum, pass through this filter unimpeded.

Exciter Filter After the light has been cooled down by the heat filter it passes through the exciter filter, which absorbs all the wavelengths except the short ones needed to excite the fluorochrome on the slide. These filters are very dark and are designed to let through only the green, blue, violet, or ultraviolet rays. If the exciter filter is intended for visible light (blue, green, or violet) transmission, it will also allow ultraviolet transmittance.

Condenser To achieve the best contrast of a fluorescent object in the microscopic field, a darkfield condenser is used. It must be kept in mind that weak fluorescence of an object in a brightfield would be difficult to see. The dark background produced by the darkfield condenser, thus, provides the desired contrast. Another bonus of this type of condenser is that

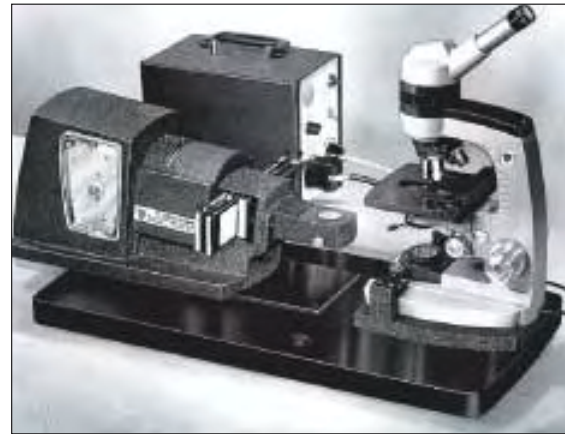


Figure 4.1 An early model American Optical fluorescence illuminator (*Fluorolume*) that could be adapted to an ordinary darkfield microscope.

the majority of the ultraviolet light rays are deflected by the condenser, protecting the observer's eyes. To achieve this, the numerical aperture of the objective is always 0.05 less than that of the condenser.

Barrier Filter This filter is situated between the objective and the eyepiece to remove all remnants of the exciting light so that only the fluorescence is seen. When ultraviolet excitation is employed with its very dark, almost black-appearing exciter filters, the corresponding barrier filters appear almost colorless. On the other hand, when blue exciter filters are used, the matching barrier filters have a yellow to deep orange color. In both instances, the significant fact is that the barrier filter should cut off precisely the shorter exciter wavelengths without affecting the longer fluorescence wavelengths.

USE OF THE MICROSCOPE

As in the case of most sophisticated equipment of this type, it is best to consult the manufacturer's instruction manual before using it. Although different makes of fluorescence microscopes are essentially alike in principle, they may differ considerably in the fine points of operation. Since it is not possible to be explicit about the operation of all makes, all that will be attempted here is to generalize.

Some Precautions To protect yourself and others it is well to outline the hazards first. Keep the following points in mind:

Fluorescence Microscopy • Exercise 4

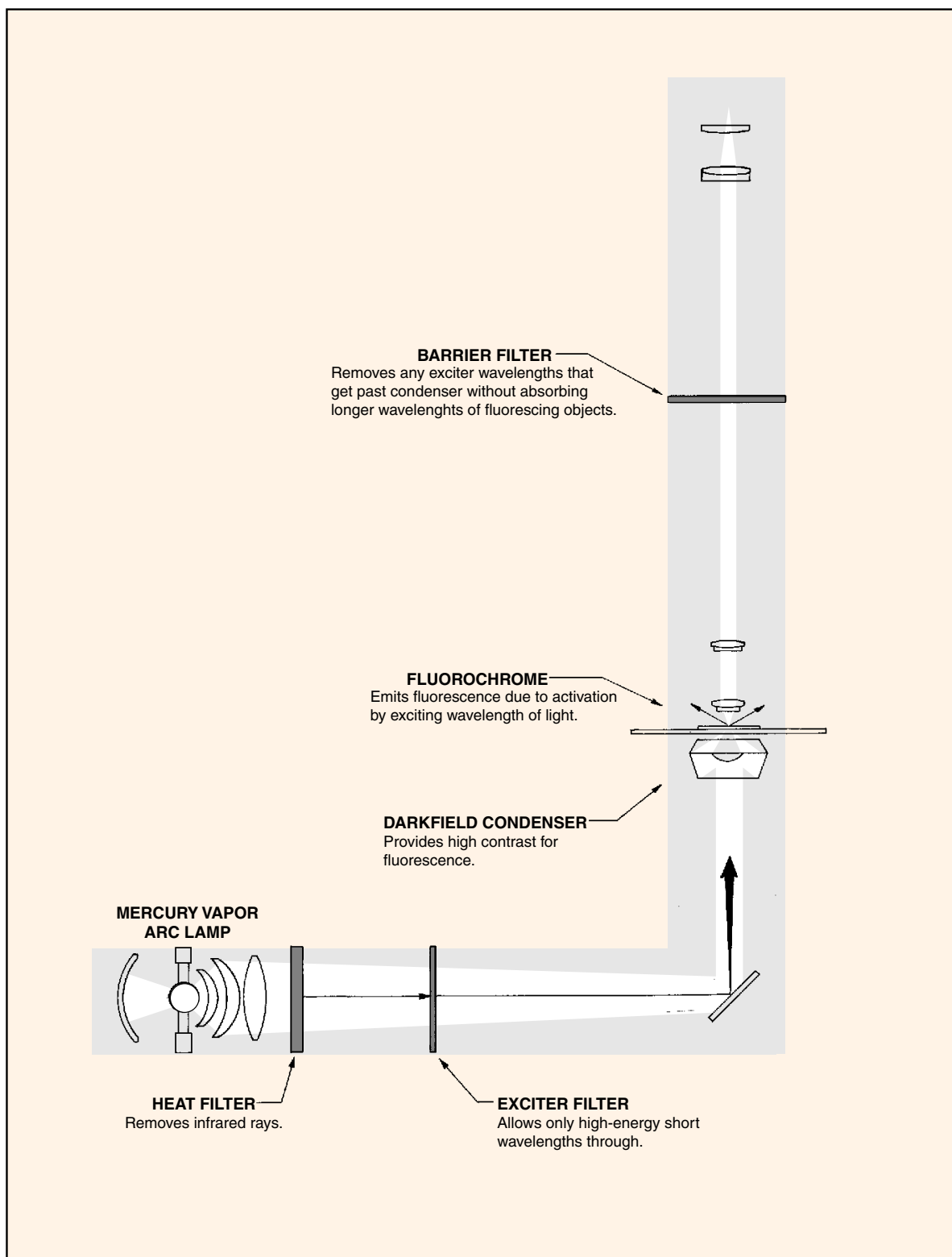


Figure 4.2 The light pathway of a fluorescence microscope

Exercise 4 • Fluorescence Microscopy

1. Remember that the pressurized mercury arc lamp is literally a potential bomb. Design of the equipment is such, however, that with good judgment, no injury should result. When these lamps are cold they are relatively safe, but when hot, the inside pressure increases to eight atmospheres, or 112 pounds per square inch.

The point to keep in mind is this—*never attempt to inspect the lamp while it is hot*. Let it cool completely before opening up the lamp housing. Usually, 15 to 20 minutes cooling time is sufficient.

2. *Never expose your eyes to the direct rays of the mercury arc lamp.* Equipment design is such that the bulb is always shielded against the scattering of its rays. Remember that the unfiltered light from one of these lamps is rich in both ultraviolet and infrared rays—both of which are damaging to the eyes. *Severe retinal burns can result from exposure to the mercury arc rays.*
3. Be sure that the barrier filter is always in place when looking down through the microscope. *Removal of the barrier filter or exciter filter or both filters while looking through the microscope could cause eye injury.* It is possible to make mistakes of this nature if one is not completely familiar with the instrument. Remember, the function of the barrier filter is to prevent traces of ultraviolet light from reaching the eyes without blocking wavelengths of fluorescence.

Warm-up Period The lamps in fluorescence microscopes require a warm-up period. When they are first turned on the illumination is very low, but it increases to maximum in about 2 minutes. *Optimum illumination occurs when the equipment has been operating for 30 minutes or more.* Most manufacturers recommend leaving the instruments turned on for an hour or more when using them. It is not considered good economy to turn the instrument on and off several times within a 2- or 3-hour period.

Keeping a Log The life expectancy of a mercury arc lamp is around 400 hours. A log should be kept of the number of hours that the instrument is used so that inspection can be made of the bulb at approximately 200 hours. A card or piece of paper should be kept conveniently near the instrument so that the individual using the instrument is reminded to record the time that the instrument is turned on and off.

Filter Selection The most frequently used filter combination is the bluish Schott BG12 (AO #702) exciter and the yellowish Schott OG1 barrier filters.

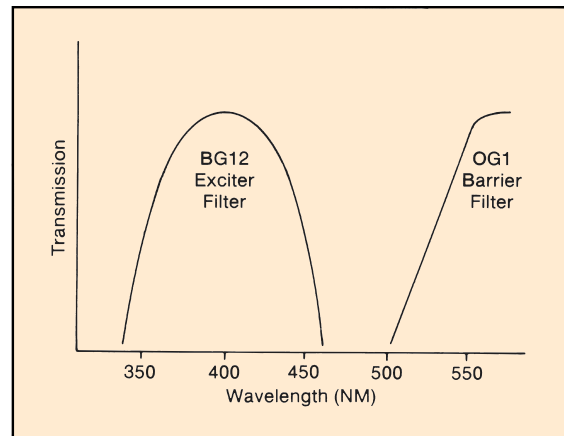


Figure 4.3 Spectral transmissions of BG12 and OG1 filters

Figure 4.3 shows the wavelength transmission of each of these filters. Note that the exciter filter gives peak emission of light in the 400 nm area of the spectrum. These rays are violet. It allows practically no green or yellow wavelengths through. The shortest wavelengths that this barrier filter lets through are green to greenish-yellow.

If a darker background is desired than is being achieved with the above filters, one may add a pale blue Schott BG38 to the system. It may be placed on either side of the heat filter, depending on the type of equipment being used. If it is placed between the lamp and heat filter, it will also function as another heat filter.

Examination When looking for material on the slide, it is best to use low- or high-power objectives. If the illuminator is a separate unit, as in figure 4.1, it may be desirable to move the illuminator out of position and use incandescent lighting for this phase of the work. Once the desirable field has been located, the mercury vapor arc illuminator can be moved into position. One problem with fluorescence microscopes is that most darkfield condensers do not illuminate well through the low-power objectives (exception: the Reichert-Toric setup used on some American Optical instruments).

Keep in mind that there is no diaphragm control on darkfield condensers. Some instruments are supplied with neutral density filters to reduce light intensity. The best system of illumination control, however, is achieved with objectives that have a built-in iris control. These objectives have a knurled ring that can be rotated to control the contrast.

Fluorescence Microscopy • Exercise 4

For optimum results it is essential that oil be used between the condenser and the slide. And, of course, if the oil immersion lens is used, the oil must also be interposed between the slide and the objective. It is also important that special low-fluorescing immersion oil be used. *Ordinary immersion oil should be avoided.*

Although the ocular of a fluorescence microscope is usually 10 $\times$, one should not hesitate to try other size oculars if they are available. With bright-field microscopes it is generally accepted that nothing is

gained by going beyond 1000 $\times$ magnification. In a fluorescence microscope, however, the image is formed in a manner quite different from its brightfield counterpart, obviating the need for following the 1000 $\times$ rule. The only loss by using the higher magnification is some brightness.

LABORATORY REPORT

Complete all the answers to the questions on Laboratory Report 3–5 that pertain to this exercise.

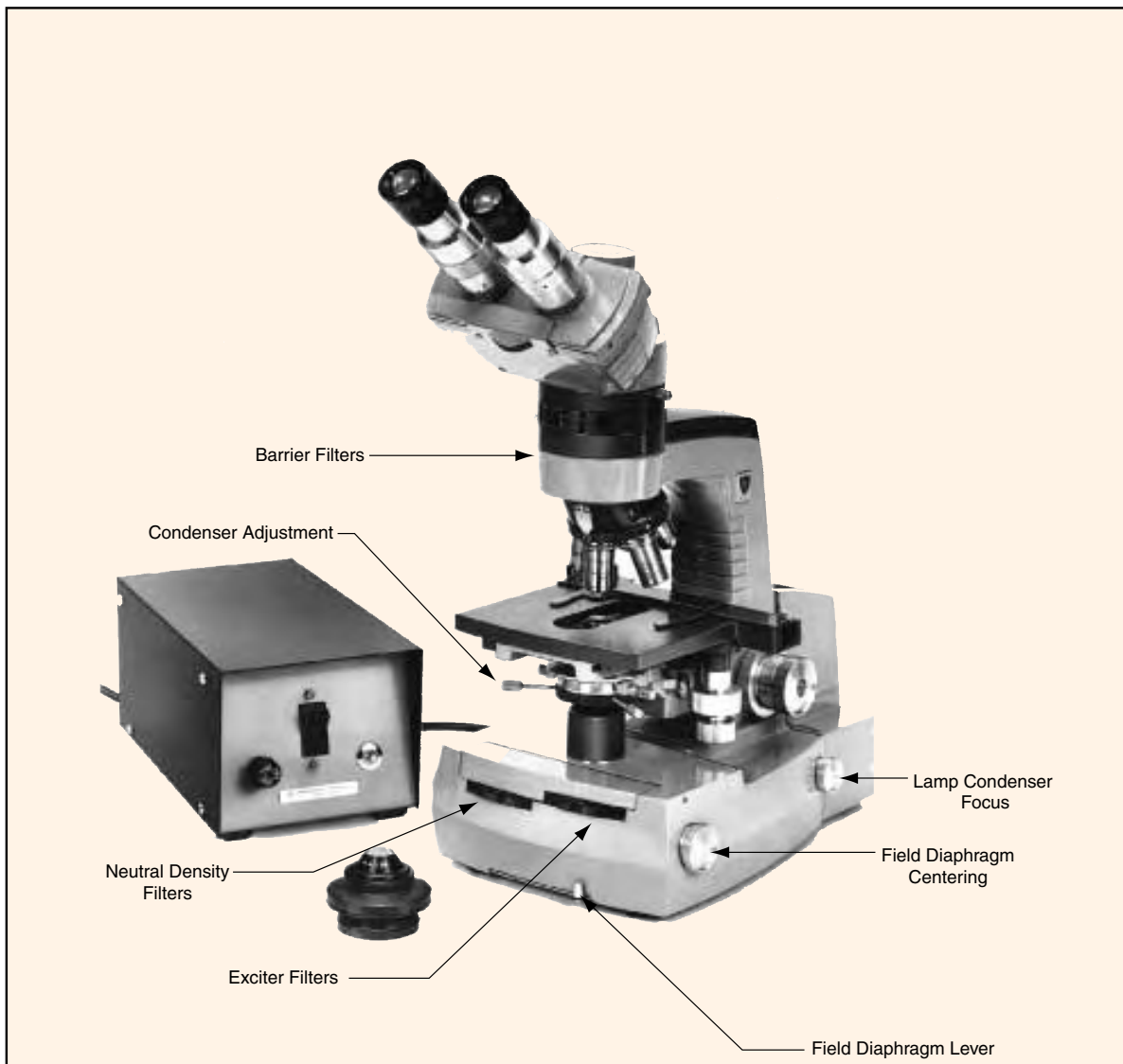


Figure 4.4 An American Optical fluorescence microscope

5

Microscopic Measurements

With an ocular micrometer properly installed in the eyepiece of your microscope, it is a simple matter to measure the size of microorganisms that are seen in the microscopic field. An **ocular micrometer** consists of a circular disk of glass that has graduations engraved on its upper surface. These graduations appear as shown in illustration B, figure 5.4. On some microscopes one has to disassemble the ocular so that the disk can be placed on a shelf in the ocular tube between the two lenses. On most microscopes, however, the ocular micrometer is simply inserted into the bottom of the ocular, as shown in figure 5.1. Before one can use the micrometer it is necessary to calibrate it for each of the objectives by using a stage micrometer.

The principal purpose of this exercise is to show you how to calibrate an ocular micrometer for the various objectives on your microscope. Proceed as follows:

CALIBRATION PROCEDURE

The distance between the lines of an ocular micrometer is an arbitrary value that has meaning only if the ocular micrometer is calibrated for the objective that is being used. A **stage micrometer** (figure 5.2), also known as an *objective micrometer*, has lines scribed on it that are exactly 0.01 mm (10 μm) apart. Illustration C, figure 5.4 reveals the appearance of these graduations.

To calibrate the ocular micrometer for a given objective, it is necessary to superimpose the two scales and determine how many of the ocular graduations coincide with one graduation on the scale of the stage micrometer. Illustration A in figure 5.4 shows how the two scales appear when they are properly aligned in the microscopic field. In this case, seven ocular divisions match up with one stage micrometer division of 0.01 mm to give an ocular value of $0.01/7$, or 0.00143 mm. Since there are 1000 micrometers in 1 millimeter, these divisions are 1.43 μm apart.

With this information known, the stage micrometer is replaced with a slide of organisms to be measured. Illustration D, figure 5.4, shows how a field of microorganisms might appear with the ocular mi-



Figure 5.1 Ocular micrometer with retaining ring is inserted into base of eyepiece.



Figure 5.2 Stage micrometer is positioned by centering small glass disk over the light source.



Figure 5.3 After calibration is completed, stage micrometer is replaced with slide for measurements.

Microscopic Measurements • Exercise 5

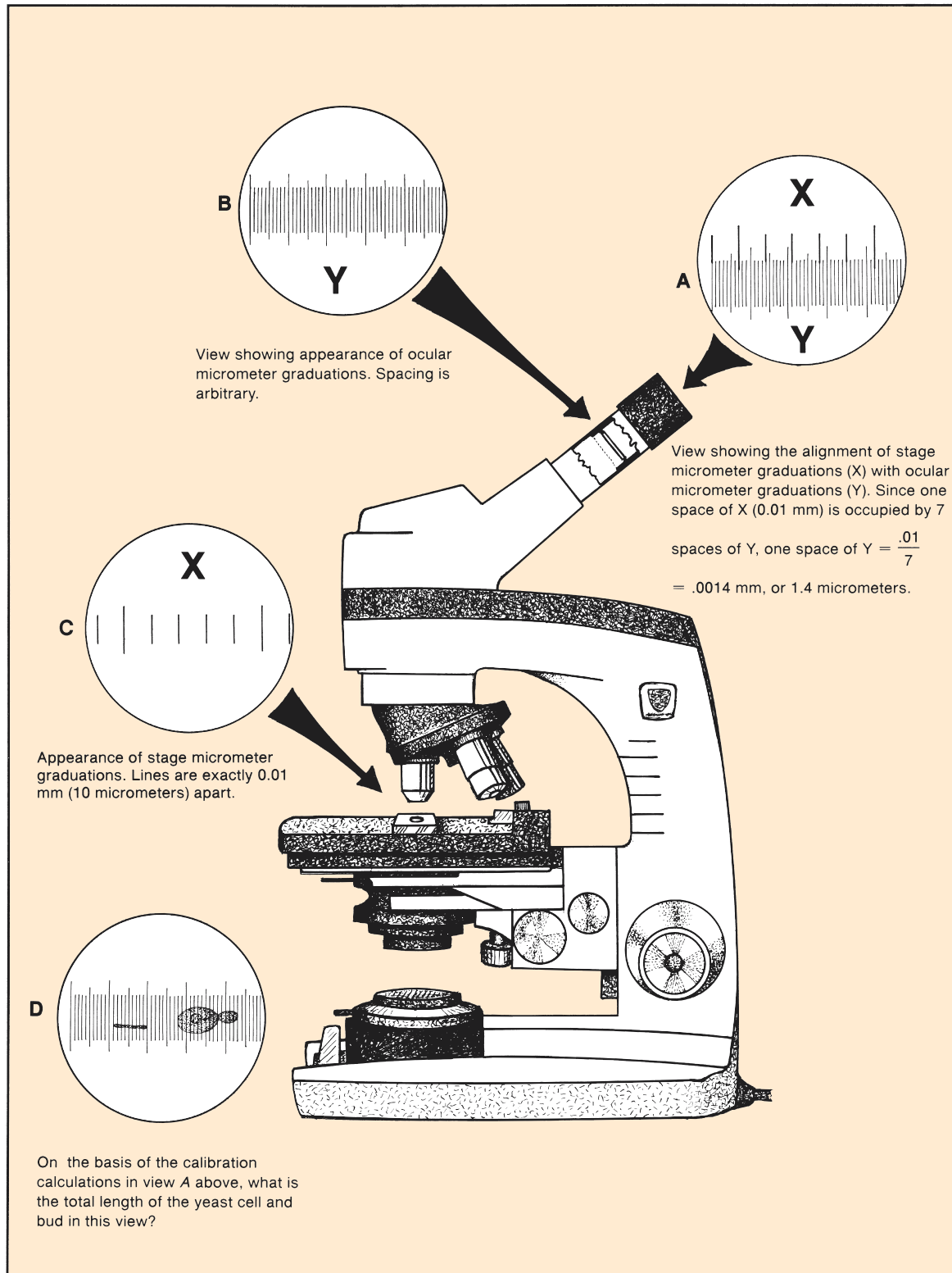


Figure 5.4 Calibration of ocular micrometer

Exercise 5 • Microscopic Measurements

chrometer in the eyepiece. To determine the size of an organism, then, it is a simple matter to count the graduations and multiply this number by the known distance between the graduations. When calibrating the objectives of a microscope, proceed as follows.

Materials:

ocular micrometer or eyepiece that contains a micrometer disk
stage micrometer

- If eyepieces are available that contain ocular micrometers, replace the eyepiece in your microscope with one of them. If it is necessary to insert an ocular micrometer in your eyepiece, find out from your instructor whether it is to be inserted below the bottom lens or placed between the two lenses within the eyepiece. In either case, great care must be taken to avoid dropping the eyepiece or reassembling the lenses incorrectly. *Only with your instructor's prior approval shall eyepieces be disassembled.* Be sure that the graduations are on the upper surface of the glass disk.
- Place the stage micrometer on the stage and center it exactly over the light source.
- With the low-power (10×) objective in position, bring the graduations of the stage micrometer into focus, *using the coarse adjustment knob. Reduce the lighting.* **Note:** If the microscope has an automatic stop, do not use it as you normally would for regular microscope slides. The stage micrometer slide is too thick to allow it to function properly.
- Rotate the eyepiece until the graduations of the ocular micrometer lie parallel to the lines of the stage micrometer.
- If the **low-power objective** is the objective to be calibrated, proceed to step 8.
- If the **high-dry objective** is to be calibrated, swing it into position and proceed to step 8.
- If the **oil immersion lens** is to be calibrated, place a drop of immersion oil on the stage micrometer, swing the oil immersion lens into position, and bring the lines into focus; then, proceed to the next step.
- Move the stage micrometer laterally until the lines at one end coincide. Then look for another line on the ocular micrometer that coincides *exactly* with one on the stage micrometer. Occasionally one stage micrometer division will include an even number of ocular divisions, as shown in illustration A. In most instances, however, several stage graduations will be involved. In this case, divide the number of stage micrometer divisions by the number of ocular divisions that coincide. The figure you get will be that part of a stage micrometer division that is seen in an ocular division. This value must then be multiplied by 0.01 mm to get the amount of each ocular division.

Example: 3 divisions of the stage micrometer line up with 20 divisions of the ocular micrometer.

$$\begin{aligned}\text{Each ocular division} &= \frac{3}{20} \times 0.01 \\ &= 0.0015 \text{ mm} \\ &= 1.5 \mu\text{m}\end{aligned}$$

- Replace the stage micrometer with slides of organisms to be measured.

MEASURING ASSIGNMENTS

Organisms such as protozoans, algae, fungi, and bacteria in the next few exercises may need to be measured. If your instructor requires that measurements be made, you will be referred to this exercise.

Later on you will be working with unknowns. In some cases measurements of the unknown organisms will be pertinent to identification.

If trial measurements are to be made at this time, your instructor will make appropriate assignments.

Important: Remove the ocular micrometer from your microscope at the end of the laboratory period.

LABORATORY REPORT

Answer the questions on combined Laboratory Report 3–5 that pertain to this exercise.

PART 2 Survey of Microorganisms

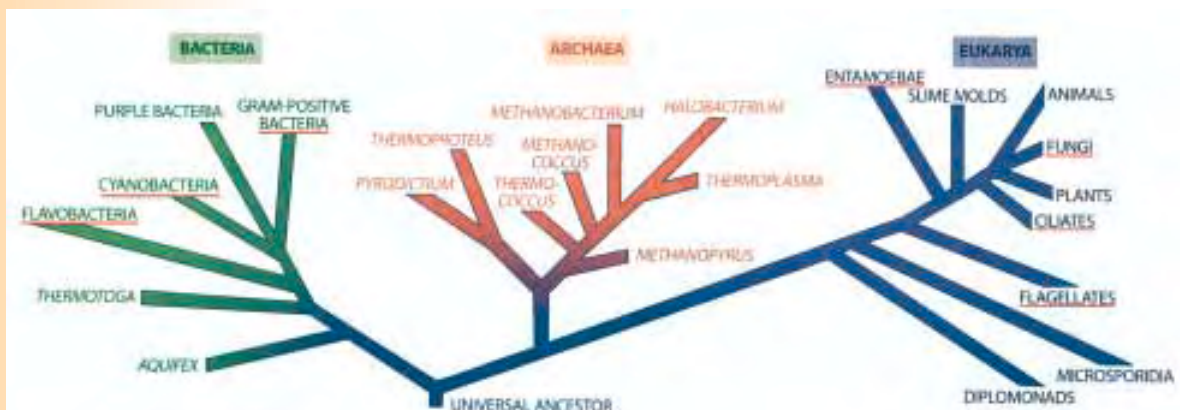
Too often, in our serious concern with the direct applications of microbiology to human welfare, we neglect the large number of interesting free-living microorganisms that abound in the water, soil, and air. It is these free-spirited forms that we will study in the four exercises of this unit. To observe these organisms we will examine samples of pond water and Petri plates with special media that have been exposed to the air and various items in our environment. The principal organisms that we will encounter are protozoans, algae, molds, yeasts, cyanobacteria, and bacteria.

The phylogenetic tree on this page illustrates where these organisms fit in the evolutionary scheme of organisms. The organisms that you are likely to encounter are underlined on the diagram. A few comments about each domain are presented here.

Domain Archaea Since the principal habitats of these organisms are extreme environments such as volcanic waters, hot springs, or waters of high salt conditions, you will not encounter any of these organisms in this study. These ancient organisms that exist in such hostile environments have often been referred to as “extremophiles.”

Domain Eukarya The protozoans, algae, and fungi fall in this domain. All members of this domain have distinct nuclei with nuclear membranes and mitochondria. Some eukaryotes, such as the algae, have chloroplasts, which puts them in the plant kingdom. The eukaryotes appear to be more closely related to the Archaea than to the Bacteria.

Domain Bacteria Members of this domain are also called “prokaryotes.” They are smaller than eukaryotes, lack distinct nuclei (no nuclear membrane), and are enclosed in a rigid cell wall with a distinct cell membrane. In this study you will encounter various species of cyanobacteria and bacteria.



From: Extremophiles. Michael T. Madigan and Barry L. Marrs in *Scientific American* Vol. 276, Number 4, pages 82–87, April 1997.

6

Protozoa, Algae, and Cyanobacteria

In this exercise a study will be made of protozoans, algae, and cyanobacteria that are found in pond water. Bottles that contain water and bottom debris from various ponds will be available for study. Illustrations and text provided in this exercise will be used to assist you in an attempt to identify the various types that are encountered. Unpigmented, moving microorganisms will probably be protozoans. Greenish or golden-brown organisms are usually algae. Organisms that appear blue-green will be cyanobacteria. Supplementary books on the laboratory bookshelf will also be available for assistance in identifying organisms that are not described in the short text of this exercise. If you encounter invertebrates and are curious as to their identification, you may refer to Exercise 7; however, keep in mind that our prime concern here is only with protozoans, algae, and cyanobacteria.

The purpose of this exercise is, simply, to provide you with an opportunity to become familiar with the differences between the three groups by comparing their characteristics. The extent to which you will be held accountable for the names of various organisms will be determined by your instructor. The amount of time available for this laboratory exercise will determine the depth of scope to be pursued.

To study the microorganisms of pond water, it will be necessary to make wet mount slides. The procedure for making such slides is relatively simple. All that is necessary is to place a drop of suspended organisms on a microscope slide and cover it with a cover glass. If several different cultures are available, the number of the bottle should be recorded on the slide with a china marking pencil. As you prepare and study your slides, observe the following guidelines:

Materials:

bottles of pond-water samples
microscope slides and cover glasses
rubber-bulbed pipettes and forceps
china marking pencil
reference books

1. Clean the slide and cover glass with soap and water, rinse thoroughly, and dry. Do not attempt to study a slide that lacks a cover glass.

2. When using a pipette, insert it into the bottom of the bottle to get a maximum number of organisms. Very few organisms will be found swimming around in middepth of the bottle.
3. To remove filamentous algae from a specimen bottle, use forceps. Avoid putting too much material on the slides.
4. Explore the slide first with the low-power objective. Reduce the lighting with the iris diaphragm. Keep the condenser at its highest point.
5. When you find an organism of interest, swing the high-dry objective into position and adjust the lighting to get optimum contrast. If your microscope has phase-contrast elements, use them.
6. Refer to Figures 6.1 through 6.6 and the text on these pages to identify the various organisms that you encounter.
7. Record your observations on the Laboratory Reports.

THE PROTISTS

Single-celled eukaryons that lack tissue specialization are called *protists*. Protozoologists group all protists in **Kingdom Protista**. Those protists that are animal-like are put in **Subkingdom Protozoa** and the protists that are plantlike fall into **Subkingdom Algae**. This system of classification includes all colonial species as well as the single-celled types.

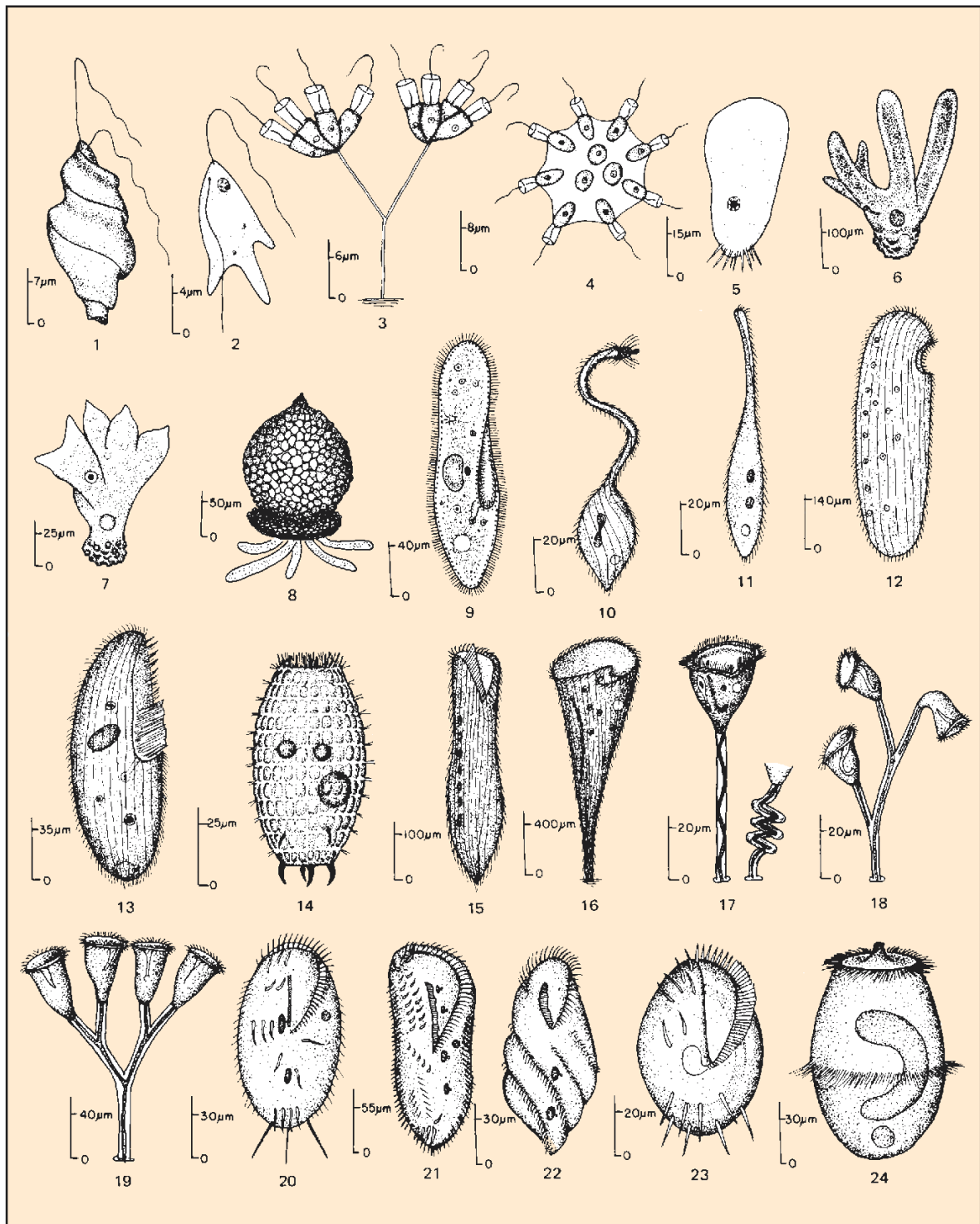
SUBKINGDOM PROTOZOA

Externally, protozoan cells are covered with a cell membrane, or pellicle; cell walls are absent; and distinct nuclei with nuclear membranes are present. Specialized organelles, such as contractile vacuoles, cytostomes, mitochondria, ribosomes, flagella, and cilia, may also be present.

All protozoa produce **cysts**, which are resistant dormant stages that enable them to survive drought, heat, and freezing. They reproduce asexually by cell division and exhibit various degrees of sexual reproduction.

The Subkingdom Protozoa is divided into three phyla: Sarcomastigophora, Ciliophora, and Apicomplexa. Type of locomotion plays an important role in classification here. A brief description of each phylum follows:

Protozoa, Algae, and Cyanobacteria • Exercise 6



- | | | | | |
|------------------------|-----------------------|-------------------------|------------------------|---------------------------|
| 1. <i>Heteronema</i> | 6. <i>Amoeba</i> | 11. <i>Lionotus</i> | 16. <i>Stentor</i> | 21. <i>Onychodromos</i> |
| 2. <i>Cercomonas</i> | 7. <i>Mayorella</i> | 12. <i>Loxodes</i> | 17. <i>Vorticella</i> | 22. <i>Hypotrichidium</i> |
| 3. <i>Codosiga</i> | 8. <i>Diffugia</i> | 13. <i>Blepharisma</i> | 18. <i>Carhesium</i> | 23. <i>Euplotes</i> |
| 4. <i>Protospongia</i> | 9. <i>Paramecium</i> | 14. <i>Coleps</i> | 19. <i>Zoothamnium</i> | 24. <i>Didinium</i> |
| 5. <i>Trichamoeba</i> | 10. <i>Lacrymaria</i> | 15. <i>Condylastoma</i> | 20. <i>Stylonychia</i> | |

Figure 6.1 Protozoans

Exercise 6 • Protozoa, Algae, and Cyanobacteria**Phylum Sarcomastigophora**

Members of this phylum have been subdivided into two subphyla: Sarcodina and Mastigophora.

Sarcodina (*Amoebae*) Members of this subphylum move about by the formation of flowing protoplasmic projections called *pseudopodia*. The formation of pseudopodia is commonly referred to as *amoeboid movement*. Illustrations 5 through 8 in figure 6.1 are representative amoebae.

Mastigophora (*Zooflagellates*) These protozoans possess whiplike structures called *flagella*. There is considerable diversity among the members of this group. Only a few representatives (illustrations 1 through 4) are seen in figure 6.1.

Phylum Ciliophora

These microorganisms are undoubtedly the most advanced and structurally complex of all protozoans. Evidence seems to indicate that they have evolved from the zooflagellates. Movement and food-getting is accomplished with short hairlike structures called *cilia*. Illustrations 9 through 24 are typical ciliates.

Phylum Apicomplexa

This phylum has only one class, the *Sporozoa*. Members of this phylum lack locomotor organelles and all are internal parasites. As indicated by their class name, their life cycles include spore-forming stages. *Plasmodium*, the malarial parasite, is a significant pathogenic sporozoan of humans.

SUBKINGDOM ALGAE

The Subkingdom Algae includes all the photosynthetic eukaryotic organisms in Kingdom Protista. Being true protists, they differ from the plants (*Plantae*) in that tissue differentiation is lacking.

The algae may be unicellular, as those shown in the top row of figure 6.2; colonial, like the four in the lower right-hand corner of figure 6.2; or filamentous, as those in figure 6.3. The undifferentiated algal structure is often referred to as a *thallus*. It lacks the stem, root, and leaf structures that result from tissue specialization.

These microorganisms are universally present where ample moisture, favorable temperature, and sufficient sunlight exist. Although a great majority of them live submerged in water, some grow on soil. Others grow on the bark of trees or on the surfaces of rocks.

Algae have distinct, visible nuclei and chloroplasts. **Chloroplasts** are organelles that contain **chlorophyll a** and other pigments. Photosynthesis takes place within these bodies. The size, shape, distribution, and number of chloroplasts vary considerably from species to species. In some instances a single chloroplast may occupy most of the cell space.

Although there are seven divisions of algae, only five will be listed here. Since two groups, the cryptomonads and red algae, are not usually encountered in freshwater ponds, they have not been included here.

Division 1 Euglenophycophyta
(*Euglenoids*)

Illustrations 1 through 6 in figure 6.2 are typical euglenoids, representing four different genera within this relatively small group. All of them are flagellated and appear to be intermediate between the algae and protozoa. Protozoanlike characteristics seen in the euglenoids are (1) the absence of a cell wall, (2) the presence of a gullet, (3) the ability to ingest food but not through the gullet, (4) the ability to assimilate organic substances, and (5) the absence of chloroplasts in some species. In view of these facts, it becomes readily apparent why many zoologists often group the euglenoids with the zooflagellates.

The absence of a cell wall makes these protists very flexible in movement. Instead of a cell wall they possess a semirigid outer **pellicle**, which gives the organism a definite form. Photosynthetic types contain **chlorophylls a** and **b**, and they always have a red **stigma** (eyespot), which is light sensitive. Their characteristic food-storage compound is a lipopolysaccharide, **paramylum**. The photosynthetic euglenoids can be bleached experimentally by various means in the laboratory. The colorless forms that develop, however, cannot be induced to revert back to phototrophy.

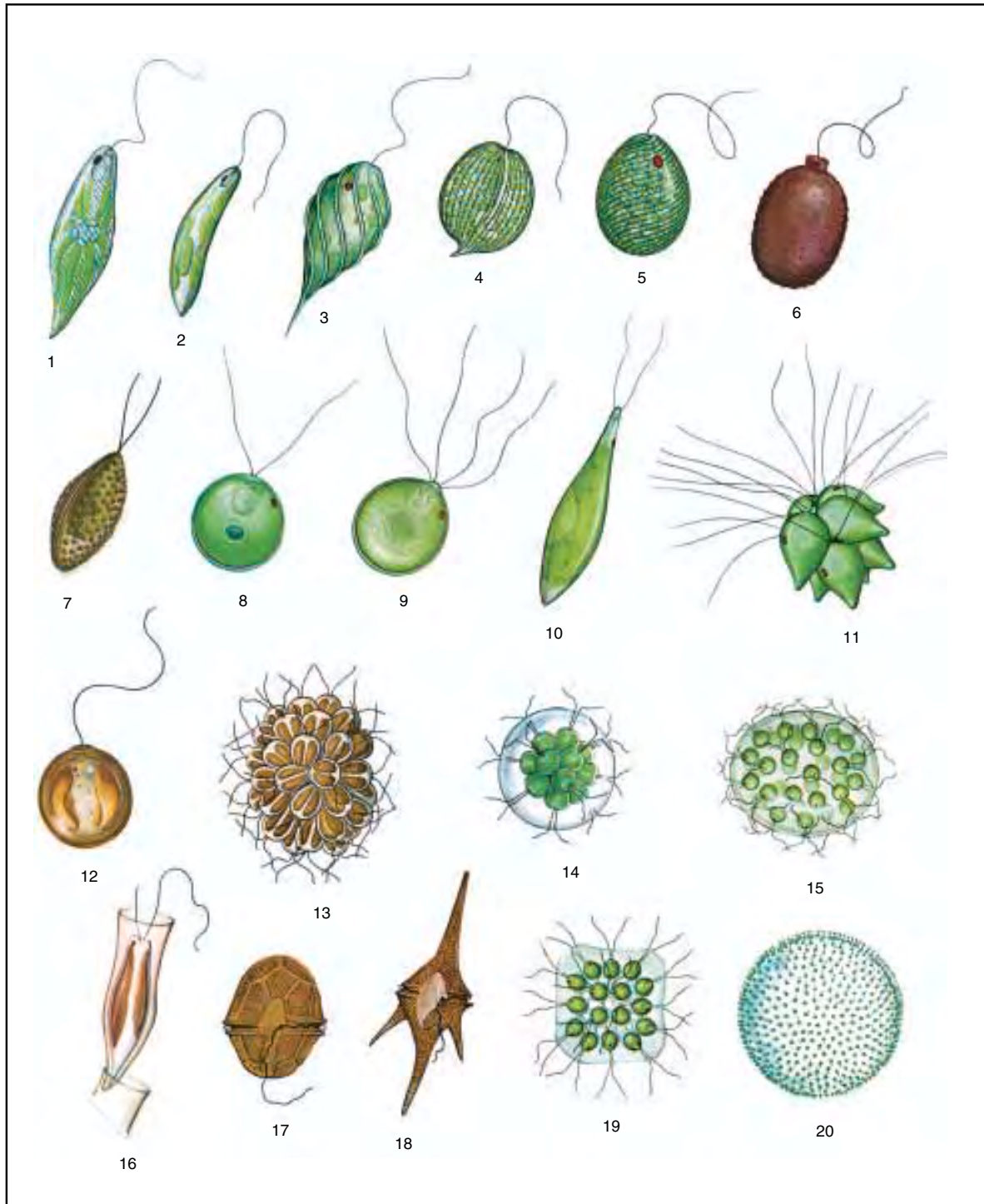
Division 2 Chlorophycophyta
(*Green Algae*)

The majority of algae observed in ponds belong to this group. They are grass-green in color, resembling the euglenoids in having **chlorophylls a** and **b**. They differ from euglenoids in that they synthesize **starch** instead of paramylum for food storage.

The diversity of this group is too great to explore its subdivisions in this preliminary study; however, the small flagellated *Chlamydomonas* (illustration 8, figure 6.2) appears to be the archetype of the entire group and has been extensively studied. Many colonial forms, such as *Pandorina*, *Eudorina*, *Gonium*, and *Volvox* (illustrations 14, 15, 19, and 20, figure 6.2), consist of organisms similar to *Chlamydomonas*. It is the general consensus that from this flagellated form all the filamentous algae have evolved.

Except for *Vaucheria* and *Tribonema*, all of the filamentous forms in figure 6.3 are Chlorophycophyta. All of the nonfilamentous, nonflagellated algae in figure 6.4 also are green algae.

Protozoa, Algae, and Cyanobacteria • Exercise 6



Courtesy of the U.S. Environmental Protection Agency, Office of Research & Development, Cincinnati, Ohio 45268.

- | | | | |
|------------------------------|---------------------------------|---------------------------------|------------------------------|
| 1. <i>Euglena</i> (700X) | 6. <i>Trachelomonas</i> (1000X) | 11. <i>Pyrobotrys</i> (1000X) | 16. <i>Dinobryon</i> (1000X) |
| 2. <i>Euglena</i> (700X) | 7. <i>Phacotus</i> (1500X) | 12. <i>Chrysococcus</i> (3000X) | 17. <i>Peridinium</i> (350X) |
| 3. <i>Phacus</i> (1000X) | 8. <i>Chlamydomonas</i> (1000X) | 13. <i>Synura</i> (350X) | 18. <i>Ceratium</i> (175X) |
| 4. <i>Phacus</i> (350X) | 9. <i>Carteria</i> (1500X) | 14. <i>Pandorina</i> (350X) | 19. <i>Gonium</i> (350X) |
| 5. <i>Lepocinclis</i> (350X) | 10. <i>Chlorogonium</i> (1000X) | 15. <i>Eudorina</i> (175X) | 20. <i>Volvox</i> (100X) |

Figure 6.2 Flagellated algae

Exercise 6 • Protozoa, Algae, and Cyanobacteria

A unique group of green algae is the **desmids** (illustrations 16 through 20, figure 6.4). With the exceptions of a few species, the cells of desmids consist of two similar halves, or semicells. The two halves usually are separated by a constriction, the *isthmus*.

Division 3 Chrysophycophyta
(Golden Brown Algae)

This large diversified division consists of over 6,000 species. They differ from the euglenoids and green algae in that (1) food storage is in the form of **oils** and **leucosin**, a polysaccharide; (2) **chlorophylls a** and **c** are present; and (3) **fucoxanthin**, a brownish pigment, is present. It is the combination of fucoxanthin, other yellow pigments, and the chlorophylls that causes most of these algae to appear golden brown.

Representatives of this division are seen in figures 6.2, 6.3, and 6.5. In figure 6.2, *Chrysococcus*, *Synura*, and *Dinobryon* are typical flagellated chrysophytes. *Vaucheria* and *Tribonema* are the only filamentous chrysophytes shown in figure 6.3.

All of the organisms in figure 6.5 are chrysophytes and fall into a special category of algae called the **diatoms**. The diatoms are unique in that they have hard cell walls of pectin, cellulose, or silicon oxide that are constructed in two halves. The two halves fit together like lid and box.

Skeletons of dead diatoms accumulate on the ocean bottom to form *diatomite*, or “diatomaceous earth,” which is commercially available as an excellent polishing compound. It is postulated by some that much of our petroleum reserves may have been formulated by the accumulation of oil from dead diatoms over millions of years.

Division 4 Phaeophycophyta
(Brown Algae)

With the exception of three freshwater species, all algal protists of this division exist in salt water (marine); thus, it is unlikely that you will encounter any phaeophytes in this laboratory experience. These algae have essentially the same pigments seen in the chrysophytes, but they appear brown because of the masking effect of the greater amount of fucoxanthin. Food storage in the brown algae is in the form of **laminarin**, a polysaccharide, and **mannitol**, a sugar alcohol. All species of brown algae are multicellular and sessile. Most seaweeds are brown algae.

Division 5 Pyrrophyphyta
(Fire Algae)

The principal members of this division are the **dinoflagellates**. Since the majority of these protists are marine, only two freshwater forms are shown in figure 6.2: *Peridinium* and *Ceratium* (illustrations 17 and

18). Most of these protists possess cellulose walls of interlocking armor plates, as in *Ceratium*. Two flagella are present: one is directed backward when swimming and the other moves within a transverse groove. Many marine dinoflagellates are bioluminescent. Some species of marine *Gymnodinium*, when present in large numbers, produce the **red tides** that cause water discoloration and unpleasant odors along our coastal shores.

These algae have **chlorophylls a** and **c** and several xanthophylls. Foods are variously stored in the form of **starch**, **fats**, and **oils**.

THE PROKARYOTES

As indicated on the first page of this unit, the prokaryotes differ from the protists in that they are considerably smaller, lack distinct nuclei with nuclear membranes, and are enclosed in rigid cell walls. Since all members of this group are bacteria, the three-domain system of classification puts them in Domain Bacteria.

Division Cyanobacteria

Division Cyanobacteria in Domain Bacteria includes a large number of microorganisms that were at one time referred to as the blue-green algae. All these prokaryotes are photosynthetic, utilizing **chlorophyll a** for photosynthesis. They differ from the green sulfur and green nonsulfur photosynthetic bacteria in that the latter use *bacteriochlorophyll* instead of chlorophyll a for photosynthesis.

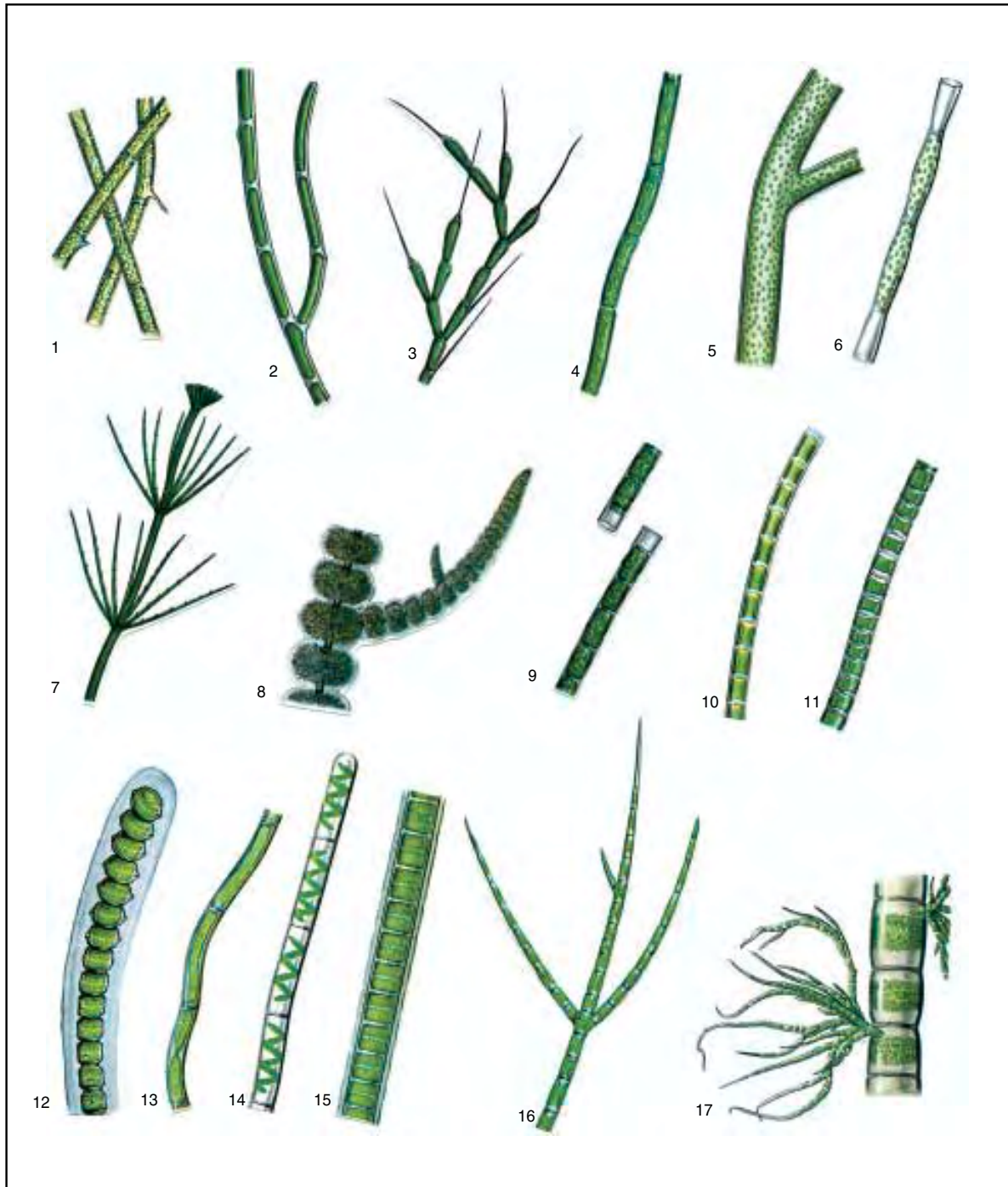
Over 1,000 species of cyanobacteria have been reported. They are present in almost all moist environments from the tropics to the poles, including both freshwater and marine. Figure 6.6 illustrates only a random few that are frequently seen.

The designation of these bacteria as “blue-green” is somewhat misleading in that many cyanobacteria are actually black, purple, red, and various shades of green instead of blue-green. These different colors are produced by the varying proportions of the numerous pigments present. These pigments are **chlorophyll a**, **carotene**, **xanthophylls**, blue **c-phycocyanin**, and red **c-phycoerythrin**. The last two pigments are unique to the cyanobacteria and red algae.

Cellular structure is considerably different from the eukaryotic algae. Although cells lack visible nuclei, nuclear material is present in the form of DNA granules in a colorless area in the center of the cell.

Unlike the algae, the pigments of the cyanobacteria are not contained in chloroplasts; instead, they are located in granules (**phycobilisomes**) that are attached to membranes (**thylakoids**) that permeate the cytoplasm.

Protozoa, Algae, and Cyanobacteria • Exercise 6



Courtesy of the U.S. Environmental Protection Agency, Office of Research & Development, Cincinnati, Ohio 45268.

1. *Rhizoclonium* (175X)
2. *Cladophora* (100X)
3. *Bulbochaete* (100X)
4. *Oedogonium* (350X)

5. *Vaucheria* (100X)
6. *Tribonema* (300X)
7. *Chara* (3X)
8. *Batrachospermum* (2X)

9. *Microspora* (175X)
10. *Ulothrix* (175X)
11. *Ulothrix* (175X)
12. *Desmidium* (175X)

13. *Mougeotia* (175X)
14. *Spirogyra* (175X)
15. *Zygnema* (175X)
16. *Stigeoclonium* (300X)
17. *Draparnaldia* (100X)

Figure 6.3 Filamentous algae

Exercise 6 • Protozoa, Algae, and Cyanobacteria



Courtesy of the U.S. Environmental Protection Agency, Office of Research & Development, Cincinnati, Ohio 45268.

- | | | | |
|--------------------------------|----------------------------------|--------------------------------|--------------------------------|
| 1. <i>Chlorococcum</i> (700X) | 6. <i>Micractinium</i> (700X) | 11. <i>Pamella</i> (700X) | 16. <i>Staurastrum</i> (700X) |
| 2. <i>Oocystis</i> (700X) | 7. <i>Scenedesmus</i> (700X) | 12. <i>Botryococcus</i> (700X) | 17. <i>Staurastrum</i> (350X) |
| 3. <i>Coelastrum</i> (350X) | 8. <i>Actinastrum</i> (700X) | 13. <i>Tetraedron</i> (1000X) | 18. <i>Closterium</i> (175X) |
| 4. <i>Chlorella</i> (350X) | 9. <i>Phytoconis</i> (700X) | 14. <i>Pediatrum</i> (100X) | 19. <i>Euastrum</i> (350X) |
| 5. <i>Sphaerocystis</i> (350X) | 10. <i>Ankistrodesmus</i> (700X) | 15. <i>Tetraspora</i> (100X) | 20. <i>Micrasterias</i> (175X) |

Figure 6.4 Nonfilamentous and nonflagellated algae

Protozoa, Algae, and Cyanobacteria • Exercise 6



Courtesy of the U.S. Environmental Protection Agency, Office of Research & Development, Cincinnati, Ohio 45268.

- | | | | |
|------------------------------|-------------------------------|-------------------------------|----------------------------------|
| 1. <i>Diatoma</i> (1000X) | 7. <i>Nitzschia</i> (1500X) | 13. <i>Synedra</i> (175X) | 19. <i>Asterionella</i> (175X) |
| 2. <i>Gomphonema</i> (175X) | 8. <i>Pinnularia</i> (175X) | 14. <i>Melosira</i> (750X) | 20. <i>Asterionella</i> (750X) |
| 3. <i>Cymbella</i> (175X) | 9. <i>Cyclotella</i> (1000X) | 15. <i>Surirella</i> (350X) | 21. <i>Navicula</i> (750X) |
| 4. <i>Cymbella</i> (1000X) | 10. <i>Tabellaria</i> (175X) | 16. <i>Stauroneis</i> (350X) | 22. <i>Stephanodiscus</i> (750X) |
| 5. <i>Gomphonema</i> (2000X) | 11. <i>Tabellaria</i> (1000X) | 17. <i>Fragillaria</i> (750X) | 23. <i>Meridion</i> (750X) |
| 6. <i>Cocconeis</i> (750X) | 12. <i>Synedra</i> (350X) | 18. <i>Fragillaria</i> (750X) | |

Figure 6.5 Diatoms

Exercise 6 • Protozoa, Algae, and Cyanobacteria



Courtesy of the U.S. Environmental Protection Agency, Office of Research & Development, Cincinnati, Ohio 45268.

- | | | | |
|----------------------------------|---------------------------------|-----------------------------------|-----------------------------|
| 1. <i>Anabaena</i> (350X) | 7. <i>Microcoleus</i> (350X) | 13. <i>Entophysalis</i> (1000X) | 19. <i>Rivularia</i> (175X) |
| 2. <i>Anabaena</i> (350X) | 8. <i>Phormidium</i> (350X) | 14. <i>Gomphosphaeria</i> (1000X) | 20. <i>Anacystis</i> (700X) |
| 3. <i>Anabaena</i> (175X) | 9. <i>Oscillatoria</i> (175X) | 15. <i>Gomphosphaeria</i> (350X) | 21. <i>Anacystis</i> (175X) |
| 4. <i>Nodularia</i> (350X) | 10. <i>Aphanizomenon</i> (175X) | 16. <i>Agmenellum</i> (700X) | 22. <i>Anacystis</i> (700X) |
| 5. <i>Cylindrospermum</i> (175X) | 11. <i>Lyngbya</i> (700X) | 17. <i>Agmenellum</i> (175X) | |
| 6. <i>Arthrospira</i> (700X) | 12. <i>Tolypothrix</i> (350X) | 18. <i>Calothrix</i> (350X) | |

Figure 6.6 Cyanobacteria

Microscopic Invertebrates

7

While looking for protozoa, algae, and cyanobacteria in pond water, one invariably encounters large, transparent, complex microorganisms that, to the inexperienced, appear to be protozoans. In most instances these moving “monsters” are rotifers (illustrations 13 through 17, figure 7.1); in some cases they are copepods, daphnia, or any one of the other forms illustrated in figure 7.1.

All of the organisms illustrated in figure 7.1 are multicellular with organ systems. If organ systems are present, then the organisms cannot be protists, because organs indicate the presence of tissue differentiation. Collectively, these microscopic forms are designated as “invertebrates.” It is to prevent you from misinterpreting some of these invertebrates as protozoans that they are described here.

In using figure 7.1 to identify what you consider might be an invertebrate, keep in mind that there are considerable size differences. A few invertebrates, such as *Dugesia* and *Hydra*, are macroscopic in adult form but microscopic when immature. Be sure to judge size differences by reading the scale beside each organism. The following phyla are listed according to the degree of complexity, the simplest first.

PHYLUM COELENTERATA

(Illustration 1)

Members of this phylum are almost exclusively marine. The only common freshwater form shown in figure 7.1 is *Hydra*. In addition, there are a few less-common freshwater genera similar to the marine hydroids.

The hydras are quite common in ponds and lakes. They are usually attached to rocks, twigs, or other substrata. Around the mouth at the free end are five tentacles of variable length, depending on the species. Smaller organisms, such as *Daphnia*, are grasped by the tentacles and conveyed to the mouth. These animals have a digestive cavity that makes up the bulk of the interior. Since no anus is present, undigested remains of food are expelled through the mouth.

PHYLUM PLATYHELMINTHES

(Illustrations 2, 3, 4, 5)

The invertebrates of this phylum are commonly referred to as **flatworms**. The phylum contains two parasitic classes and one class of free-living organisms, the *Turbellaria*. It is the organisms in this class that are encountered in fresh water. The four genera of this class shown in figure 7.1 are *Dugesia*, *Planaria*, *Macrostomum*, and *Provortex*. The characteristics common to all these organisms are dorso-ventral flatness, a ciliated epidermis, a ventral mouth, and eyespots on the dorsal surface near the anterior end. As in the coelenterates, undigested food must be ejected through the mouth since no anus is present. Reproduction may be asexual by fission or fragmentation; generally, however, reproduction is sexual, each organism having both male and female reproductive organs. Species identification of the turbellarians is exceedingly difficult and is based to a great extent on the details of the reproductive system.

PHYLUM NEMATODA

(Illustration 6)

The members of this phylum are the **roundworms**. They are commonly referred to as *nemas* or *nematodes*. They are characteristically round in cross section, have an external cuticle without cilia, lack eyes, and have a tubular digestive system complete with mouth, intestine, and anus. The males are generally much smaller than the females and have a hooked posterior end. The number of named species is only a fraction of the total nematodes in existence. Species identification of these invertebrates requires very detailed study of many minute anatomical features, which requires complete knowledge of anatomy.

PHYLUM ASCHELMINTHES

This phylum includes classes *Gastrotricha* and *Rotifera*. Most of the members of this phylum are microscopic. Their proximity to the nematodes in

Exercise 7 • Microscopic Invertebrates

classification is due to the type of body cavity (*pseudocoel*) that is present in both phyla.

The **gastrotrichs** (illustrations 7, 8, 9, 10) range from 10 to 540 μm in size. They are very similar to the ciliated protozoans in size and habits. The typical gastrotrich is elongate, flexible, forked at the posterior end, and covered with bristles. The digestive system consists of an anterior mouth surrounded by bristles, a pharynx, intestine, and posterior anus. Species identification is based partially on the shape of the head, tail structure and size, and distribution of spines. Overall length is also an important identification characteristic. They feed primarily on unicellular algae.

The **rotifers** (illustrations 13, 14, 15, 16, and 17) are most easily differentiated by the wheel-like arrangement of cilia at the anterior end and the presence of a chewing pharynx within the body. They are considerably diversified in food habits: some feed on algae and protozoa, others on juices of plant cells, and some are parasitic. They play an important role in keeping waters clean. They also serve as food for small worms and crustaceans, being an important link in the food chain of fresh waters.

PHYLUM ANNELIDA

(Illustration 18)

This phylum includes three classes: *Oligochaeta*, *Polychaeta*, and *Hirudinea*. Since polychaetes are primarily marine and the leeches (*Hirudinea*) are mostly macroscopic and parasitic, only the oligochaete is represented in figure 7.1. Some oligochaetes are marine, but the majority are found in fresh water and soil. These worms are characterized by body segmentation, bristles (*setae*) on each segment, an anterior mouth, and a roundish protrusion—the *prostomium*—anterior to the mouth. Although most oligochaetes breathe through the skin, some aquatic forms possess gills at the posterior end or along the sides of the segments. Most oligochaetes feed on vegetation; some feed on the muck of the bottoms of polluted waters, aiding in purifying such places.

PHYLUM TARDIGRADA

(Illustrations 11 and 12)

These invertebrates are of uncertain taxonomic position. They appear to be closely related to both the

Annelida and Arthropoda. They are commonly referred to as the **water bears**. They are generally no more than 1 mm long, with a head, four trunk segments, and four pairs of legs. The ends of the legs may have claws, fingers, or disklike structures. The anterior end has a retractable snout with teeth. Eyes are often present. Sexes are separate, and females are oviparous. They are primarily herbivorous. Locomotion is by crawling, not swimming. During desiccation of their habitat they contract to form barrel-shaped *tuns* and are able to survive years of dryness, even in extremes of heat and cold. Widespread distribution is due to dispersal of the tuns by the wind.

PHYLUM ARTHROPODA

(Illustrations 19, 20, 21, 22, 23)

This phylum contains most of the known Animalia, almost a million species. Representatives of three groups of the Class *Crustacea* are shown in figure 7.1: *Cladocera*, *Ostracoda*, and *Copepoda*. The characteristics these three have in common are jointed appendages, an exoskeleton, and gills.

The **cladocera** are represented by *Daphnia* and *Latanopsis* in figure 7.1. They are commonly known as **water fleas**. All cladocera have a distinct head. The body is covered by a bivalvelike carapace. There is often a distinct cervical notch between the head and body. A compound eye may be present; when present, it is movable. They have many appendages: antennules, antennae, mouth parts, and four to six pairs of legs.

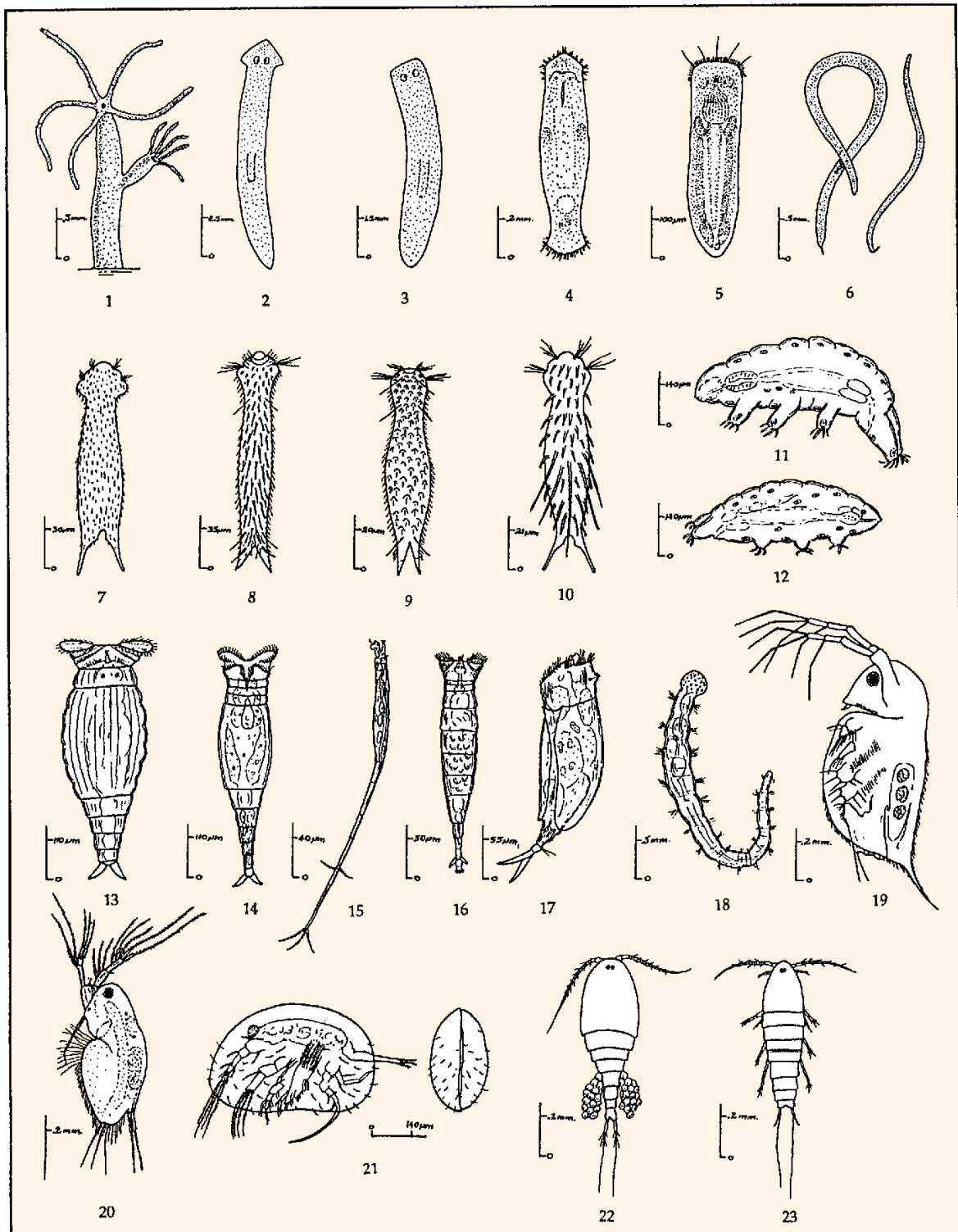
The **ostracods** are bivalved crustaceans that are distinguished from minute clams by the absence of lines of growth on the shell. Their bodies are not distinctly segmented. They have seven pairs of appendages. The end of the body terminates with a pair of *caudal furca*.

The **copepods** represented here are *Cyclops* and *Canthocamptus*. They lack the shell-like covering of the ostracods and cladocera; instead, they exhibit distinct body segmentation. They may have three simple eyes or a single median eye. Eggs are often seen attached to the abdomen on females.

LABORATORY REPORT

There is no Laboratory Report for this exercise.

Microscopic Invertebrates • Exercise 7



- | | | | |
|-----------------------|------------------------------|------------------------|--------------------------|
| 1. <i>Hydra</i> | 6. <i>Nematodes</i> | 15, 16. <i>Rotaria</i> | 21. <i>Ostracod</i> |
| 2. <i>Dugesia</i> | 7. <i>Lepidermella</i> | 17. <i>Euchlanis</i> | 22. <i>Cyclops</i> |
| 3. <i>Planaria</i> | 8, 9, 10. <i>Chaetonotus</i> | 18. <i>Oligochaete</i> | 23. <i>Canthocamptus</i> |
| 4. <i>Macrostomum</i> | 11, 12. <i>Hypsibius</i> | 19. <i>Daphnia</i> | |
| 5. <i>Provortex</i> | 13, 14. <i>Philodina</i> | 20. <i>Latonopsis</i> | |

Figure 7.1 Microscopic invertebrates

Aseptic Technique

8

Exercises 9 and 10 of this survey involve the use of sterile media in culture tubes and Petri plates. The proper handling of these materials requires special skills that you must master. It is the purpose of this exercise to provide you with procedures that will become routine as you progress through this course. It's all about aseptic technique.

The procedures put forth in the section of this manual entitled Laboratory Protocol outline some of the specifics to be observed to ensure that you understand what is required in maintaining an aseptic environment when handling cultures of microorganisms. In this exercise you will have an opportunity to actually work with cultures and different kinds of media to develop those skills that are required to maintain asepsis.

Aseptic transfer of a culture from one culture vessel to another is successful only if no contaminating microorganisms are introduced in the process. A transfer may involve the transport of organisms from an isolated colony on a plate of solid medium to a broth tube, or inoculating various media (solid or liquid) from a broth culture for various types of tests. The general procedure is as follows:

Work Area Disinfection. The work area is first treated with a disinfectant to kill any microorganisms that may be present. This step destroys vegetative cells and viruses; endospores, however, are not destroyed in this brief application of disinfectant.

Loops and Needles. The transport of organisms will be performed with an inoculating loop or needle. To sterilize the loop or needle prior to picking up the organisms, heat must be applied with a Bunsen burner flame, rendering them glowing red-hot.

Culture Tube Flaming. Before inserting the cooled loop or needle into a tube of culture, the tube cap is removed and the mouth of the culture tube flamed. Once the organisms have been removed from the tube, the tube mouth must be flamed again before returning the cap to the tube.

Liquid Medium Inoculation. If a tube of liquid medium is to be inoculated, the tube mouth must be flamed before inserting the loop into the tube. To disperse the organisms on the loop, the loop should be twisted back and forth in the medium.

If an inoculating needle is used for stabbing a solid medium, the needle is inserted deep into the medium.

Final Flaming. Once the inoculation is completed, the loop or needle is removed from the tube, flamed as before, and returned to a receptacle. These tools should never be placed on the tabletop. The inoculated tube is also flamed before placing the cap on the tube.

Petri Plate Inoculation. To inoculate a Petri plate, no heat is applied to the plate and a loop is used for the transfer. When streaking the surface of the medium, the cover should be held diagonally over the plate bottom to prevent air contamination of the medium.

Final Disinfection. When all work is finished, the work area is treated with disinfectant to ensure that any microorganisms deposited during any of the procedures are eliminated.

To gain some practice in aseptic transfer of bacterial cultures, three simple transfers will be performed here in this exercise: (1) broth culture to broth, (2) agar slant culture to agar slant, and (3) agar plate to agar slant. Proceed as follows:

TRANSFER FROM BROTH CULTURE TO ANOTHER BROTH

Do a broth tube to broth tube inoculation, using the following technique. Figure 8.1 illustrates the procedure for removing organisms from a culture, and figure 8.2 shows how to inoculate a tube of sterile broth.

Materials:

broth culture of *Escherichia coli*
tubes of sterile nutrient broth
inoculating loop
Bunsen burner
disinfectant for desktop and sponge
china marking pencil

1. Prepare your desktop by swabbing down its surface with a disinfectant. Use a sponge.
2. With a china marking pencil, label a tube of sterile nutrient broth with your initials and *E. coli*.
3. Sterilize your inoculating loop by holding it over the flame of a Bunsen burner **until it becomes bright red**. The entire wire must be heated. See illustration 1, figure 8.1.
4. Using your free hand, gently shake the tube to disperse the culture (illustration 2, figure 8.1).

Exercise 8 • Aseptic Technique

5. Grasp the tube cap with the little finger of your hand holding the inoculating loop and remove it from the tube. Flame the mouth of the tube as shown in illustration 3, figure 8.1.
6. Insert the inoculating loop into the culture (illustration 4, figure 8.1).
7. Remove the loop containing the culture, flame the mouth of the tube again (illustration 5, figure 8.1), and recap the tube (illustration 6). Place the culture tube back on the test-tube rack.
8. Grasp a tube of sterile nutrient broth with your free hand, carefully remove the cap with your little finger, and flame the mouth of this tube (illustration 1, figure 8.2).
9. Without flaming the loop, insert it into the sterile broth, inoculating it (illustration 2, figure 8.2). To disperse the organisms into the medium, move the loop back and forth in the tube.

10. Remove the loop from the tube and flame the mouth (illustration 3, figure 8.2). Replace the cap on the tube (illustration 4, figure 8.2).
11. Sterilize the loop by flaming it (illustration 5, figure 8.2). Return the loop to its container.
12. Incubate the culture you just inoculated at 37° C for 24–48 hours.

TRANSFER OF BACTERIA FROM SLANT TO SLANT

To inoculate a sterile nutrient agar slant from an agar slant culture, use the following procedure. Figure 8.4 illustrates the entire process.

Materials:

agar slant culture of *E. coli*
sterile nutrient agar slant
inoculating loop
Bunsen burner
china marking pencil

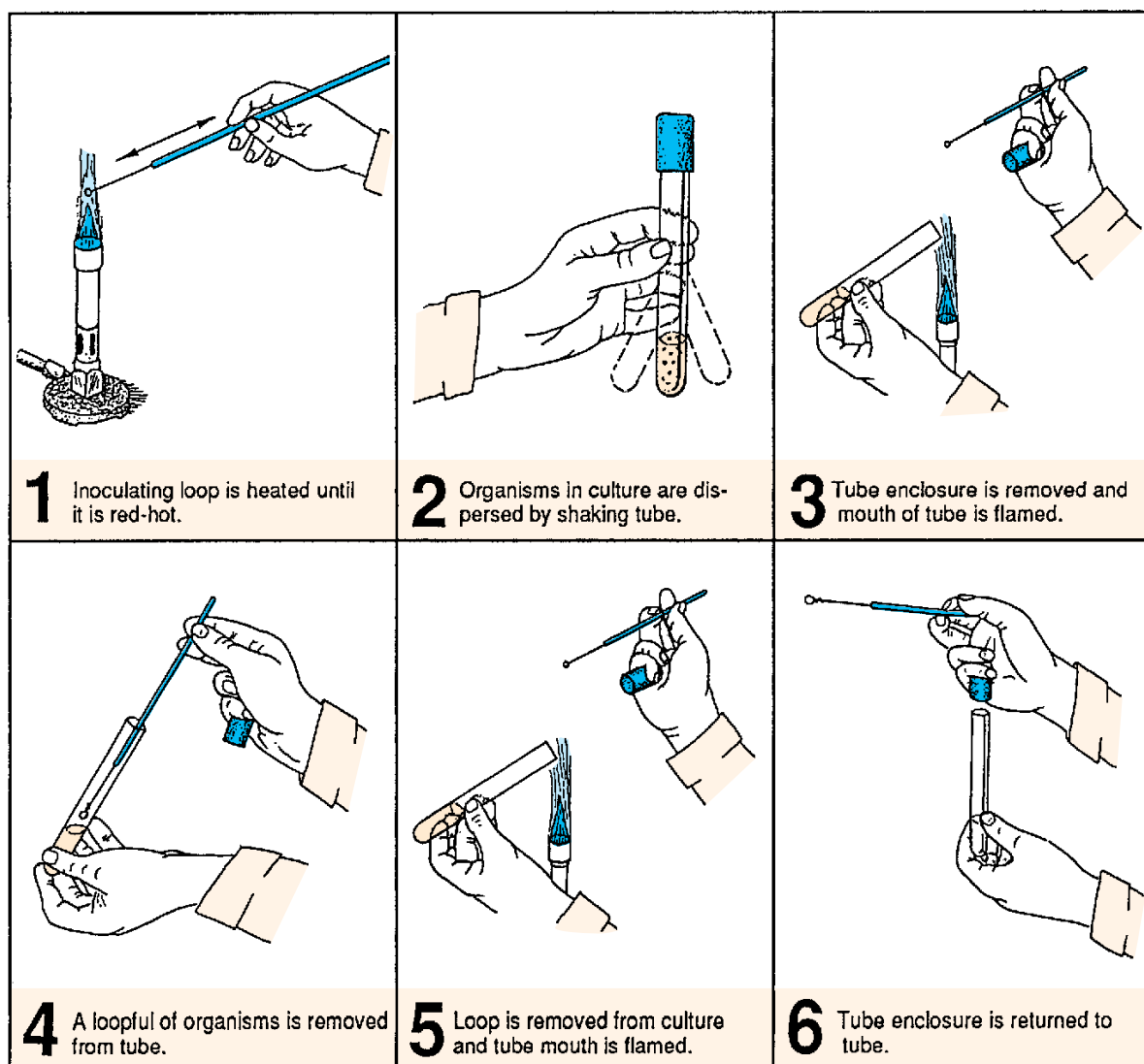


Figure 8.1 Procedure for removing organisms from a broth culture with inoculating loop

Aseptic Technique • Exercise 8

1. If you have not already done so, prepare your desktop by swabbing down its surface with a disinfectant.
2. With a china marking pencil label a tube of nutrient agar slant with your initials and *E. coli*.
3. Sterilize your inoculating loop by holding it over the flame of a Bunsen burner **until it becomes bright red** (illustration 1, figure 8.4). The entire wire must be heated. Allow the loop to cool completely.
4. Using your free hand, pick up the slant culture of *E. coli* and remove the cap using the little finger of the hand that is holding the loop (illustration 2, figure 8.4).
5. Flame the mouth of the tube and insert the cooled loop into the tube. Pick up some of the culture on the loop (illustration 3, figure 8.4) and remove the loop from the tube.
6. Flame the mouth of the tube (illustrations 4 and 5, figure 8.4) and replace the cap, being careful not to burn your hand. Return tube to rack.
7. Pick up a sterile nutrient agar slant with your free hand, remove the cap with your little finger as before, and flame the mouth of the tube (illustration 6, figure 8.4).
8. Without flaming the loop containing the culture, insert the loop into the tube and gently inoculate the surface of the slant by moving the loop back and forth over the agar surface, while moving up the surface of the slant (illustration 7, figure 8.4). This should involve a type of serpentine motion.
9. Remove the loop, flame the mouth of the tube, and recap the tube (illustration 8, figure 8.4). Replace the tube in the rack.

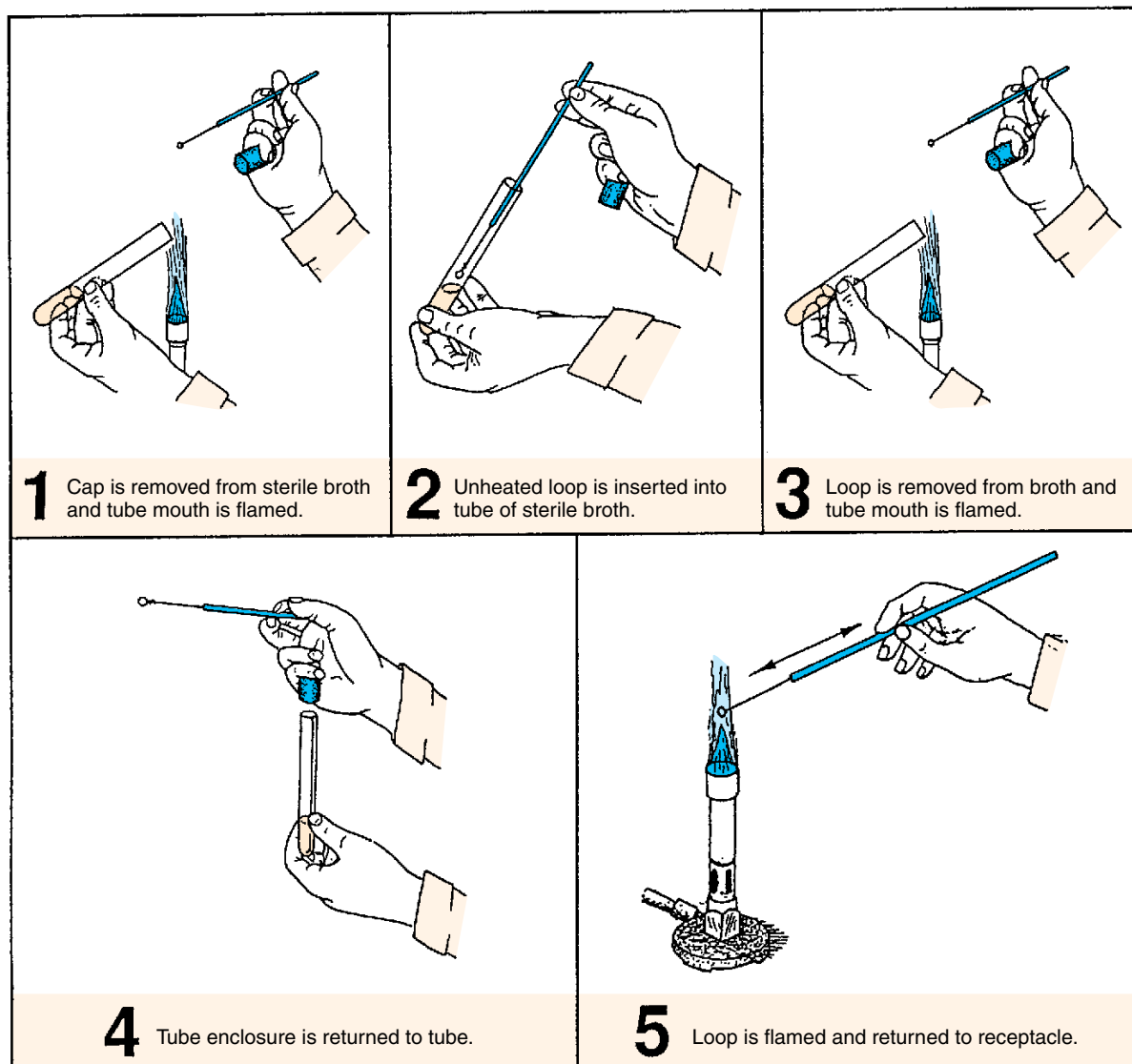


Figure 8.2 Procedure for inoculating a nutrient broth

Exercise 8 • Aseptic Technique

10. Flame the loop, heating the entire wire to red-hot (illustration 9, figure 8.4), allow to cool, and place the loop in its container.
11. Incubate the inoculated agar slant at 30° C for 24–48 hours.

WORKING WITH AGAR PLATES (Inoculating a slant from a Petri plate)

The transfer of organisms from colonies on agar plates to slants or broth tubes is very similar to the procedures used in the last two transfers (broth to broth and slant to slant). The following rules should be observed.

Loops vs. Needles In some cases a loop is used. In other situations a needle is preferred. When a large inoculum is needed in the transfer, a loop will be used. Needles are preferred, however, when making transfers in pure culture isolations and making stab cultures. In

pure culture isolations, a needle is inserted into the center of a colony for the transfer. This technique is used, primarily, when working with mixed cultures.

Plate Handling Media in plates must always be protected against contamination. To prevent exposure to air contamination, covers should always be left closed. When organisms are removed from a plate culture, the cover should be only partially opened as shown in illustration 2, figure 8.3.

Flaming Procedures Inoculating loops or needles must be flamed in the same manner that you used when working with previous tubes. One difference when working with plates is that plates are never flamed!

Plate Labeling Petri plates with media in them are always labeled on the bottom. Inoculated plates are preferably stored upside down.

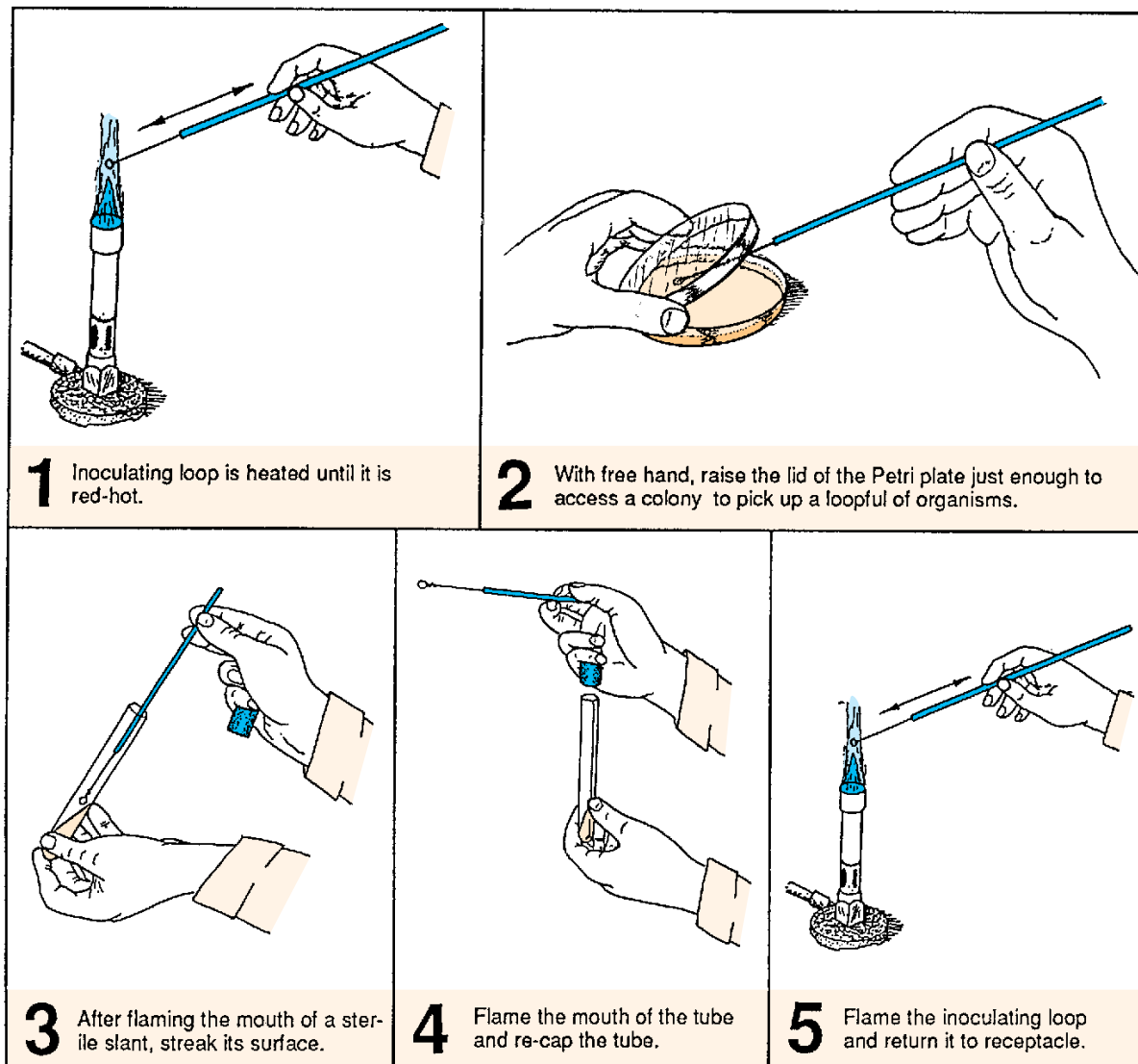


Figure 8.3 Procedure for inoculating a nutrient agar slant from an agar plate

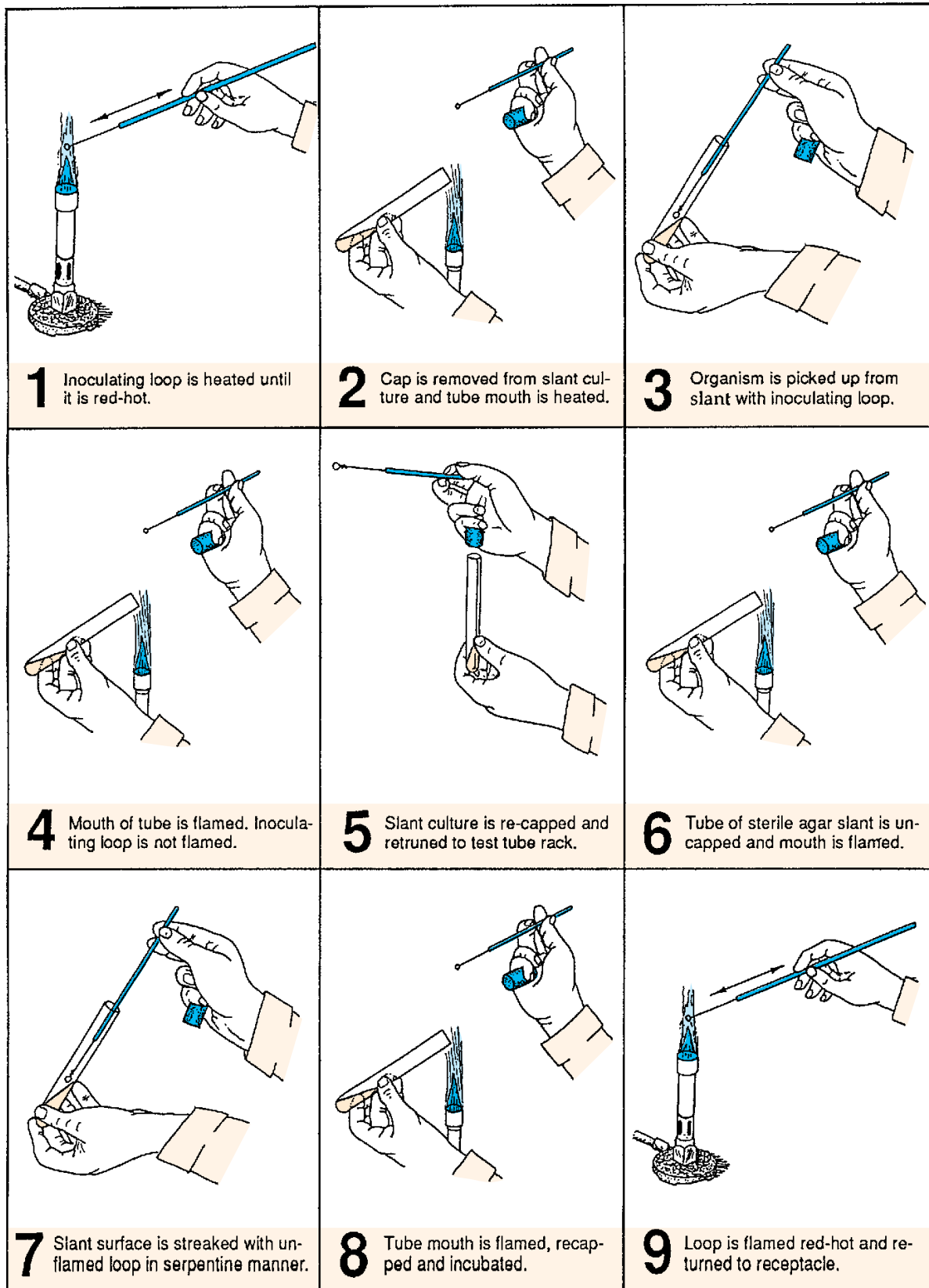


Figure 8.4 Procedure for inoculating a nutrient agar slant from a slant culture

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Aseptic Technique • Exercise 8

To transfer organisms from a Petri plate to an agar slant, use the following procedure:

Materials:

nutrient agar plate with bacterial colonies
sterile nutrient agar slant
inoculating loop
china marking pencil

1. If you have not done so, swab your work area with disinfectant. Allow area to dry.
2. Label a sterile nutrient agar slant with your name and organism to be transferred.
3. Flame an inoculating loop until it is red-hot (illustration 1, figure 8.3). Allow the loop to cool.
4. As shown in illustration 2, figure 8.3, raise the lid of a Petri plate sufficiently to access a colony with your sterile loop.

Do not gouge into the agar with your loop as you pick up organisms, and do not completely remove the lid, exposing the surface to the air. Close the lid once you have picked up the organisms.

5. With your free hand, pick up the sterile nutrient agar slant tube. Remove the cap by grasping the cap with the little finger of the hand that is holding the loop.
6. Flame the mouth of the tube and insert the loop into the tube to inoculate the surface of the slant, using a serpentine motion (illustration 3, figure 8.3). Avoid disrupting the agar surface with the loop.
7. Remove the loop from the tube and flame the mouth of the tube. Replace the cap on the tube (illustration 4, figure 8.3).
8. Flame the loop (illustration 5, figure 8.3) and place it in its container.
9. Incubate the nutrient agar slant at 37° C for 24–48 hours.

SECOND PERIOD

Examine all three tubes and record your results on Laboratory Report 8,9.

9

The Bacteria

Of all the microorganisms studied so far, the bacteria are the most widely distributed, the simplest in morphology, the smallest in size, the most difficult to classify, and the hardest to identify. It is even difficult to provide a descriptive definition of what a bacterial organism is because of considerable diversity in the group. About the only generalization that can be made for the entire group is that they are prokaryotic and are seldom photosynthetic. The few that are photosynthetic utilize a pigment that is chemically different from chlorophyll *a*. It is called *bacteriochlorophyll*. Probably the simplest definition that one can construct from these facts is: ***Bacteria are prokaryons without chlorophyll *a*.***

Although the bacteria are generally smaller than the cyanobacteria, some cyanobacteria are in the size range of bacteria. Most bacteria are only 0.5 to 2.0 micrometers in diameter.

Figure 9.1 illustrates most of the shapes of bacteria that one would encounter. Note that they can be grouped into three types: rod, spherical, and helical or curved. Rod-shaped bacteria may vary considerably in length; have square, round, or pointed ends; and be motile or nonmotile. The spherical, or coccus-shaped, bacteria may occur singly, in pairs, in tetrads, in chains, and in irregular masses. The helical and curved bacteria exist as slender spirochaetes, spirillum, and bent rods (vibrios).

In this exercise an attempt will be made to demonstrate the ubiquitousness of these organisms. No attempt will be made to study detailed bacterial anatomy or physiology. Many exercises related to staining, microscopy, and physiology in subsequent laboratory periods will provide a clear understanding of these microorganisms.

Our concern here relates primarily to the widespread distribution of bacteria in our environment. Being thoroughly aware of their existence all around us is of prime importance if we are to develop those laboratory skills that we recognize as *aseptic technique*. The awareness that bacteria are everywhere must be constantly in our minds when handling bacterial cultures. In the next laboratory period you will be handling tube cultures of bacteria, much in the

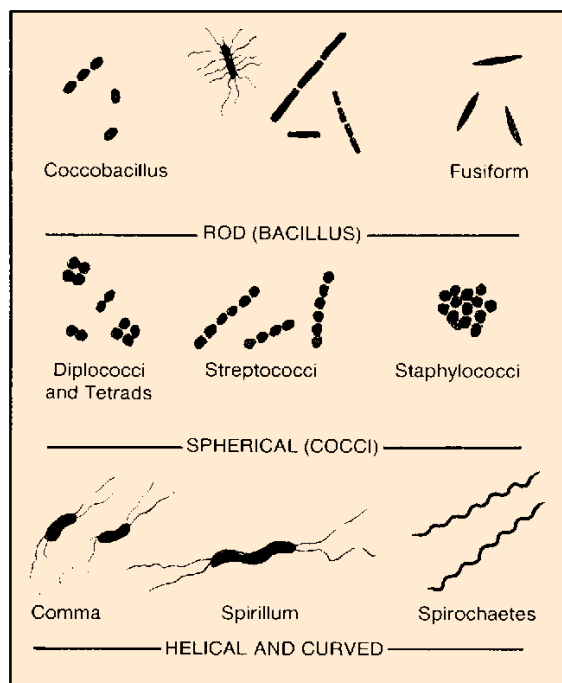


Figure 9.1 Bacterial morphology

same manner that you did in Exercise 8 (Aseptic Technique). A continuing constant effort will always be made to develop the proper routine. Remember: *without pure cultures the study of bacteriology becomes a hopeless endeavor.*

During this laboratory period you will be provided with three kinds of sterile bacteriological media that you will expose to the environment in various ways. To ensure that these exposures cover as wide a spectrum as possible, specific assignments will be made for each student. In some instances a moistened swab will be used to remove bacteria from some object; in other instances a Petri plate of medium will be exposed to the air or a cough. You will be issued a number that will enable you to determine your specific assignment from the chart on the next page.

The Bacteria • Exercise 9

Materials:

per student:

- 1 tube of nutrient broth
- 1 Petri plate of trypticase soy agar (TSA)
- 1 sterile cotton swab
- china marking pencil

per two or more students:

- 1 Petri plate of blood agar
1. Scrub down your desktop with a disinfectant in the same manner you used in Exercise 8 (Aseptic Technique).
 2. Expose your TSA plate according to your assignment in the table below. *Label the bottom* of your plate with your initials, your assignment number, and the date.
 3. Moisten a sterile swab by immersing it into a tube of nutrient broth and expressing most of the broth out of it by pressing the swab against the inside wall of the tube.
 4. Rub the moistened swab over a part of your body such as a finger or ear, or some object such as a doorknob or telephone mouthpiece, and return the swab to the tube of broth. It may be necessary to break off the stick end of the swab so that you can replace the cap on the tube.
 5. Label the tube with your initials and the source of the bacteria.

6. Expose the blood agar plate by coughing onto it. Label the bottom of the plate with the initials of the individuals that cough onto it. Be sure to date the plate also.
7. Incubate the plates and tube at 37° C for 48 hours.

EVALUATION

After 48 hours incubation, examine the tube of nutrient broth and two plates. Shake the tube vigorously without wetting the cap. Is it cloudy or clear? Compare it with an uncontaminated tube of broth. What is the significance of cloudiness? Do you see any colonies growing on the blood agar plate? Are the colonies all the same size and color? If not, what does this indicate? Group together a set of TSA plates representing all nine types of exposure. Record your results on the Laboratory Report.

Your instructor will indicate whether these tubes and plates are to be used for making slides in Exercise 13 (Simple Staining). If the plates and tubes are to be saved, containers will be provided for their storage in the refrigerator. Place the plates and tubes in the designated containers.

LABORATORY REPORT

Record your results on the last portion of Laboratory Report 8,9.

Exposure Method for TSA Plate	Student Number
1. To the air in laboratory for 30 minutes	1, 10, 19, 28
2. To the air in room other than laboratory for 30 minutes	2, 11, 20, 29
3. To the air outside of building for 30 minutes	3, 12, 21, 30
4. Blow dust onto exposed medium	4, 13, 22, 31
5. Moist lips pressed against medium	5, 14, 23, 32
6. Fingertips pressed lightly on medium	6, 15, 24, 33
7. Several coins pressed temporarily on medium	7, 16, 25, 34
8. Hair is combed over exposed medium (10 strokes)	8, 17, 26, 35
9. Optional: Any method not listed above	9, 18, 27, 36

10

The Fungi: Yeasts and Molds

The fungi comprise a large group of eukaryotic non-photosynthetic organisms that include such diverse forms as slime molds, water molds, mushrooms, puff-balls, bracket fungi, yeasts, and molds. Fungi belong to Kingdom **Myceteae**. The study of fungi is called **mycology**.

Myceteae consist of three divisions: Gymnomycota (slime molds), Mastigomycota (water molds and others), and Amastigomycota (yeasts, molds, bracket fungi, and others). It is the last division that we will study in this exercise.

Fungi may be saprophytic or parasitic and unicellular or filamentous. Some organisms, such as the slime molds (Exercise 25), are borderline between fungi and protozoa in that amoeboid characteristics are present and fungi like spores are produced.

The distinguishing characteristics of the group as a whole are that they (1) are eukaryotic, (2) are nonphotosynthetic, (3) lack tissue differentiation, (4) have cell walls of chitin or other polysaccharides, and (5) propagate by spores (sexual and/or asexual).

In this study we will examine prepared stained slides and slides made from living cultures of yeasts and molds. Molds that are normally present in the air will be cultured and studied macroscopically and microscopically. In addition, an attempt will be made to identify the various types that are cultured.

Before attempting to identify the various molds, familiarize yourself with the basic differences between molds and yeasts. Note in figure 10.1 that

yeasts are essentially unicellular and molds are multicellular.

MOLD AND YEAST DIFFERENCES

Species within the Amastigomycota may have cottony (moldlike) appearance or moist (yeasty) characteristics that set them apart. As pronounced as these differences are, we do not classify the various fungi in this group on the basis of their being mold or yeast. The reason that this type of division doesn't work is that some species exist as molds under certain conditions and as yeasts under other conditions. Such species are said to be **dimorphic**, or **biphasic**.

The principal differences between molds and yeasts are as follows:

Molds

Hyphae Molds have microscopic filaments called *hyphae* (hypha, singular). As shown in figure 10.1, if the filament has crosswalls, it is referred to as having **septate hyphae**. If no crosswalls are present, the coenocytic filament is said to be **nonseptate**, or **aseptate**. Actually, most of the fungi that are classified as being septate are incompletely septate since the septae have central openings that allow the streaming of cytoplasm from one compartment to the next. A mass of intermeshed hyphae, as seen macroscopically, is a *mycelium*.

Asexual Spores Two kinds of asexual spores are seen in molds: sporangiospores and conidia. **Spor-**

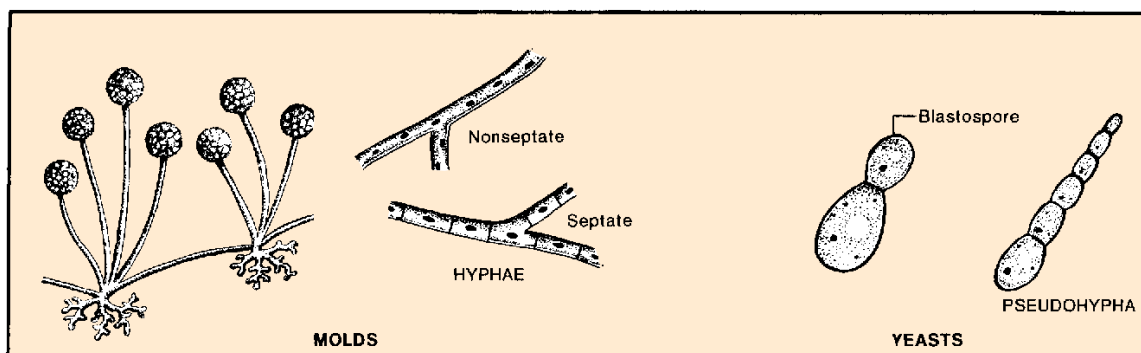


Figure 10.1 Structural differences between molds and yeasts

The Fungi: Yeasts and Molds • Exercise 10

angiospores are spores that form within a sac called a *sporangium*. The sporangia are attached to stalks called *sporangiophores*. See illustration 1, figure 10.2.

Conidia are asexual spores that form on specialized hyphae called *conidiophores*. If the conidia are small they are called *microconidia*; large multicellular conidia are known as *macroconidia*. The following four types of conidia are shown in figure 10.2:

- **Phialospores:** Conidia of this type are produced by vase-shaped cells called *phialides*. Note in figure 10.2 that *Penicillium* and *Gliocadium* produce this type.
- **Blastoconidia:** Conidia of this type are produced by budding from cells of preexisting conidia, as in *Cladosporium*, which typically has lemon-shaped spores.
- **Arthrospores:** This type of conidia forms by separation from preexisting hyphal cells. Example: *Oospora*.
- **Chlamydospores:** These spores are large, thick-walled, round, or irregular structures formed within or on the ends of a hypha. Common to most fungi, they generally form on old cultures. Example: *Candida albicans*.

Sexual Spores Three kinds of sexual spores are seen in molds: zygospores, ascospores, and basidiospores. Figure 10.3 illustrates the three types.

Zygospores are formed by the union of nuclear material from the hyphae of two different strains. **Ascospores**, on the other hand, are sexual spores produced in enclosures, which may be oval sacs or elongated tubes. **Basidiospores** are sexually produced on club-shaped bodies called *basidia*. A basidium is considered by some to be a modified type of ascus.

Yeasts

Hyphae Unlike molds, yeasts do not have true hyphae. Instead they form multicellular structures called **pseudohyphae**. See figure 10.1.

Asexual Spores The only asexual spore produced by yeasts is called a **blastospore**, or **bud**. These spores form as an outpouching of a cell by a budding process. It is easily differentiated from the parent cell by its small size. It may separate from the original cell or remain attached. If successive buds remain attached in the budding process, the result is the formation of a **pseudohypha**.

SUBDIVISIONS OF THE AMASTIGOMYCOTA

Division Amastigomycota consists of four subdivisions: Zygomycotina, Ascomycotina, Basidiomycotina, and Deuteromycotina. They are separated on the basis of the type of sexual reproductive spores as follows:

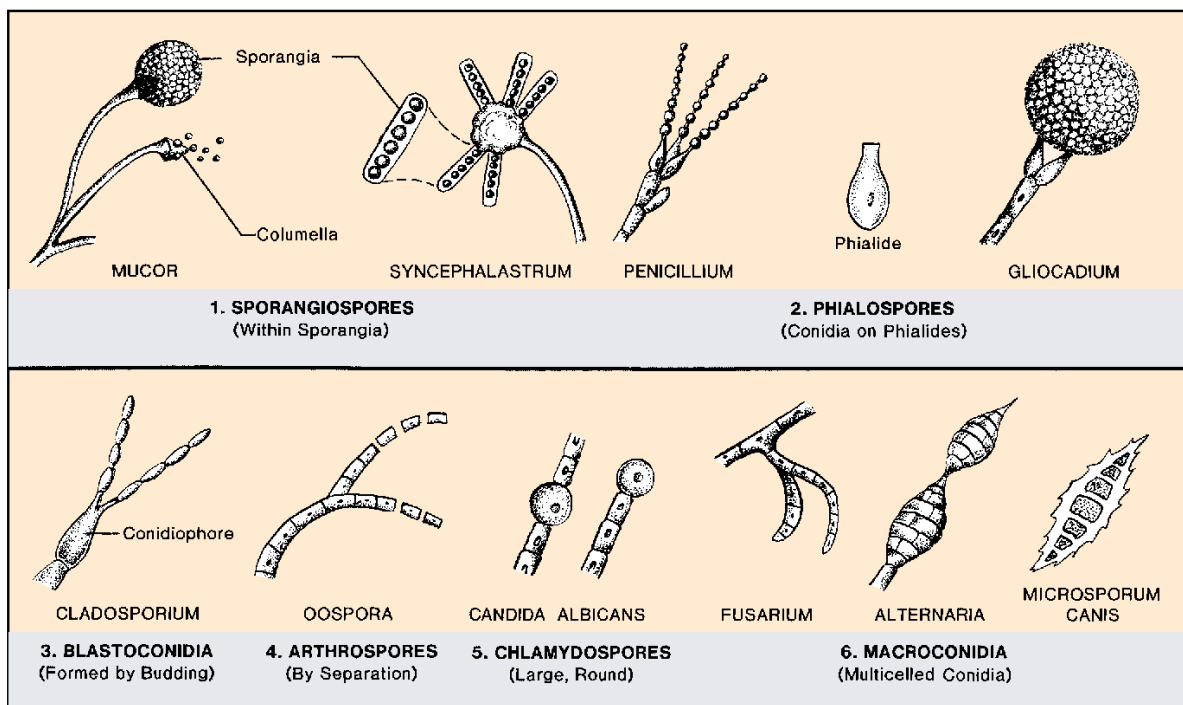


Figure 10.2 Types of asexual spores seen in fungi

Exercise 10 • The Fungi: Yeasts and Molds**Zygomycotina**

These fungi have nonseptate hyphae and produce zygospores. They also produce sporangiospores. *Rhizopus*, *Mucor*, and *Syncephalastrum* are representative genera of this subdivision.

Ascomycotina

Since all the fungi in this subdivision produce ascospores, they are grouped into one class, the *Ascomycetes*. They are commonly referred to as the “ascomycetes” and are also called “sac fungi.” All of them have septate hyphae and most of them have chitinous walls.

Fungi in this group that produce a single ascus are called *ascomycetes yeasts*. Other ascomycetes produce numerous asci in complex flask-shaped fruiting bodies called **perithecia** or **pseudothecia**, in cup-shaped structures, or in hollow spherical bodies, as in powdery mildews, *Eupenicillium* or *Talaromyces* (the sexual stages for *Penicillium*).

Basidiomycotina

All fungi in this subdivision belong to one class, the *Basidiomycetes*. Puffballs, mushrooms, smuts, rust, and shelf fungi on tree branches are also basidiomycetes. The sexual spores of this class are **basidiospores**.

Deuteromycotina

This fourth division of the Amastigomycota is an artificial group that was created to place any fungus that has not been shown to have some means of sexual reproduction. Often, species that are relegated to this division remain here for only a short period of time: as soon as the right conditions have been provided for sexual spores to form, they are reclassified into one of the first three subdivisions. Sometimes, however, the asexual and sexual stages of a fungus are discovered and named separately by different mycologists, with the result that a single species acquires two different names. Although generally there is a switch over to the sexual-stage name, not all mycologists conform to this practice.

Members of this group are commonly referred to as the *fungi imperfecti* or *deuteromycetes*. It is a large group, containing over 15,000 species.

LABORATORY PROCEDURES

Several options are provided here for the study of molds and yeasts. The procedures to be followed will be outlined by your instructor.

Yeast Study

The organism *Saccharomyces cerevisiae*, which is used in bread making and alcohol fermentation, will be used for this study. Either prepared slides or living organisms may be used.

Materials:

- prepared slides of *Saccharomyces cerevisiae*
- broth cultures of *Saccharomyces cerevisiae*
- methylene blue stain
- microscope slides and cover glasses

Prepared Slides If prepared slides are used, they may be examined under high-dry or oil immersion. One should look for typical **blastospores** and **ascospores**. Space is provided on the Laboratory Report for drawing the organisms.

Living Material If broth cultures of *Saccharomyces cerevisiae* are available they should be examined on a wet mount slide with phase-contrast or brightfield optics. Two or three loopfuls of the organisms should be placed on the slide with a drop of methylene blue stain. Oil immersion will reveal the greatest amount of detail. Look for the **nucleus** and **vacuole**. The nucleus is the smaller body. Draw a few cells on the Laboratory Report.

Mold Study

Examine a Petri plate of Sabouraud's agar that has been exposed to the air for about an hour and incubated at room temperature for 3–5 days. This medium

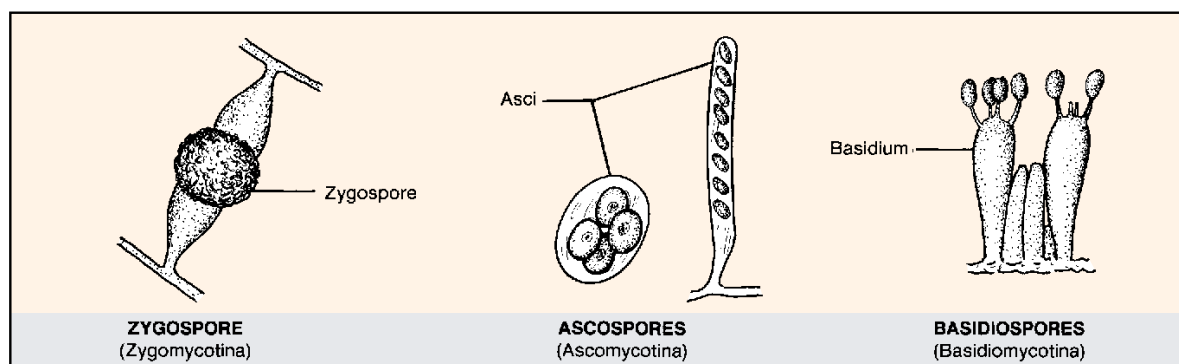


Figure 10.3 Types of sexual spores seen in the Amastigomycota

The Fungi: Yeasts and Molds • Exercise 10

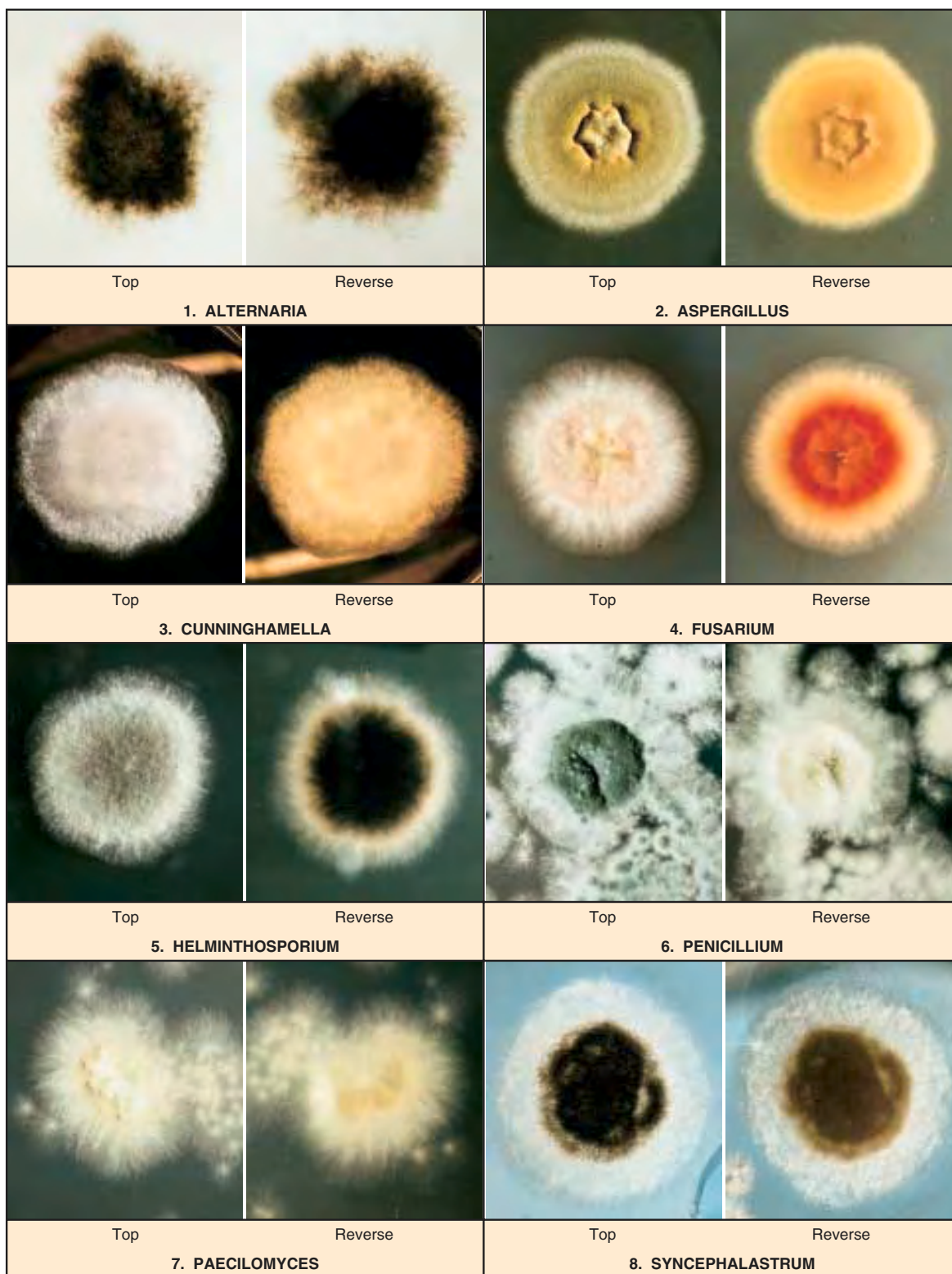


Figure 10.4 Colony characteristics of some of the more common molds

Exercise 10 • The Fungi: Yeasts and Molds

has a low pH, which makes it selective for molds. A good plate will have many different-colored colonies. Note the characteristic “cottony” nature of the colonies. Also, look at the bottom of the plate and observe how the colonies differ in color here. The identification of molds is based on surface color, backside color, hyphal structure, and types of spores.

Figure 10.4 reveals how some of the more common molds appear when grown on Sabouraud’s agar. Keep in mind when using figure 10.4 that the appearance of a mold colony can change appreciably as it gets older. The photographs in figure 10.4 are of colonies that are 10 to 21 days old.

Conclusive identification cannot be made unless a microscope slide is made to determine the type of hyphae and spores that are present. Figure 10.5 reveals, diagrammatically, the microscopic differences that one looks for when identifying mold genera.

Two Options In making slides from mold colonies, one can make either wet mounts directly from the colonies by the procedure outlined here or make cultured slides as outlined in Exercise 26. The following steps should be used for making stained slides directly from the colonies. Your instructor will indicate the number of identifications that are to be made.

Materials:

mold cultures on Sabouraud’s agar
microscope slides and cover glasses
lactophenol cotton blue stain
sharp-pointed scalpels or dissecting needles

1. Place the uncovered plate on the stage of your microscope and examine the edge of a colored colony with the low-power objective. Look for hyphal structure and spore arrangement. Ignore the white colonies since they generally lack spores and are difficult to identify.

CAUTION

Avoid leaving the cover off the mold culture plates or disturbing the colonies very much. Dispersal of mold spores to the air must be kept to a minimum.

2. Consult figures 10.4 and 10.5 to make a preliminary identification based on colony characteristics and low-power magnification of hyphae and spores.
3. Make a wet mount slide by transferring a small amount of the culture with a sharp scalpel or dissecting needle to a drop of lactophenol cotton blue stain on a slide. Cover with a cover glass and examine under low-power and high-dry objectives. Refer again to figure 10.5 to confirm any conclusions drawn from your previous examination of the edge of the colony.
4. Repeat the above procedure for each different colony.

LABORATORY REPORT

After recording your results on the Laboratory Report, answer all the questions.

Figure 10.5 Legend

- | | |
|---|---|
| 1. <i>Penicillium</i> —bluish-green; brush arrangement of phialospores. | 12. <i>Pullularia</i> —black, shiny, leathery surface; thick-walled; budding spores. |
| 2. <i>Aspergillus</i> —bluish-green with sulfur-yellow areas on the surface. <i>Aspergillus niger</i> is black. | 13. <i>Diplosporium</i> —buff-colored woolly surface; reverse side has red center surrounded by brown. |
| 3. <i>Verticillium</i> —pinkish-brown, elliptical microconidia. | 14. <i>Oospora</i> (<i>Geotrichum</i>)—buff-colored surface; hyphae break up into thin-walled rectangular arthrospores. |
| 4. <i>Trichoderma</i> —green, resemble <i>Penicillium</i> macroscopically. | 15. <i>Fusarium</i> —variants of yellow, orange, red, and purple colonies; sickle-shaped macroconidia. |
| 5. <i>Gliocadium</i> —dark green; conidia (phialospores) borne on phialides, similar to <i>Penicillium</i> ; grows faster than <i>Penicillium</i> . | 16. <i>Trichothecium</i> —white to pink surface; two-celled conidia. |
| 6. <i>Cladosporium</i> (<i>Hormodendrum</i>)—light green to grayish surface; gray to black back surface; blastoconidia. | 17. <i>Mucor</i> —a zygomycete; sporangia with a slimy texture; spores with dark pigment. |
| 7. <i>Pleospora</i> —tan to green surface with brown to black back; ascospores shown are produced in sacs borne within brown, flask-shaped fruiting bodies called pseudothecia. | 18. <i>Rhizopus</i> —a zygomycete; spores with dark pigment. |
| 8. <i>Scopulariopsis</i> —light brown; rough-walled microconidia. | 19. <i>Syncephalastrum</i> —a zygomycete; sporangioophores bear rod-shaped sporangioles, each containing a row of spherical spores. |
| 9. <i>Paecilomyces</i> —yellowish-brown; elliptical microconidia. | 20. <i>Nigrospora</i> —conidia black, globose, one-celled, borne on a flattened, colorless vesicle at the end of a conidiophore. |
| 10. <i>Alternaria</i> —dark greenish-black surface with gray periphery; black on reverse side; chains of macroconidia. | 21. <i>Montospora</i> —dark gray center with light gray periphery; yellow-brown conidia. |
| 11. <i>Bipolaris</i> —black surface with grayish periphery; macroconidia shown. | |

The Fungi: Yeasts and Molds • Exercise 10



Figure 10.5 Microscopic appearance of some of the more common molds (legend on opposite page)

PART 3 Microscope Slide Techniques (Bacterial Morphology)

The nine exercises in this unit include the procedures for ten slide techniques that one might employ in morphological studies of bacteria. A culture method in Exercise 18 also is included as a substitute for slide techniques when pathogens are encountered.

These exercises are intended to serve two equally important functions: (1) to help you to develop the necessary skills in making slides and (2) to introduce you to the morphology of bacteria. Although the title of each exercise pertains to a specific technique, the organisms chosen for each method have been carefully selected so that you can learn to recognize certain morphological features. For example, in the exercise on simple staining (Exercise 13), the organisms selected exhibit metachromatic granules, pleomorphism, and palisade arrangement of cells. In Exercise 15, (Gram Staining) you will observe the differences between cocci and bacilli, as well as learn how to execute the staining routine.

The importance of the mastery of these techniques cannot be overemphasized. Although one is seldom able to make species identification on the basis of morphological characteristics alone, it is a very significant starting point. This fact will become increasingly clear with subsequent experiments.

Although the steps in the various staining procedures may seem relatively simple, student success is often quite unpredictable. Unless your instructor suggests a variation in the procedure, try to follow the procedures exactly as stated, without improvisation. Photomicrographs in color have been provided for many of the techniques; use them as a guide to evaluate the slides you have prepared. Once you have mastered a specific technique, feel free to experiment.

11

Negative Staining

The simplest way to make a slide of bacteria is to prepare a wet mount, much in the same manner that was used for studying protozoa and algae. Although this method will quickly produce a slide, finding the bacteria on the slide may be difficult, especially for a beginner. The problem one encounters is that bacteria are quite colorless and transparent. Unless the diaphragm is carefully adjusted, the beginner usually has considerable difficulty bringing the organisms into focus.

A better way to observe bacteria for the first time is to prepare a slide by a process called **negative**, or **background, staining**. This method consists of mixing the microorganisms in a small amount of nigrosine or india ink and spreading the mixture over the surface of a slide. (Incidentally, nigrosine is far superior to india ink.)

Since these two pigments are not really bacterial stains, they do not penetrate the microorganisms. Instead they obliterate the background, leaving the organisms transparent and visible in a darkened field.

Although this technique has limitations, it can be useful for determining cell morphology and size. Since no heat is applied to the slide, there is no shrinkage of the cells, and, consequently, more accurate cell-size determinations result than with some other methods. This method is also useful for studying spirochaetes that don't stain readily with ordinary dyes.

THREE METHODS

Negative staining can be done by one of three different methods. Figure 11.1 illustrates the more commonly used method in which the organisms are mixed in a drop of nigrosine and spread over the slide with another slide. The goal is to produce a smear that is thick at one end and feather-thin at the other end. Somewhere between the too thick and too thin areas will be an ideal spot to study the organisms.

Figure 11.2 illustrates a second method, in which organisms are mixed in only a loopful of nigrosine instead

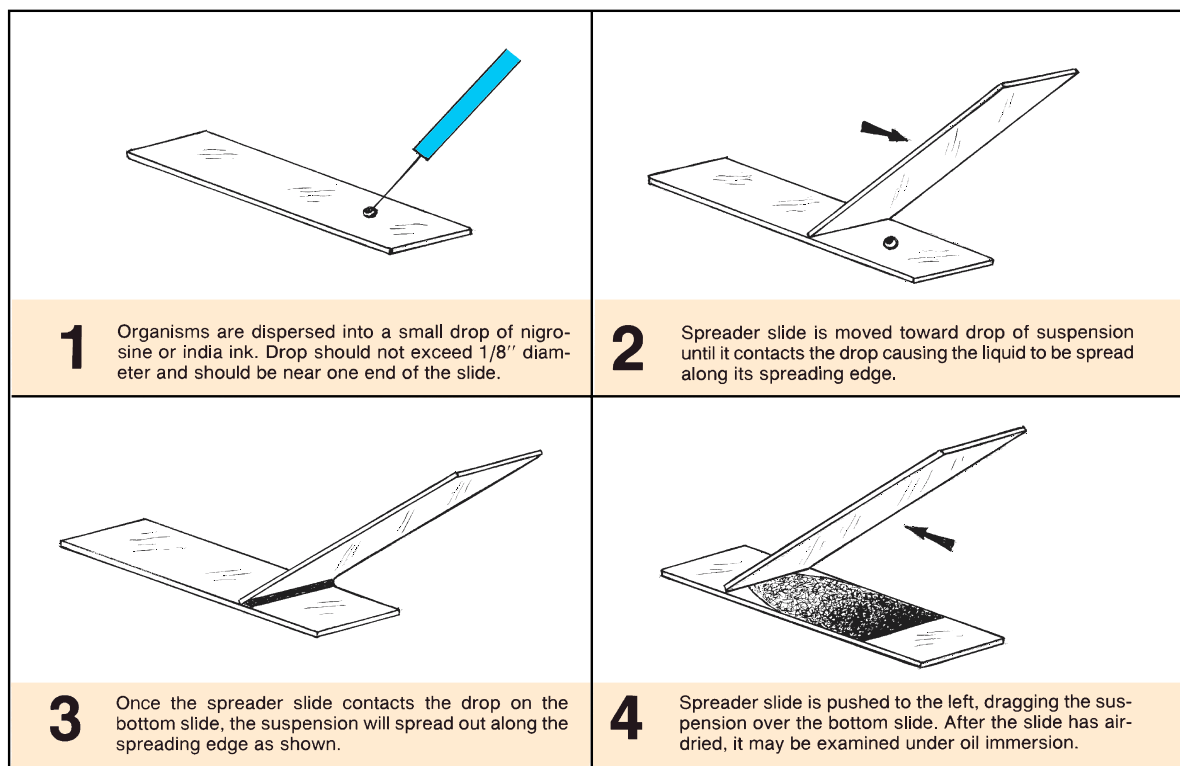


Figure 11.1 Negative staining technique, using a spreader slide

Negative Staining • Exercise 11

of a full drop. In this method the organisms are spread over a smaller area in the center of the slide with an inoculating needle. No spreader slide is used in this method.

The third procedure (Woeste-Demchick's method), which is not illustrated here, involves applying ink to a conventional smear with a black felt marking pen. If this method is used, it should be done on a smear prepared in the manner described in the next exercise. Simply put, the technique involves applying a *single coat* of felt-pen ink over a smear.

Note in the procedure below that slides may be made from organisms between your teeth or from specific bacterial cultures. Your instructor will indicate which method or methods you should use and demonstrate some basic aseptic techniques. Various options are provided here to ensure success.

Materials:

microscope slides (with polished edges)
nigrosine solution or india ink
slant cultures of *S. aureus* and *B. megaterium*
inoculating straight wire and loop
sterile toothpicks
Bunsen burner
china marking pencil
felt marking pen (see Instructor's Handbook)

1. Swab down your tabletop with disinfectant in preparation for making slides.

2. Clean two or three microscope slides with Bon Ami to rid them of all dirt and grease.
3. By referring to figure 11.1 or 11.2, place the proper amount of stain on the slide.
4. **Oral Organisms:** Remove a small amount of material from between your teeth with a sterile straight toothpick or inoculating needle and mix it into the stain on the slide. Be sure to break up any clumps of organisms with the wire or toothpick. When using a wire, *be sure to flame it first to make it sterile.*

CAUTION

If you use a toothpick, discard it into a beaker of disinfectant.

5. **From Cultures:** With a *sterile* straight wire, transfer a very small amount of bacteria from the slant to the center of the stain on the slide.
6. Spread the mixture over the slide according to the procedure used in figure 11.1 or 11.2.
7. Allow the slide to air-dry and examine with an oil immersion objective.

LABORATORY REPORT

Draw a few representative types of organisms on Laboratory Report 11–14. If slide is of oral organisms, look for yeasts and hyphae as well as bacteria. Spirochaetes may also be present.

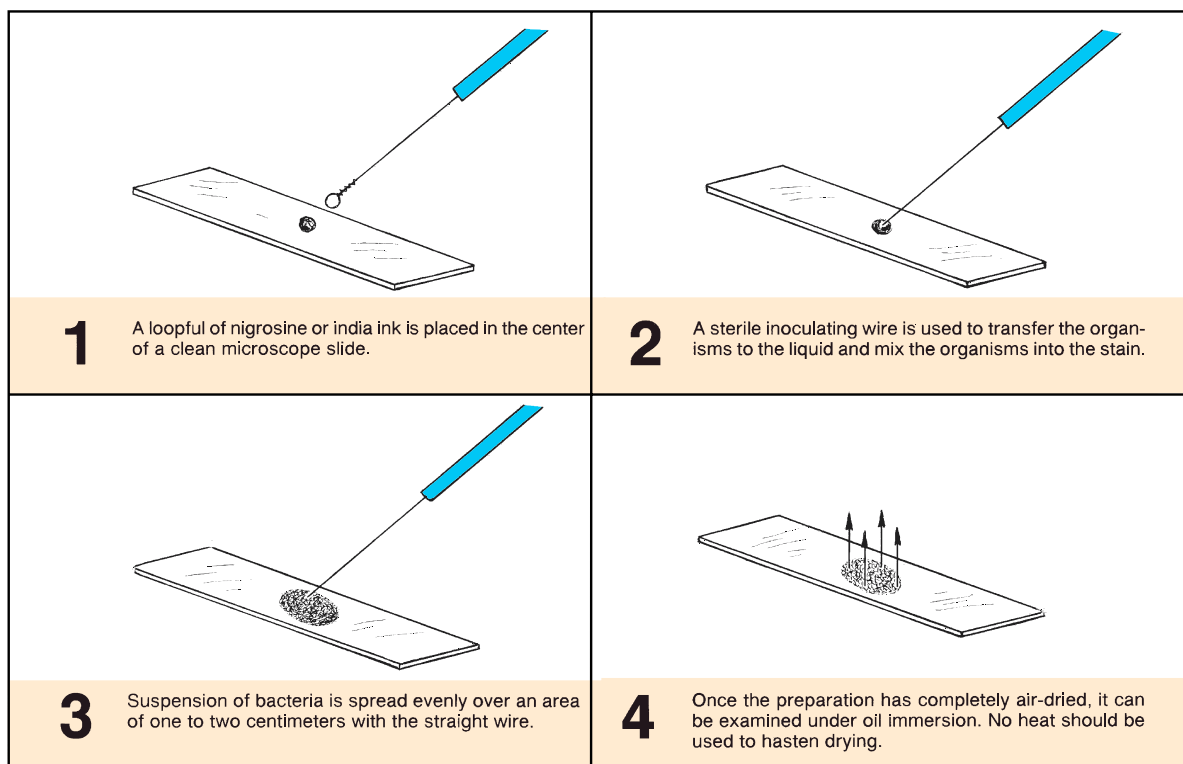


Figure 11.2 A second method for negative staining

12

Smear Preparation

While negative staining is a simple enough process to make bacteria more visible with a brightfield microscope, it is of little help when one attempts to observe anatomical microstructures such as flagella, granules, and endospores. Only by applying specific bacteriological stains to organisms can such organelles be seen. However, success at bacterial staining depends first of all on the preparation of a suitable *smear* of the organisms. A properly prepared bacterial smear is one that withstands one or more washings during staining without loss of organisms, is not too thick, and does not result in excessive distortion due to cell shrinkage. The procedure for making such a smear is illustrated in figure 12.1.

The first step in preparing a bacteriological smear differs according to the source of the organisms. If the bacteria are growing in a liquid medium (broths, milk, saliva, urine, etc.), one starts by placing one or two loopfuls of the liquid medium directly on the slide.

From solid media such as nutrient agar, blood agar, or some part of the body, one starts by placing one or two loopfuls of water on the slide and then uses a straight inoculating wire to disperse the organisms in the water. Bacteria growing on solid media tend to cling to each other and must be dispersed sufficiently by dilution in water; unless this is done, the smear will be too thick. *The most difficult concept for students to understand about making slides from solid media is that it takes only a very small amount of material to make a good smear.* When your instructor demonstrates this step, pay very careful attention to the amount of material that is placed on the slide.

The organisms to be used for your first slides may be from several different sources. If the plates from Exercise 9 were saved, some slides may be made from them. If they were discarded, the first slides may be made for Exercise 13, which pertains to simple staining. Your instructor will indicate which cultures to use.

FROM LIQUID MEDIA (Broths, saliva, milk, etc.)

If you are preparing a bacterial smear from liquid media, follow this routine, which is depicted on the left side of figure 12.1.

Materials:

microscope slides
Bunsen burner
wire loop
china marking pencil
slide holder (clothespin), optional

1. Wash a slide with soap or Bon Ami and hot water, removing all dirt and grease. Handle the clean slide by its edges.
2. Write the initials of the organism or organisms on the left-hand side of the slide with a china marking pencil.
3. To provide a target on which to place the organisms, make a $\frac{1}{2}$ " circle on the *bottom* side of the slide, centrally located, with a marking pencil. Later on, when you become more skilled, you may wish to omit the use of this "target circle."
4. Shake the culture vigorously and transfer two loopfuls of organisms to the center of the slide over the target circle. Follow the routine for inoculations shown in figure 12.2. *Be sure to flame the loop after it has touched the slide.*

CAUTION

Be sure to cool the loop completely before inserting it into a medium. A loop that is too hot will spatter the medium and move bacteria into the air.

5. Spread the organisms over the area of the target circle.
6. Allow the slide to dry by normal evaporation of the water. Don't apply heat.
7. After the smear has become completely dry, pass the slide over a Bunsen burner flame to heat-kill the organisms and fix them to the slide.

Note that in this step one has the option of using or not using a clothespin to hold the slide. *Use the option preferred by your instructor.*

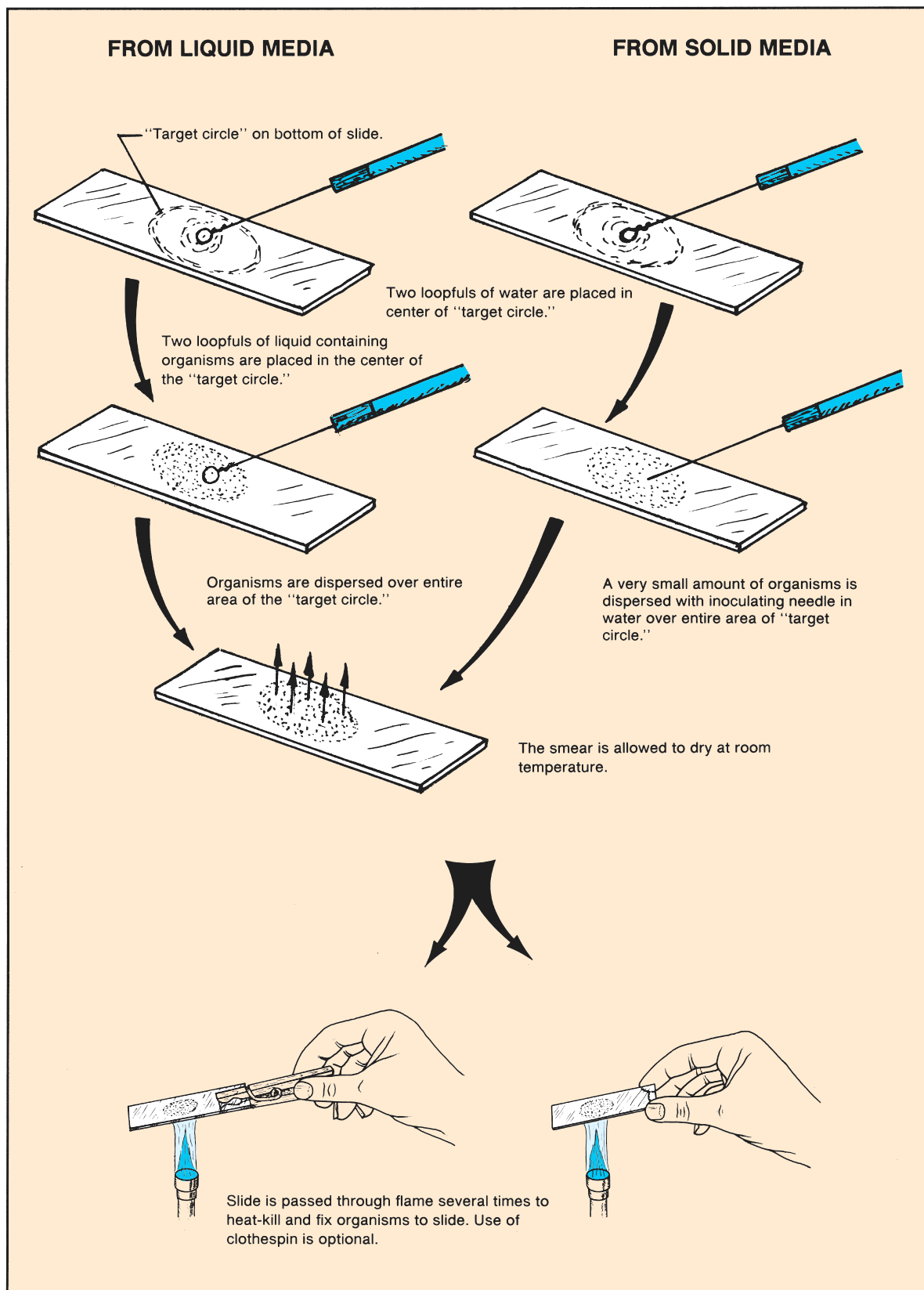


Figure 12.1 Procedure for making a bacterial smear

Exercise 12 • Smear Preparation

FROM SOLID MEDIA

When preparing a bacterial smear from solid media, such as nutrient agar or a part of the body, follow this routine, which is depicted on the right side of figure 12.1.

Materials:

microscope slides
inoculating needle and loop
china marking pencil
slide holder (clothespin), optional
Bunsen burner

1. Wash a slide with soap or Bon Ami and hot water, removing all dirt and grease. Handle the clean slide by its edges.
2. Write the initials of the organism or organisms on the left-hand side of the slide with a china marking pencil.
3. Mark a “target circle” on the bottom side of the slide with a china marking pencil. (See comments in step 3 on page 58.)
4. Flame an inoculating loop, let it cool, and transfer two loopfuls of water to the center of the target circle.
5. Flame an inoculating needle then let it cool. Pick up *a very small amount of the organisms*, and mix it into the water on the slide. Disperse the mixture over the area of the target circle. Be certain that the organisms have been well emulsified in the liquid. *Be sure to flame the inoculating needle before placing it aside.*
6. Allow the slide to dry by normal evaporation of the water. Don’t apply heat.
7. Once the smear is completely dry, pass the slide over the flame of a Bunsen burner to heat-kill the organisms and fix them to the slide. Use a clothespin to hold the slide if it is preferred by your instructor. Some workers prefer to hold the slide with their fingers so that they can monitor the temperature of the slide (to prevent overheating).

LABORATORY REPORT

Answer the questions on Laboratory Report 11–14 that relate to this exercise.

Smear Preparation • Exercise 12

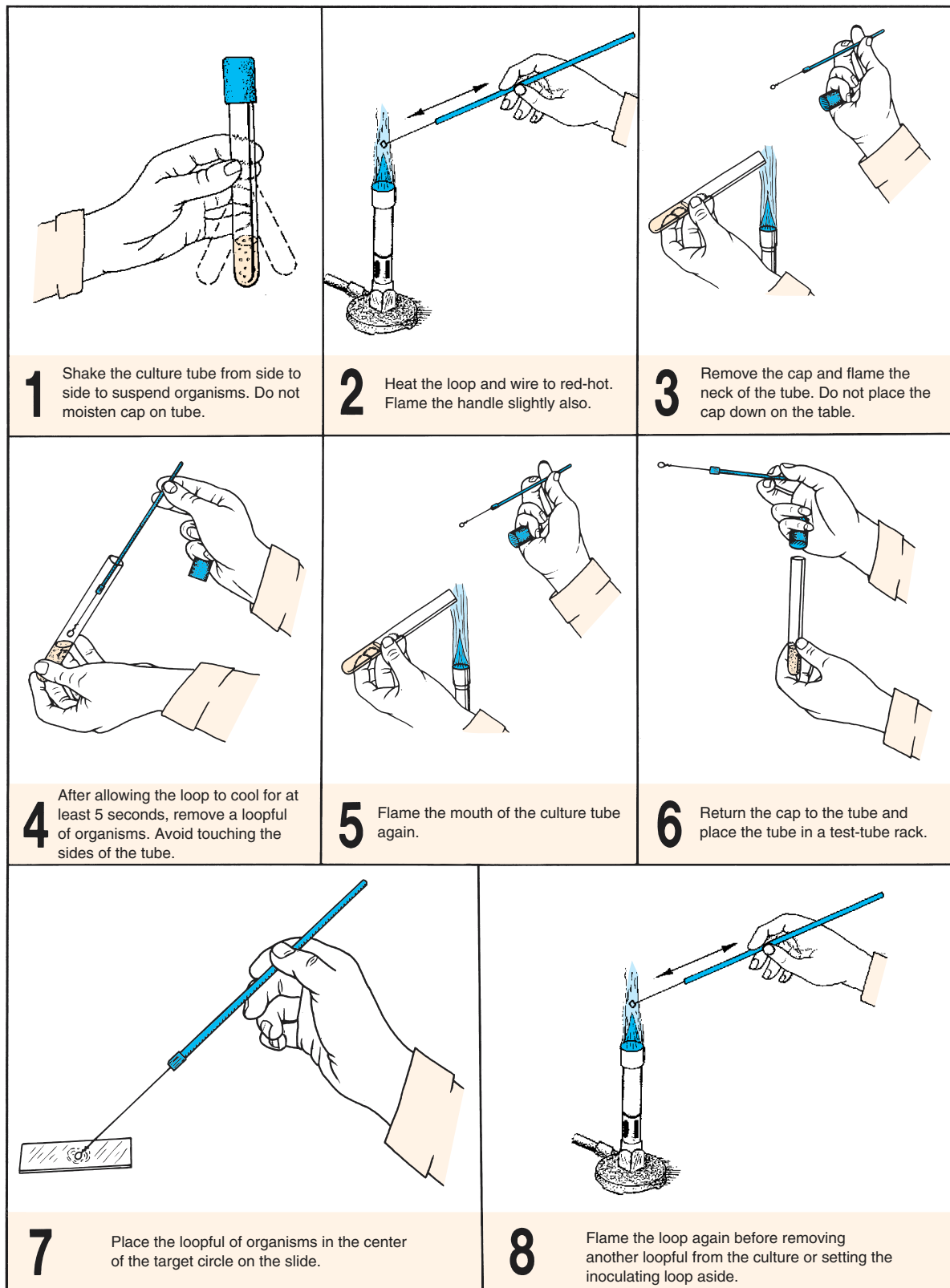


Figure 12.2 Aseptic procedure for organism removal

13

Simple Staining

The use of a single stain to color a bacterial organism is commonly referred to as **simple staining**. Some of the most commonly used dyes for simple staining are methylene blue, basic fuchsin, and crystal violet. All of these dyes work well on bacteria because they have color-bearing ions (*chromophores*) that are positively charged (cationic).

The fact that bacteria are slightly negatively charged produces a pronounced attraction between these cationic chromophores and the organism. Such dyes are classified as **basic dyes**. The basic dye methylene blue (methylene⁺ chloride⁻) will be used in this exercise. Those dyes that have anionic chromophores are called **acidic dyes**. Eosin (sodium⁺ eosinate⁻) is such a dye. The anionic chromophore, eosinate⁻, will not stain bacteria because of the electrostatic repelling forces that are involved.

The staining times for most simple stains are relatively short, usually from 30 seconds to 2 minutes, depending on the affinity of the dye. After a smear has been stained for the required time, it is washed off gently, blotted dry, and examined directly under oil immersion. Such a slide is useful in determining basic morphology and the presence or absence of certain kinds of granules.

An avirulent strain of *Corynebacterium diphtheriae* will be used here for simple staining. In its pathogenic form, this organism is the cause of diphtheria, a very serious disease. One of the steps in identifying this pathogen is to do a simple stain of it to demonstrate the following unique characteristics: pleomorphism, metachromatic granules, and palisade arrangement of cells.

Pleomorphism pertains to irregularity of form: i.e., demonstrating several different shapes. While *C. diphtheriae* is basically rod-shaped, it also appears club-shaped, spermlike, or needle-shaped. *Bergey's Manual* uses the terms "pleomorphic" and "irregular" interchangeably.

Metachromatic granules are distinct reddish-purple granules within cells that show up when the organisms are stained with methylene blue. These granules are considered to be masses of *volutin*, a polymetaphosphate.

Palisade arrangement pertains to parallel arrangement of rod-shaped cells. This characteristic, also called "picket fence" arrangement, is common to many corynebacteria.

PROCEDURE

Prepare a slide of *C. diphtheriae*, using the procedure outlined in figure 13.1. It will be necessary to refer back to Exercise 12 for the smear preparation procedure.

Materials:

- slant culture of avirulent strain of
Corynebacterium diphtheriae
- methylene blue (Loeffler's)
- wash bottle
- bibulous paper

After examining the slide, compare it with the photomicrograph in illustration 1, figure 15.3 (page 66). Record your observations on Laboratory Report 11–14.

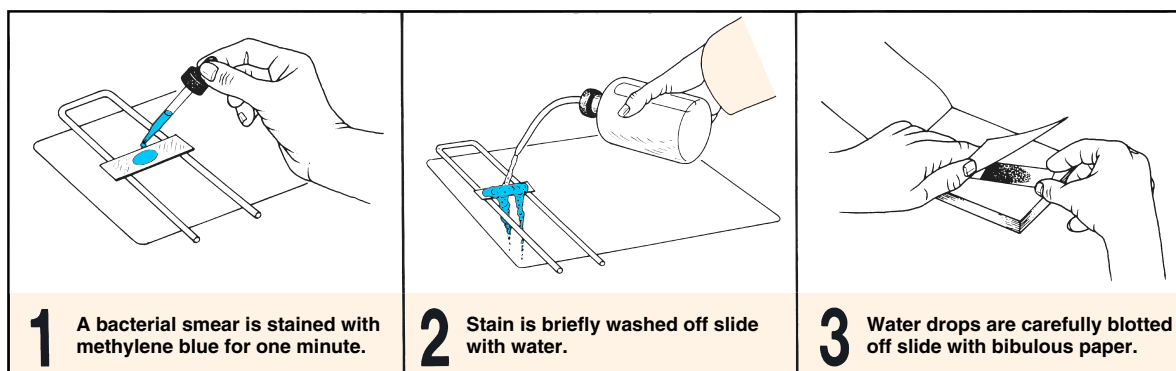


Figure 13.1 Procedure for simple staining

Capsular Staining

14

Some bacterial cells are surrounded by a pronounced gelatinous or slimy layer called a **capsule**. There is considerable evidence to support the view that all bacteria have some amount of slime material surrounding their cells. In most instances, however, the layer is not of sufficient magnitude to be readily discernible. Although some capsules appear to be made of glycoprotein, others contain polypeptides. All appear to be water-soluble.

Staining the bacterial capsule cannot be accomplished by ordinary simple staining procedures. The problem with trying to stain capsules is that if you prepare a heat-fixed smear of the organism by ordinary methods, you will destroy the capsule; and, if you do not heat-fix the slide, the organism will slide off the slide during washing. In most of our bacteriological studies our principal concern is simply to demonstrate

the presence or absence of a pronounced capsule. This can be easily achieved by combining negative and simple staining techniques, as in figure 14.1. To learn about this technique prepare a capsule “stained” slide of *Klebsiella pneumoniae*, using the procedure outlined in figure 14.1.

Materials:

36–48 hour milk culture of *Klebsiella pneumoniae*
india ink
crystal violet

Observation: Examine the slide under oil immersion and compare your slide with illustration 2, figure 15.3 on page 66. Record your results on Laboratory Report 11–14.

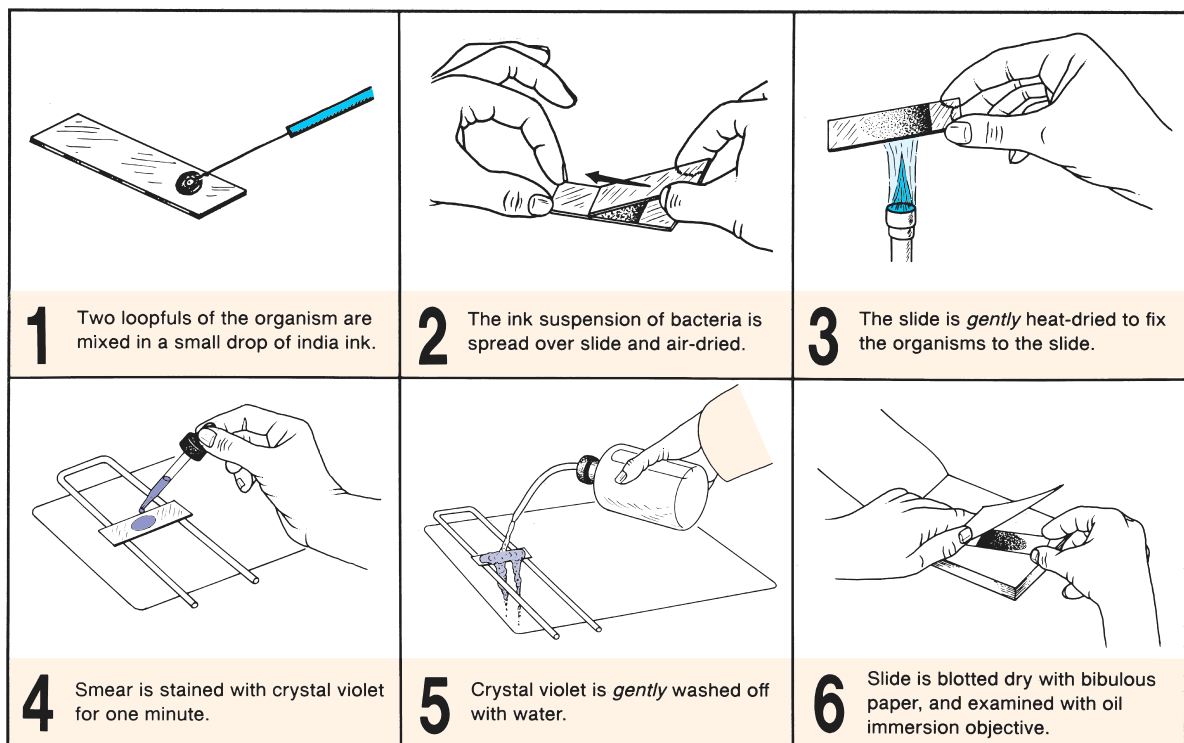


Figure 14.1 Procedure for demonstration of capsule presence

15

Gram Staining

In 1884 the Danish bacteriologist Christian Gram developed a staining technique that separates bacteria into two groups: those that are gram-positive and those that are gram-negative. The procedure is based on the ability of microorganisms to retain the purple color of crystal violet during decolorization with alcohol. Gram-negative bacteria are decolorized by the alcohol, losing the purple color of crystal violet. Gram-positive bacteria are not decolorized and remain purple. After decolorization, safranin, a red counterstain, is used to impart a pink color to the decolorized gram-negative organisms.

Figure 15.1 illustrates the effects of the various reagents on bacterial cells at each stage in the process. Note that crystal violet, the **primary stain**, causes both gram-positive and gram-negative organisms to become purple after 20 seconds of staining. When Gram's iodine, the **mordant**, is applied to the cells for one minute, the color of gram-positive and gram-negative bacteria remains the same: purple. The function of the mordant here is to combine with crystal violet to form a relatively insoluble compound in the gram-positive bacteria. When the **decolorizing agent**, 95% ethanol, is added to the cells for 10–20 seconds, the gram-negative bacteria are leached colorless, but the gram-positive bacteria remain purple. In the final step a **counterstain**, safranin, adds a pink color to the decolorized gram-negative bacteria without affecting the color of the purple gram-positive bacteria.

Of all the staining techniques you will use in the identification of unknown bacteria, Gram staining is, undoubtedly, the most important tool you will use. Although this technique seems quite simple, performing it with a high degree of reliability is a goal that requires some practice and experience. Here are two suggestions that can be helpful: first, don't make your smears too thick, and second, pay particular attention to the comments in step 4 on the next page that pertain to decolorization.

When working with unknowns keep in mind that old cultures of gram-positive bacteria tend to decolorize more rapidly than young ones, causing them to appear gram-negative instead of gram-positive. For reliable results one should use cultures that are approximately 16 hours old. Another point to remember is that some species of *Bacillus* tend to be gram-

REAGENT	GRAM-POS.	GRAM-NEG.
NONE (Heat-fixed Cells)		
CRYSTAL VIOLET (20 seconds)		
GRAM'S IODINE (1 minute)		
ETHYL ALCOHOL (10–20 seconds)		
SAFRANIN (20 seconds)		

Figure 15.1 Color changes that occur at each step in the gram-staining process

variable; i.e., sometimes positive and sometimes negative.

During this laboratory period you will be provided an opportunity to stain several different kinds of bacteria to see if you can achieve the degree of success that is required. Remember, if you don't master this technique now, you will have difficulty with your unknowns later.

STAINING PROCEDURE

Materials:

- slides with heat-fixed smears
- gram-staining kit and wash bottle
- bibulous paper

Gram Staining • Exercise 15

1. Cover the smear with **crystal violet** and let stand for *20 seconds*.
2. Briefly wash off the stain, using a wash bottle of distilled water. Drain off excess water.
3. Cover the smear with **Gram's iodine** solution and let it stand for *one minute*. (Your instructor may prefer only 30 seconds for this step.)
4. Pour off the Gram's iodine and flood the smear with **95% ethyl alcohol** for *10 to 20 seconds*. This step is critical. Thick smears will require more time than thin ones. *Decolorization has occurred when the solvent flows colorlessly from the slide.*
5. Stop action of the alcohol by rinsing the slide with water from wash bottle for a *few seconds*.
6. Cover the smear with **safranin** for *20 seconds*. (Some technicians prefer more time here.)
7. Wash gently for a few seconds, blot dry with bibulous paper, and air-dry.
8. Examine the slide under oil immersion.

ASSIGNMENTS

The organisms that will be used here for Gram staining represent a diversity of form and staining characteristics. Some of the rods and cocci are gram-positive;

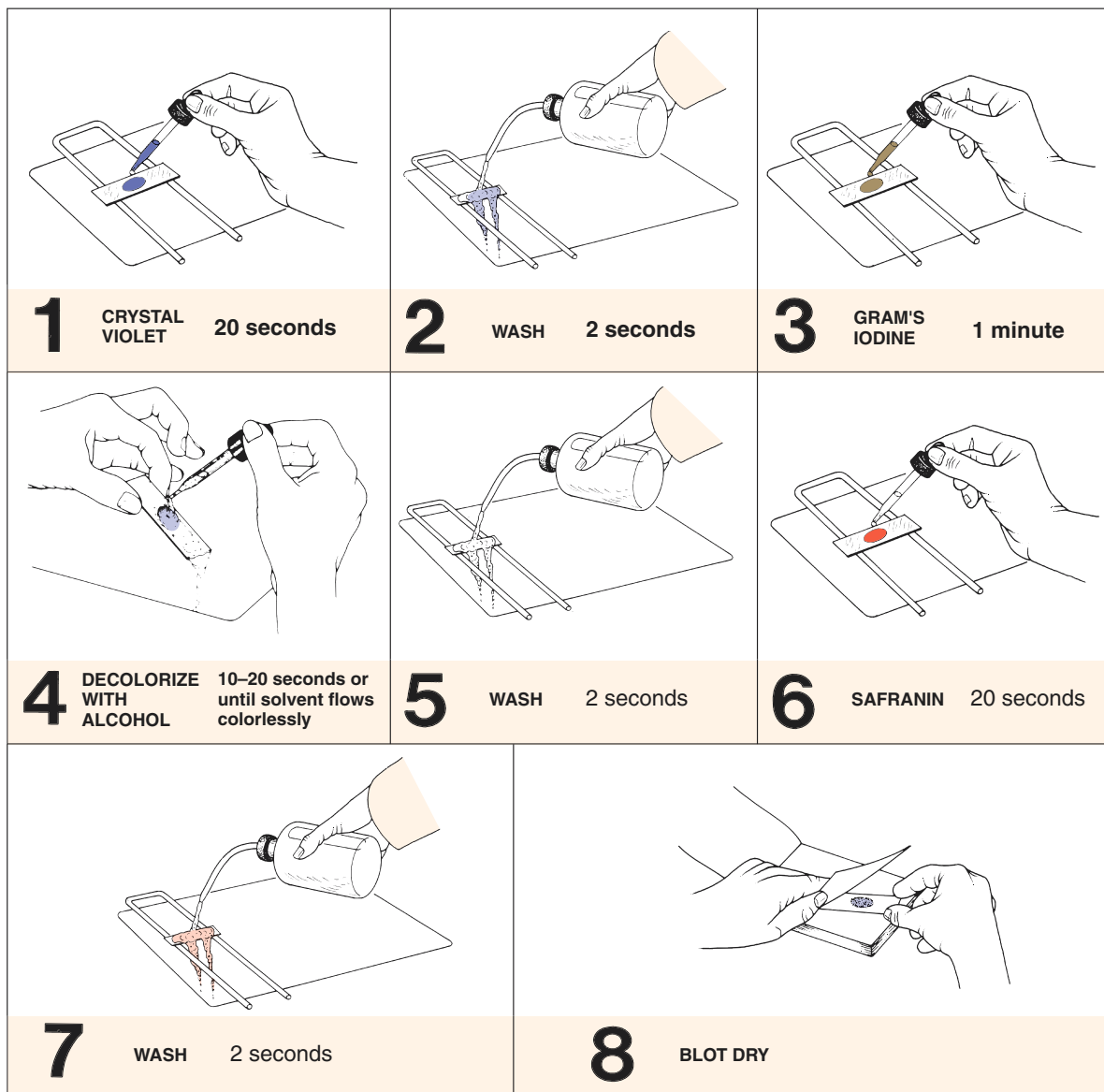


Figure 15.2 The gram-staining procedure

Exercise 15 • Gram Staining

others are gram-negative. One rod-shaped organism is a spore-former and another is acid-fast. The challenge here is to make gram-stained slides of various combinations that reveal their differences.

Materials:

broth cultures of *Staphylococcus aureus*,
Pseudomonas aeruginosa, and *Moraxella*
(*Branhamella*) *catarrhalis*
nutrient agar slant cultures of *Bacillus*
megaterium and *Mycobacterium smegmatis*

Mixed Organisms I (Triple Smear Practice Slides) Prepare three slides with three smears on each slide. On the left portion of each slide make a smear of *Staphylococcus aureus*. On the right portion of each slide make a smear of *Pseudomonas aeruginosa*. In the middle of the slide make a smear that is a mixture of both organisms, using two loopfuls of each organism. **Be sure to flame the loop sufficiently to avoid contaminating cultures.**

Gram stain one slide first, saving the other two for later. Examine the center smear. If done properly, you should see purple cocci and pink rods as shown in illustration 3, figure 15.3.

Call your instructor over to evaluate your slide. If the slide is improperly stained, the instructor will be able to tell what went wrong by examining all three

smears. He or she will inform you how to correct your technique when you stain the next triple smear reserve slide.

Record your results on Laboratory Report 15–18 by drawing a few cells in the appropriate circle.

Mixed Organisms II Make a gram-stained slide of a mixture of *Bacillus megaterium* and *Moraxella* (*Branhamella*) *catarrhalis*.

This mixture differs from the previous slide in that the rods (*B. megaterium*) will be purple and the cocci (*M.B. catarrhalis*) will be large pink diplococci. See illustration 4, figure 15.3.

As you examine this slide look for clear areas on the rods, which represent endospores. Since endospores are refractile and impermeable to crystal violet they will appear as transparent holes in the cells.

Draw a few cells in the appropriate circle on your Laboratory Report sheet.

Acid-Fast Bacteria To see how acid-fast mycobacteria react to Gram's stain, make a gram-stained slide of *Mycobacterium smegmatis*. If your staining technique is correct, the organisms should appear gram-positive.

Draw a few cells in the appropriate circle on your Laboratory Report sheet.

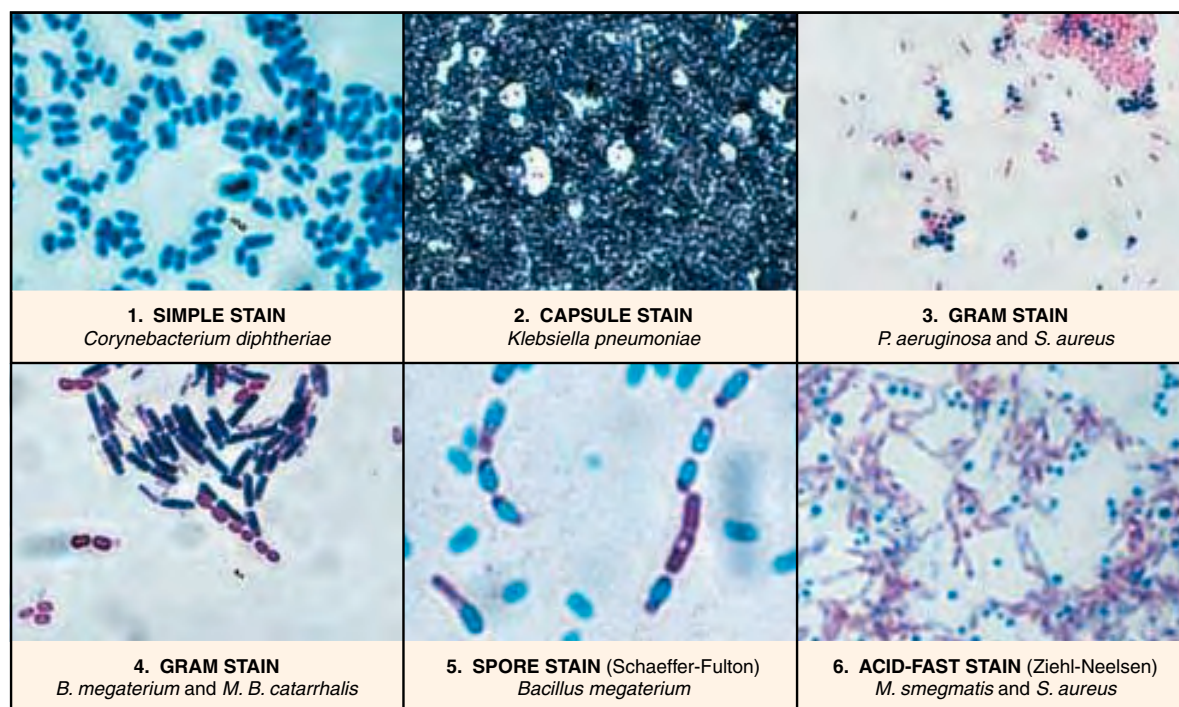


Figure 15.3 Photomicrographs of representative staining techniques (8000x)

Spore Staining: Two Methods

16

Species of bacteria, belonging principally to the genera *Bacillus* and *Clostridium*, produce extremely heat-resistant structures called **endospores**. In addition to being heat-resistant, they are very resistant to many chemicals that destroy non-spore-forming bacteria. This resistance to heat and chemicals is due primarily to a thick, tough spore coat.

It was observed in Exercise 15 that Gram staining will not stain endospores. Only if considerable heat is applied to a suitable stain can the stain penetrate the spore coat. Once the stain has entered the spore, however, it is not easily removed with decolorizing agents or water.

Several methods are available that employ heat to provide stain penetration. However, since the Schaeffer-Fulton and Dorner methods are the principal ones used by most bacteriologists, both have been included in this exercise. Your instructor will indicate which procedure is preferred in this laboratory.

SCHAEFFER-FULTON METHOD

This method, which is depicted in figure 16.1, utilizes malachite green to stain the endospore and safranin to stain the vegetative portion of the cell. Utilizing this technique, a properly stained spore-former will have a green endospore contained in a pink sporangium. Illustration 5, figure 15.3 on page 66 reveals what such a slide looks like under oil immersion.

After preparing a smear of *Bacillus megaterium*, follow the steps outlined in figure 16.1 to stain the spores.

Materials:

24–36 hour nutrient agar slant culture of
Bacillus megaterium
electric hot plate and small beaker (25 ml size)
spore-staining kit consisting of a bottle each of
5% malachite green and safranin

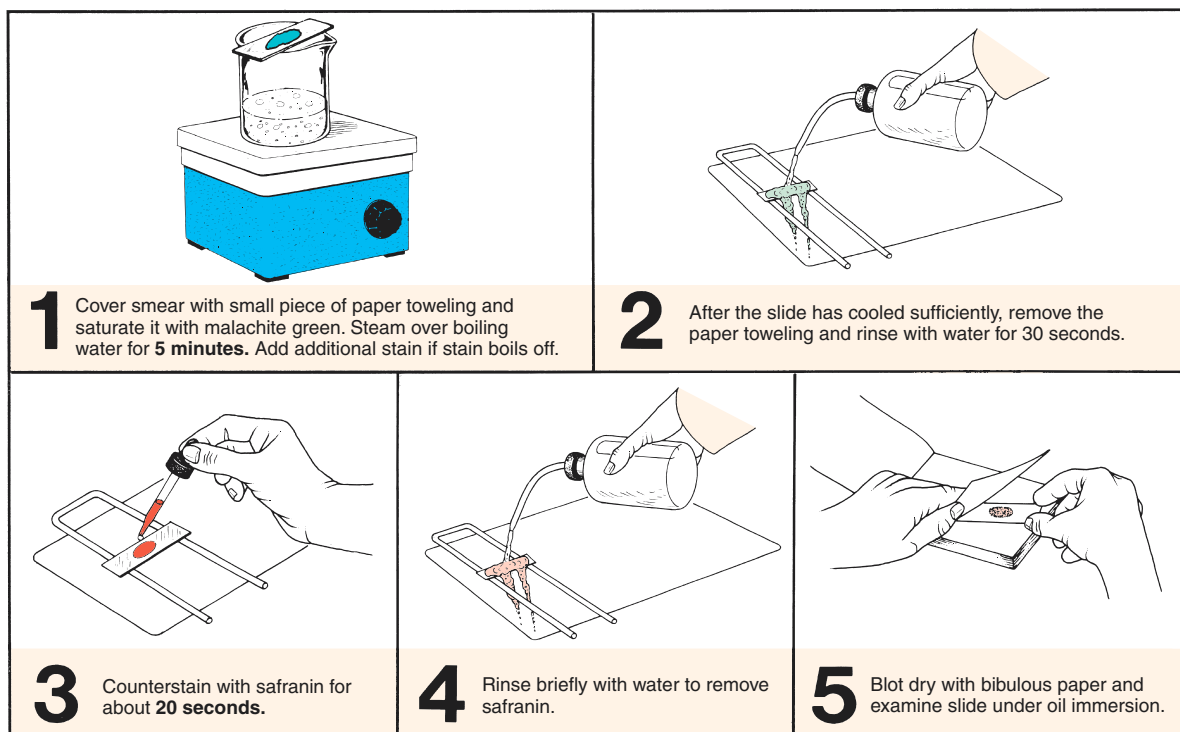


Figure 16.1 The Schaeffer-Fulton spore stain method

Exercise 16 • Spore Staining: Two Methods

DORNER METHOD

The Dorner method for staining endospores produces a red spore within a colorless sporangium. Nigrosine is used to provide a dark background for contrast. The six steps involved in this technique are shown in figure 16.2. Although both the sporangium and endospore are stained during boiling in step 3, the sporangium is decolorized by the diffusion of safranin molecules into the nigrosine.

Prepare a slide of *Bacillus megaterium* that utilizes the Dorner method. Follow the steps in figure 16.2.

Materials:

nigrosine
electric hot plate and small beaker (25 ml size)
small test tube (10 × 75 mm size)
test-tube holder
24–36 hour nutrient agar slant culture of
Bacillus megaterium

LABORATORY REPORT

After examining the organisms under oil immersion, draw a few cells in the appropriate circles on Laboratory Report 15–18.

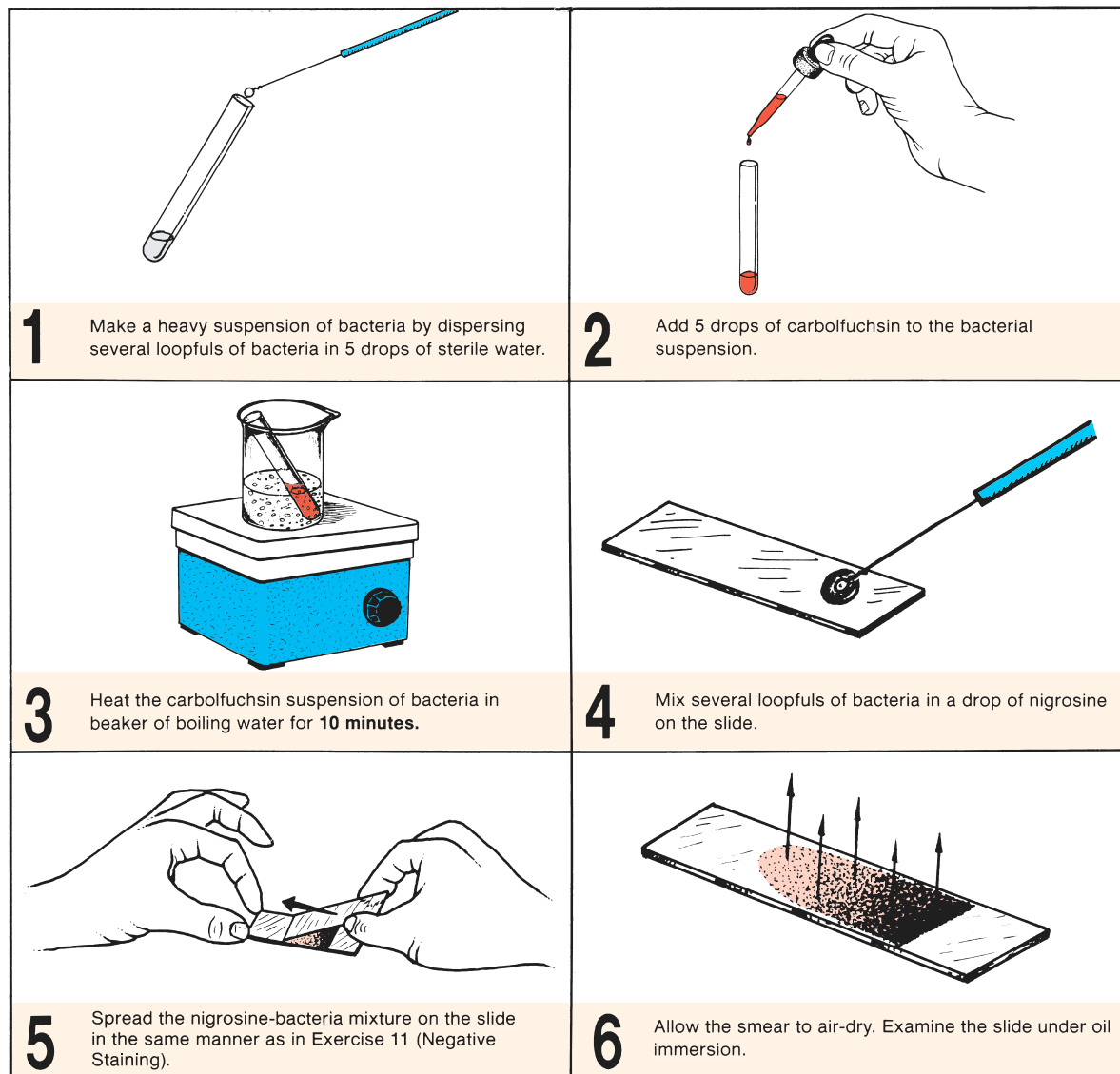


Figure 16.2 The Dorner spore stain method

Acid-Fast Staining: Ziehl-Neelsen Method

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Most bacteria in the genus *Mycobacterium* contain considerable amounts of waxlike lipoidal material, which affects their staining properties. Unlike most other bacteria, once they are properly stained with carbolfuchsin, they resist decolorization with acid-alcohol. Since they are not easily decolorized they are said to be **acid-fast**. This property sets them apart from many other bacteria.

This stain is used primarily in the identification of the tuberculosis bacillus, *Mycobacterium tuberculosis*, and the leprosy organism, *Mycobacterium leprae*. After decolorization, methylene blue is added to the organisms to counterstain any material that is not acid-fast; thus, a properly stained slide of a mixture of acid-fast organisms, tissue cells, and non-acid-fast bacteria will reveal red acid-fast rods with bluish tissue cells and bacteria. An example of acid-fast staining is shown in illustration 6 of figure 15.3.

The two organisms used in this staining exercise are *Mycobacterium smegmatis*, a nonpathogenic acid-fast rod found in soil and on external genitalia, and *Staphylococcus aureus*, a non-acid-fast coccus.

Materials:

nutrient agar slant culture of *Mycobacterium smegmatis* (48-hour culture)
nutrient broth culture of *S. aureus*
electric hot plate and small beaker
acid-fast staining kit (carbolfuchsin, acid alcohol, and methylene blue)

Smear Preparation Prepare a mixed culture smear by placing two loopfuls of *S. aureus* on a slide and transferring a small amount of *M. smegmatis* to the broth on the slide with an inoculating needle. Since the smegma bacilli are waxy and tend to cling to each other in clumps, break up the masses of organisms with the inoculating needle. After air-drying the smear, heat-fix it.

Staining Follow the staining procedure outlined in figure 17.1.

Examination Examine under oil immersion and compare your slide with illustration 6, figure 15.3.

Laboratory Report Record your results on Laboratory Report 15–18.

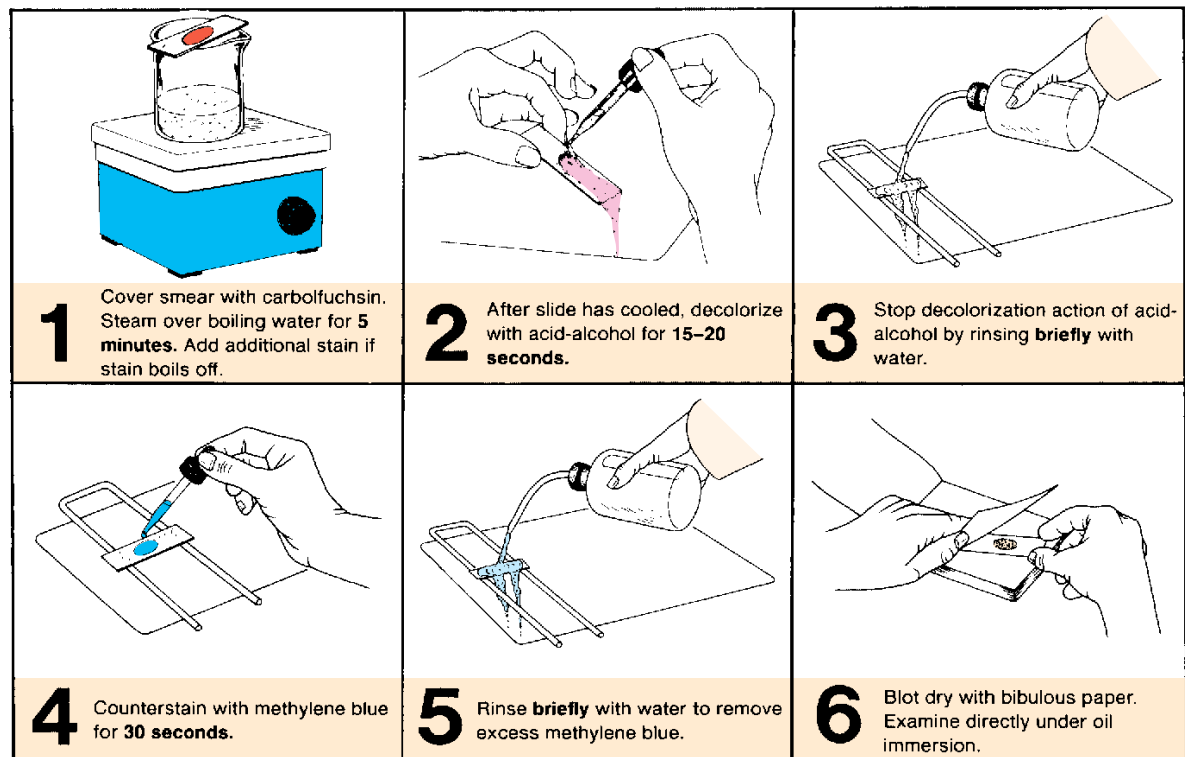


Figure 17.1 Ziehl-Neelsen acid-fast staining procedure

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Acid-Fast Staining: Fluorescence Method

In laboratories where large numbers of sputum, gastric washings, urine, and other body fluid samples are tested for pathogenic mycobacteria, **fluorochrome acid-fast staining** is used in conjunction with the Ziehl-Neelsen technique. The advantage of using a fluorescence method is that fluorochrome-stained slides can be scanned under lower magnification. While a Ziehl-Neelsen prepared slide must be examined under oil immersion (1000× magnification), fluorochrome-stained slides can be examined with 60× or 100× magnification. In only a few minutes an entire fluorochrome prepared slide can be scanned. Because of this fact, many laboratories use this faster technique as a screening tool. When they encounter a positive slide with this method, they use a Ziehl-Neelsen prepared slide as a means of confirmation. The fact that dead or noncultivable mycobacteria may fluoresce makes it necessary to use a confirmatory technique.

The **Truant method** of fluorochrome staining (figure 18.1) consists of staining smears with auramine-rhodamine for 20 minutes, decolorizing with acid-alcohol, and “counterstaining” with potassium permanganate. As soon as the slides are dry, they are examined with a fluorescence microscope. Bacteria that are acid-fast will fluoresce as yellow-orange rods in a dark field. Areas of fluorescence that show up during scanning can be examined more critically under high-dry or oil immersion.

In this exercise you will stain a mixture of *Staphylococcus aureus* and *Mycobacterium phlei* by the Truant method. It will be examined with a fluorescence microscope. If the slide is prepared properly, only the acid-fast rod-shaped mycobacteria will fluoresce.

Materials:

broth culture of *S. aureus*
slant culture of *M. phlei* (Lowenstein-Jensen medium)
auramine-rhodamine stain
acid-alcohol (for fluorochrome staining)
potassium permanganate (0.5% solution)
microscope slides, inoculating loop, Bunsen burner, wash bottle

1. Prepare a mixed smear of *S. aureus* and *M. phlei* by adding a small amount of *M. phlei* to two loopfuls of *S. aureus* on a clean slide. The organisms should be well dispersed on the slide by vigorous manipulation of the inoculating loop on the clumps of organisms.
2. Allow the smear to air-dry completely.
3. Flame-fix the slide over a Bunsen burner. Avoid overheating.

In diagnostic work where pathogens are being stained, the smear is usually heat-fixed on a slide warmer (65° C) for 2 hours.

4. Cover the smear with **auramine-rhodamine** stain and let stand for **20 minutes** at room temperature.
5. Rinse off the stain with wash bottle.
6. Decolorize with **acid-alcohol** (2.5% HCl in 70% ethanol) for **7–10 seconds**.
7. Rinse thoroughly with wash bottle.
8. Cover the smear with **potassium permanganate** and let stand for **3 minutes**. This solution eliminates background fluorescence (“quenching”). Although this step is often referred to as “counterstaining,” in actuality it is not. Excessive counterstaining must be avoided because fluorescence will be completely eliminated.
9. Rinse with water and air-dry.

OBSERVATION

The fluorescence microscope should be equipped with a BG12 exciter filter and an OG1 barrier filter. Scan the slide with the lowest magnification that is possible on the microscope. On some instruments this may be high-dry, not low power. If high-dry is used, it is necessary to place a cover glass on the smear with immersion oil between the slide and cover glass. Be sure to use only oil that is specific for fluorescence viewing.

Once you have located areas of fluorescence (yellow-orange spots), add oil and swing the oil immersion lens into position for more critical observation. Record your results on Laboratory Report 15–18.

Acid-Fast Staining: Fluorescence Method • Exercise 18

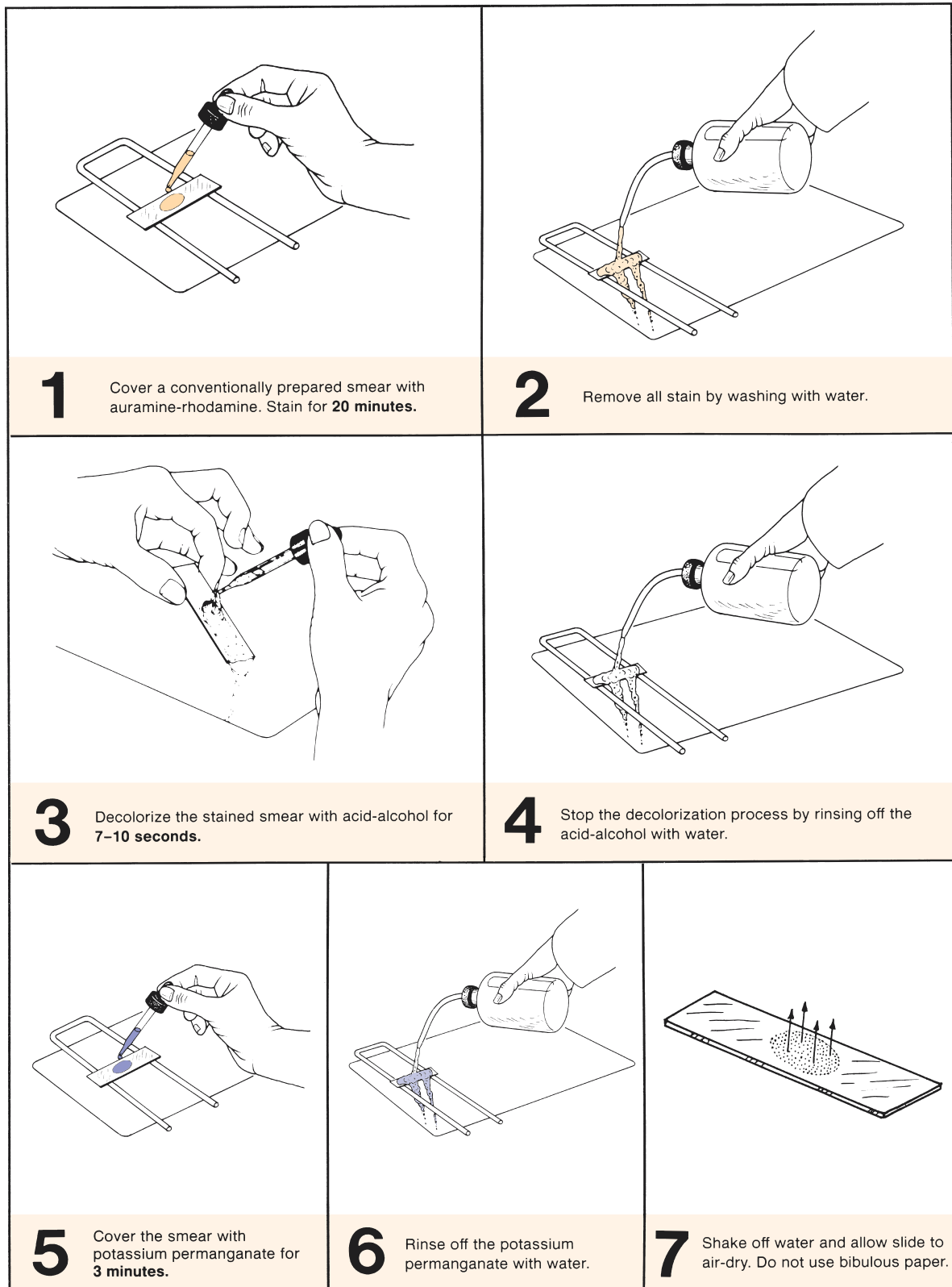


Figure 18.1 Fluorochrome acid-fast staining routine

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Motility Determination

When attempting to identify an unknown bacterium it is usually necessary to determine whether the microorganism is motile. Although one might think that this determination would be easily arrived at, such is not always the case. For the beginner there are many opportunities to err.

FOUR METHODS

For nonpathogens, there are two slide techniques that one might use. For pathogens, one tube and one Petri plate method can be used. Each method has its advantages and limitations. The method you use will depend on which one is most suitable for the situation at hand. A discussion of each procedure follows.

The Wet Mount Slide

When working with nonpathogens, the simplest way to determine motility is to place a few loopfuls of the organism on a clean slide and cover it with a cover glass. In addition to being able to determine the presence or absence of motility, this method is useful in determining cellular shape (rod, coccus, or spiral) and arrangement (irregular clusters, packets, pairs, or long chains). A wet mount is especially useful if phase optics are used. Unlike stained slides that are heat-fixed for staining, there is no distortion of cells on a wet mount.

One problem for beginners is the difficulty of being able to see the organisms on the slide. Since bacteria are generally colorless and very transparent, the novice has to learn how to bring them into focus.

The Hanging Drop Slide

If it is necessary to study viable organisms on a microscope slide for a longer period of time than is possible with a wet mount, one can resort to a hanging drop slide. As shown in illustration 4 of figure 19.1, organisms are observed in a drop that is suspended under a cover glass in a concave depression slide. Since the drop lies within an enclosed glass chamber, drying out occurs very slowly.

Tube Method

When working with pathogenic microorganisms such as the typhoid bacillus, it is too dangerous to attempt to de-

termine motility with slide techniques. A much safer method is to culture the organisms in a special medium that can demonstrate the presence of motility. The procedure is to inoculate a tube of semisolid or SIM medium that can demonstrate the presence of motility. Both media have a very soft consistency that allows motile bacteria to migrate readily through them causing cloudiness. Figure 19.2 illustrates the inoculation procedure.

Soft Agar Plate Method

Although the tube method is the generally accepted procedure for determining motility of pathogens, it is often very difficult for beginners to interpret. Richard Roller at the University of Iowa suggests that incubating a Petri plate of soft agar that has been stab inoculated with a motile organism will show up motility more clearly than an inoculated tube. This method will also be tried here in this laboratory period.

FIRST PERIOD

During the first period you will make wet mount and hanging drop slides of two organisms: *Proteus vulgaris* and *Micrococcus luteus*. Tube media (semisolid medium or SIM medium) and a soft agar plate will also be inoculated. The media inoculations will have to be incubated to be studied in the next period. Proceed as follows:

Materials:

- microscope slides and cover glasses
- depression slide
- 2 tubes of semisolid or SIM medium
- 1 Petri plate of soft nutrient agar (20–25 ml of soft agar per plate)
- nutrient broth cultures of *Micrococcus luteus* and *Proteus vulgaris* (young cultures)
- inoculating loop and needle
- Bunsen burner

Wet Mounts Prepare wet mount slides of each of the organisms, using several loopfuls of the organism on the slides. Examine under an oil immersion objective. Observe the following guidelines:

- Use only scratch-free, clean slides and cover glasses. This is particularly important when using phase-contrast optics.

Motility Determination • Exercise 19

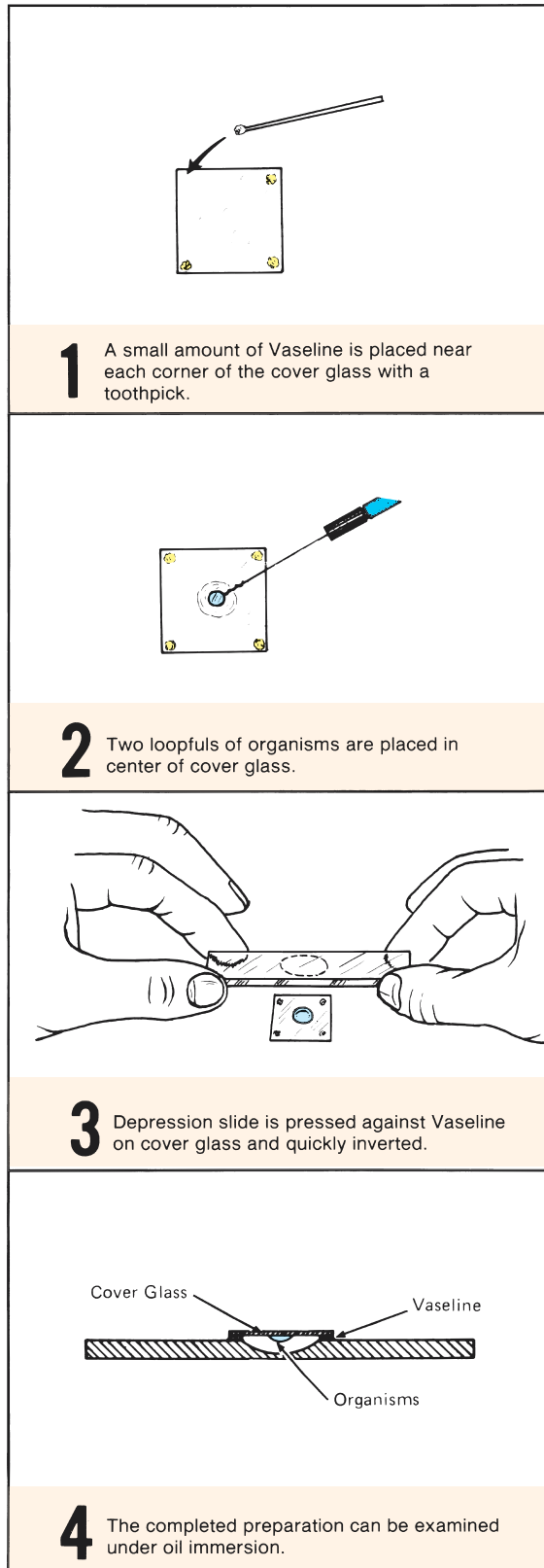


Figure 19.1 The hanging drop slide

- Label each slide with the name of the organism.
- By manipulating the diaphragm and voltage control, reduce the lighting sufficiently to make the organisms visible. Unstained bacteria are very transparent and difficult to see.
- For proof of true motility, look for directional movement that is several times the long dimension of the bacterium. The movement will also occur in different directions in the same field.
- Ignore Brownian movement. *Brownian movement* is vibrational movement caused by invisible molecules bombarding bacterial cells. If the only movement you see is vibrational and not directional, the organism is nonmotile.
- If you see only a few cells exhibiting motility, consider the organism to be motile. Characteristically, only a few of the cells will be motile at a given moment.
- Don't confuse water current movements with true motility. Water currents are due to capillary action caused by temperature changes and drying out. All objects move in a straight line in one direction.
- And, finally, always *examine a wet mount immediately*, once it has been prepared, because motility decreases with time after preparation.

Hanging Drop Slides By referring to figure 19.1 prepare hanging drop slides of each organism. Be sure to use clean cover glasses and label each slide with a china marking pencil. When placing loopfuls of organisms on the cover glass, be sure to flame the loop between applications. Once the slide is placed on the microscope stage, do as follows:

1. Examine the slide first with the low-power objective. If your microscope is equipped with an automatic stop, avoid using the stop; instead, use the coarse adjustment knob for bringing the image into focus. The greater thickness of the depression slide prevents one from being able to focus at the stop point.
2. Once the image is visible under low power, swing the high-dry objective into position and readjust the lighting. Since most bacteria are drawn to the edge of the drop by surface tension, **focus near the edge of the drop.**
3. If your microscope has phase-contrast optics, switch to high-dry phase. Although a hanging drop does not provide the shallow field desired for phase-contrast, you may find that it works fairly well.
4. If you wish to use oil immersion, simply rotate the high-dry objective out of position, add immersion oil to the cover glass, and swing the oil immersion lens into position.